

Random Signals and Noise

Probability, Random Processes and Communication Applications

An academic course with diagrams and fully worked lesson exercises.

The course develops the GJWHC material and the concepts of Chapter 5, Probability and Random Processes, in John G. Proakis and Masoud Salehi, Fundamentals of Communication Systems, second edition. Application lessons supply the additional prerequisites needed to solve the uploaded modulation, quantization and detection questions.



From a probability experiment to a filtered random signal.

The supplied Chapter 2 establishes the notation. Taylor Series provides the typography and presentation reference. Its mathematical content is not used in this course.

All solved questions appear at the end of the lesson that develops their main prerequisites. Overlapping screenshots are combined; numerical specializations remain identifiable.

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Notation and conventions

Symbol	Definition
X, x	Random variable and one possible value; $X(t)$ is a random process and $x(t)$ one realization.
$F_X(x), p_X(x), f_X(x)$	CDF, probability mass function, and probability density, respectively.
μ_X, σ_X^2, C_X	Mean, variance, and centered covariance.
$R_X(\tau)$	$E[X(t+\tau)X(t)]$ for a real WSS process.
$R_{XY}(\tau)$	$E[X(t+\tau)Y(t)]$; the first process carries the lag.
$S_X(f), S_{XY}(f)$	Fourier transforms of the corresponding correlation functions. Frequency f is in hertz.
$H(f), h(t)$	Frequency response and impulse response of an LTI filter.
$N_0/2$	Two-sided white-noise PSD. It is a spectral density, not a finite point variance.
B, W_{BP}	Lowpass positive edge B ; full positive-lobe bandpass width $W_{BP} = 2B$ when centered at a carrier.

The Fourier convention is $S_X(f) = \int R_X(\tau) e^{-j2\pi f\tau} d\tau$, with a positive exponential in the inverse transform. The normalized function is $\text{sinc}(u) = \sin(\pi u)/(\pi u)$, with value one at zero.

Powers are mean-square values on a unit resistance unless a load is specified. Independence conclusions involving Gaussian quantities require a joint Gaussian law. Assumptions that affect a worked answer are stated with that question.

Reference coverage

Chapter 5 section	Course lessons
5.1.1–5.1.2: events and conditional probability	01
5.1.3: random variables and distributions	02
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Exercise concepts and CP5.1–CP5.6	Relevant theory lessons and 15

The reference theory occupies printed pp. 190–236; its regular and computer problems continue through p. 254. Lessons 12–14 are communication applications required by the supplied questions. Repeated GJWHC versions contribute one combined set of concepts.

Lesson 01. Events, Conditional Probability, and Bayesian Inference

Reference: Proakis and Salehi, Chapter 5, Sections 5.1.1–5.1.2, pp. 190–194; GJWHC probability foundation.

1.1 Probability models in communication engineering

Communication systems carry information that the receiver does not know beforehand. Received signals also contain disturbances whose precise values cannot be predicted. Probability provides a quantitative description of this uncertainty. A probability model identifies the possible outcomes, the events of interest, and the probabilities assigned to those events. It can describe a transmitted bit, a channel error, a noise voltage, or the outcome of a measurement.

A random experiment is a procedure with an uncertain outcome. Its sample space, denoted by Ω , is the set of all possible outcomes. An individual outcome is denoted by $\omega \in \Omega$. The sample space of one die throw is $\Omega = \{1, 2, 3, 4, 5, 6\}$. A sample space is discrete when it is finite or countably infinite. Choosing a real number in an interval produces an uncountable sample space. Countability concerns the outcomes themselves, rather than the precision with which a measuring instrument records them.

An event A is a specified set of outcomes. For a die, the event that the result is even is $A = \{2, 4, 6\}$. The union $A \cup B$ means that at least one event occurs; the intersection $A \cap B$ means that both occur. The complement A^c means that A does not occur. The empty set $\emptyset$ represents the impossible event. Events are disjoint when $A \cap B = \emptyset$.

1.2 Probability axioms and their consequences

A probability measure assigns a number $P(A)$ to each admissible event and satisfies

$$P(A) \geq 0, \quad P(\Omega) = 1,$$

and, for a countable collection of pairwise disjoint events,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

These axioms imply $0 \leq P(A) \leq 1$ and

$$P(\emptyset) = 0, \quad P(A^c) = 1 - P(A).$$

For two events that may overlap, separate their common outcomes before adding probabilities. The result is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) .$$

If $A \subseteq B$, every occurrence of A is also an occurrence of B , so $P(A) \leq P(B)$. For a finite sample space with equally likely outcomes,

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} .$$

Equal likelihood must be stated or justified; it does not follow merely from a finite sample space. For a fair die, $P(A) = 3/6 = 1/2$ for the even event.

A probability can also be interpreted through repeated independent trials. If N_A occurrences of A are observed in N trials, N_A/N is an empirical relative frequency. Under the usual repeated-trial assumptions, it approaches $P(A)$ as the number of trials increases. For a finite record it fluctuates, so it should be treated as an estimate rather than an exact probability.

1.3 Conditional probability

Information about an observed event changes the relevant sample space. For $P(B) > 0$, the probability of A given that B occurred is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} .$$

The numerator counts outcomes compatible with both statements; the denominator normalizes over outcomes compatible with the observation. For a fair die, let $A = \{4, 5, 6\}$ and $B = \{2, 4, 6\}$. Then

$$P(A | B) = \frac{2/6}{3/6} = \frac{2}{3} ,$$

whereas $P(A) = 1/2$. Knowledge that the outcome was even changes the probability of obtaining a number greater than three.

The definition gives the multiplication rules

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A) ,$$

whenever the corresponding conditioning probabilities are positive. The elementary ratio does not define conditioning on an event of probability zero. Conditioning on an exact continuous value is handled later through conditional densities.

1.4 Independence and disjointness

Events A and B are independent when

$$P(A \cap B) = P(A) P(B) .$$

If $P(B) > 0$, this is equivalent to $P(A | B) = P(A)$: learning that B occurred does not change the probability of A . Independence describes an absence of probabilistic influence. Disjointness describes an impossibility of simultaneous occurrence. Two disjoint events with positive probabilities cannot be independent, because their intersection has probability zero while $P(A) P(B) > 0$.

For several events, mutual independence requires the product rule for every nonempty subcollection. Pairwise independence alone does not establish mutual independence. In communication models, independent symbol generation and independent channel errors are assumptions about the mechanism and must not be inferred solely from a limited set of observed coincidences.

1.5 Total probability and Bayes' rule

Events $E_1, \dots, E_m$ form a partition of Ω when they are disjoint and their union is the entire sample space. Any event A can then be separated into the alternatives $A \cap E_i$. Consequently,

$$P(A) = \sum_{i=1}^m P(A | E_i) P(E_i) .$$

This is the law of total probability. It expresses an unconditional probability as a weighted sum of conditional probabilities.

If $P(A) > 0$, Bayes' rule reverses the conditioning:

$$P(E_i | A) = \frac{P(A | E_i) P(E_i)}{\sum_{j=1}^m P(A | E_j) P(E_j)} .$$

Here $P(E_i)$ is the prior probability, $P(A | E_i)$ is the likelihood of the observation under alternative E_i , and $P(E_i | A)$ is the posterior probability. The denominator ensures that the posterior probabilities sum to one.

To see the engineering meaning, let X be the transmitted bit and Y the received bit. Suppose

$$P(X = 0) = 0.3, \quad P(X = 1) = 0.7,$$

$$P(Y = 1 | X = 0) = 0.01, \quad P(Y = 0 | X = 1) = 0.10.$$

A received one can arise from a correct transmitted one or an erroneous transmitted zero. Total probability gives

$$P(Y = 1) = 0.01 (0.3) + 0.90 (0.7) = 0.633.$$

After observing $Y = 1$, Bayes's rule gives

$$P(X = 1 \mid Y = 1) = \frac{0.90 (0.7)}{0.633} \approx 0.99526.$$

The probability of a received value and the probability of its transmitted origin answer different questions. Keeping their conditioning explicit is essential when deriving error probabilities and receiver decisions.

Lesson 02. Random Variables, Distribution Functions, and Standard Probability Laws

Reference: Proakis and Salehi, Section 5.1.3, pp. 194–201; Poisson enrichment from Problem 5.16; exponential laws from Problem 5.21; GJWHC Gaussian/PAM material.

2.1 Random variables and their numerical values

A random variable X assigns a real number to each outcome of a random experiment:

$$X: \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto X(\omega).$$

The capital letter X denotes the random variable; the lowercase letter x denotes a possible numerical value. For example, a binary source can assign $X = 1$ to one outcome and $X = 0$ to another. A discrete random variable takes values in a finite or countable set. The continuous variables used here possess probability densities. A mixed variable has both probability masses and a continuous part.

2.2 The cumulative distribution function

The cumulative distribution function, or CDF, provides a common description for all these types:

$$F_X(x) = P(X \leq x).$$

It lies between zero and one, is nondecreasing and right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

Write $F_X(a^-) = \lim_{x \uparrow a} F_X(x)$. Then

$$P(a < X \leq b) = F_X(b) - F_X(a),$$

$$P(a \leq X \leq b) = F_X(b) - F_X(a^-),$$

$$P(X = a) = F_X(a) - F_X(a^-).$$

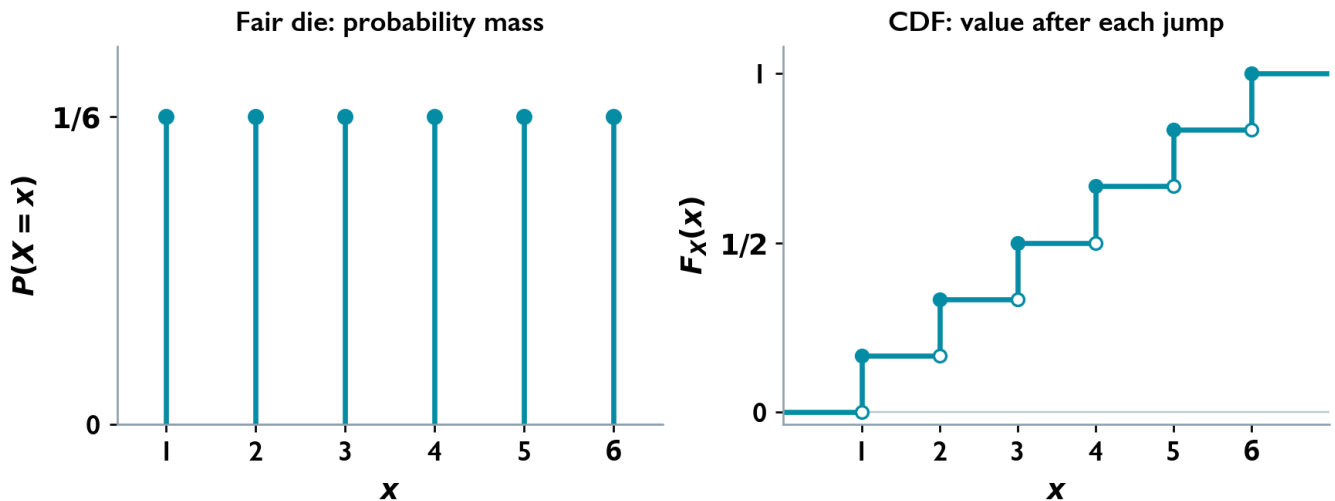
A jump therefore represents probability concentrated at one value. Endpoint distinctions matter for discrete and mixed variables even though they disappear for density-bearing continuous variables.

2.3 Probability mass and probability density

For a discrete variable with possible values x_k , the probability mass function, or PMF, is

$$p_X(x_k) = P(X = x_k), \quad p_X(x_k) \geq 0, \quad \sum_k p_X(x_k) = 1.$$

Its CDF is $F_X(x) = \sum_{x_k \leq x} p_X(x_k)$. Thus a discrete CDF is a staircase whose jump sizes equal the probability masses.



Probability masses and right-continuous CDF of a fair die.

For an absolutely continuous variable, the probability density function, or PDF, satisfies

$$F_X(x) = \int_{-\infty}^x f_X(u) du, \quad f_X(x) = \frac{dF_X(x)}{dx}$$

where the derivative exists. The density is nonnegative and integrates to one. Probabilities are areas:

$$P(a < X < b) = \int_a^b f_X(x) dx.$$

A density value is not a point probability. It may exceed one, and its units are the reciprocal of the units of X . A continuous variable has $P(X = x) = 0$ at every individual value. The derivative representation applies to absolutely continuous laws; a continuous CDF alone does not guarantee an ordinary density.

A mixed example assigns probability 1/2 to $X = 0$ and probability 1/2 to a uniform draw in $(0, 1)$. Its CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2} + \frac{x}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

The jump at zero contributes mass 1/2. The ordinary derivative equals 1/2 on $(0, 1)$ and accounts for only the continuous half of the probability. Both parts must be retained.

2.4 Bernoulli, binomial, and Poisson laws

A Bernoulli variable takes values zero and one:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad 0 \leq p \leq 1.$$

It models a binary symbol or an error indicator. If n independent Bernoulli trials share the same success probability p , the total number of successes has the binomial PMF

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

The combinatorial factor counts the positions of the k successes. Without independence or a common probability, this formula need not apply.

A Poisson count has PMF

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots, \quad \lambda > 0.$$

It describes many count models and is also the rare-event limit of a binomial law as $n \rightarrow \infty$, $p \rightarrow 0$, and $np \rightarrow \lambda$. Using this approximation for a finite binomial model requires that the rare-event assumptions be appropriate.

2.5 Uniform and exponential laws

For $X \sim \mathcal{U}[a, b]$, $a < b$, equal-length subintervals have equal probabilities:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise,} \end{cases}$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

For $a \leq c < d \leq b$, the probability of (c, d) is $(d - c) / (b - a)$. Random carrier phase is frequently modeled by a uniform law on one full period.

An exponential variable with rate $\alpha > 0$ has

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\alpha x}, & x \geq 0. \end{cases}$$

Its tail is $P(X > x) = e^{-\alpha x}$ for $x \geq 0$. Therefore

$$P(X > s + t \mid X > s) = e^{-\alpha t} = P(X > t), \quad s, t \geq 0.$$

This memoryless property is useful in waiting-time models. If X is measured in seconds, α is measured in inverse seconds.

2.6 Gaussian variables and standardization

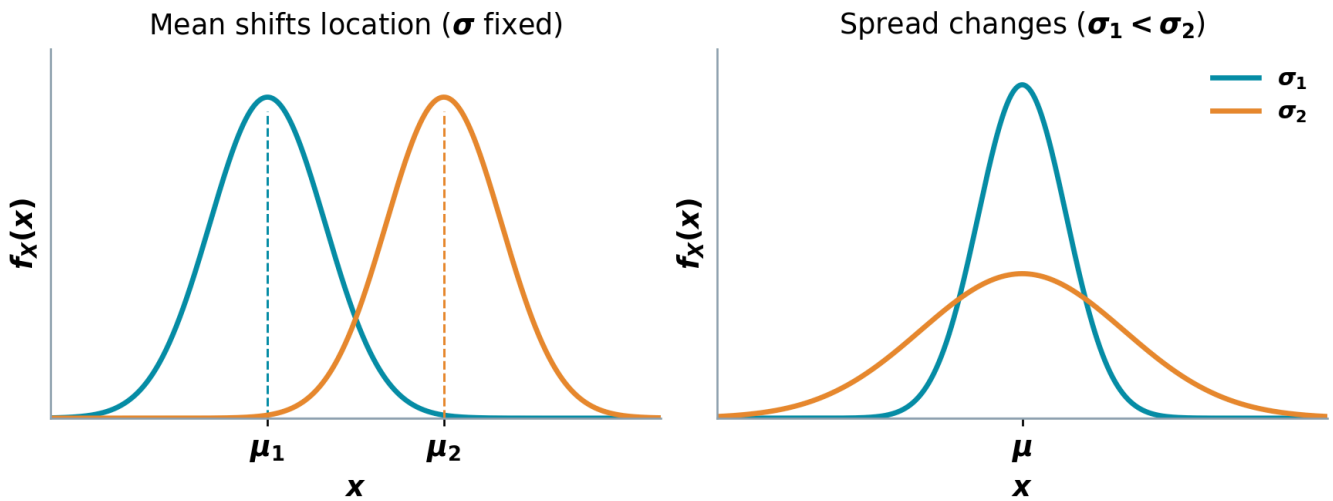
A Gaussian or normal variable $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$, with $\sigma_X > 0$, has density

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} \right], \quad -\infty < x < \infty.$$

The mean sets its location, and the standard deviation sets its spread. Define

$$Z = \frac{X - \mu_X}{\sigma_X}.$$

Then $Z \sim \mathcal{N}(0, 1)$. This change of variables reduces every Gaussian probability calculation to the same standard law.



Standardization: $Z = (X - \mu)/\sigma$

Location, spread, and standardization of Gaussian distributions.

Let $\Phi(z) = P(Z \leq z)$ and define the upper-tail function

$$Q(z) = 1 - \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-u^2/2} du.$$

Symmetry gives $Q(-z) = 1 - Q(z)$, while $Q(0) = 1/2$ and $Q(z) \rightarrow 0$ as $z \rightarrow \infty$. In particular,

$$P(X > x) = Q\left(\frac{x - \mu_X}{\sigma_X}\right),$$

$$P(a < X < b) = Q\left(\frac{a - \mu_X}{\sigma_X}\right) - Q\left(\frac{b - \mu_X}{\sigma_X}\right).$$

Useful bounds are

$$Q(z) \leq \frac{1}{2}e^{-z^2/2}, \quad z \geq 0,$$

$$Q(z) < \frac{e^{-z^2/2}}{z\sqrt{2\pi}}, \quad z > 0,$$

$$Q(z) > \frac{1 - z^{-2}}{z\sqrt{2\pi}}e^{-z^2/2}, \quad z > 1.$$

For binary antipodal signaling, receive $Y = \pm A + N$, where $A > 0$ and $N \sim \mathcal{N}(0, \sigma^2)$. With equally likely symbols and a zero decision threshold,

$$P(\text{error} \mid +A) = P(N < -A) = Q(A/\sigma).$$

The other conditional error probability is identical by symmetry, so $P_e = Q(A/\sigma)$. This result connects a Gaussian tail probability to a measurable communication-system error rate.

Solved Exercises for Lesson 02

Exercise 2.1. Recovering a density from a cubic CDF

IMG_8623.pdf, PDF page(s) 1.

Problem. A random variable X has cumulative distribution function

$$F_X(x) = \begin{cases} 0, & x < 0, \\ Ax^3, & 0 \leq x \leq 10, \\ B, & x > 10. \end{cases}$$

Determine A and B , then obtain and sketch the probability density function.

Solution.

A CDF satisfies $\lim_{x \rightarrow \infty} F_X(x) = 1$, so $B = 1$. Right continuity at $x = 10$ gives $1000A = B = 1$; hence $A = 10^{-3}$. The CDF is continuous and contains no probability atoms.

Differentiating on each open interval gives

$$f_X(x) = \begin{cases} 3 \times 10^{-3} x^2, & 0 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

The density is a rising parabola on $(0, 10)$, approaching height 0.3 at the upper endpoint, and is zero outside this interval. Its normalization verifies the result:

$$\int_0^{10} 3 \times 10^{-3} x^2 dx = 10^{-3} [x^3]_0^{10} = 1.$$

Individual endpoint values of the ordinary density do not affect any probability.

Exercise 2.2. Testing candidate probability densities

IMG_8623.pdf, PDF page(s) 2.

Problem. Determine whether the nonnegative function $g(x) = e^{-x/2}$ for $x \geq 0$, and zero for $x < 0$, is a valid probability density.

Solution.

A valid density must be nonnegative and have unit integral. Nonnegativity holds, but

$$\int_{-\infty}^{\infty} g(x) dx = \int_0^{\infty} e^{-x/2} dx = \left[-2e^{-x/2} \right]_0^{\infty} = 2.$$

Therefore g is not a normalized probability density. The normalized function would be $f_X(x) = \frac{1}{2}e^{-x/2}u(x)$.

Exercise 2.3. Normalizing a symmetric exponential law

IMG_8623.pdf, PDF page(s) 2.

Problem. Find the constant k that makes $f_X(x) = ke^{-|x|}$ a probability density on the real line.

Solution.

Since $e^{-|x|} > 0$, a density requires $k \geq 0$. Use $|x| = -x$ for $x < 0$ and $|x| = x$ for $x \geq 0$:

$$1 = k \int_{-\infty}^0 e^x dx + k \int_0^{\infty} e^{-x} dx = k + k = 2k.$$

Thus $k = \frac{1}{2}$, and $f_X(x) = \frac{1}{2}e^{-|x|}$. The symmetry about the origin provides an equivalent check: each half of the real line has probability $1/2$.

Exercise 2.4. A trapezoidal density and its CDF

IMG_8623.pdf, PDF page(s) 2, 3, 4.

Problem. Consider

$$f_X(x) = \begin{cases} ax, & 0 \leq x \leq 1, \\ a, & 1 < x \leq 2, \\ -ax + 3a, & 2 < x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Determine a , find and sketch $F_X(x)$, and calculate $P(X \leq 1.5)$.

Solution.

Normalization yields

$$1 = a \int_0^1 x dx + a \int_1^2 dx + a \int_2^3 (3-x) dx = \frac{a}{2} + a + \frac{a}{2} = 2a,$$

so $a = 1/2$. **Obtain the CDF by accumulating the area from the left endpoint. On $0 \leq x < 1$,**

$F_X(x) = x^2/4$. **On $1 \leq x < 2$, include the first triangular area:**

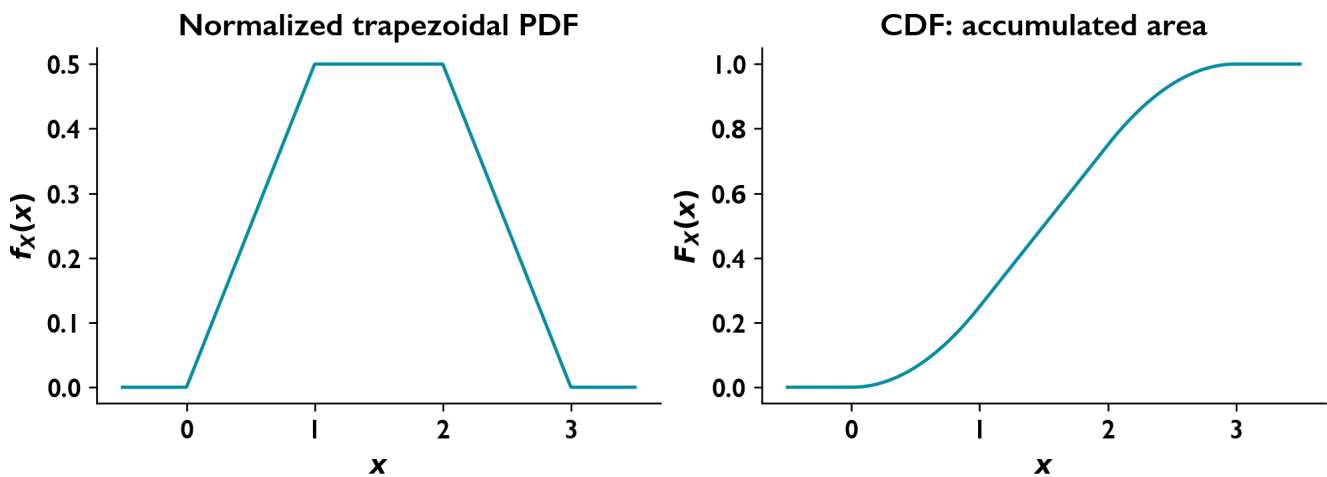
$F_X(x) = 1/4 + (x-1)/2 = x/2 - 1/4$. **On $2 \leq x < 3$, it is convenient to subtract the remaining triangular tail from one: $F_X(x) = 1 - (3-x)^2/4$. Thus**

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2/4, & 0 \leq x < 1, \\ x/2 - 1/4, & 1 \leq x < 2, \\ 1 - (3-x)^2/4, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

The formulas agree at $x = 1$ and $x = 2$, so the CDF is continuous. Finally,

$$P(X \leq 1.5) = F_X(1.5) = \frac{1.5}{2} - \frac{1}{4} = \frac{1}{2}.$$

The PDF is a symmetric trapezoid with height $1/2$. The CDF rises quadratically, then linearly, then approaches one along a concave quadratic segment.



The normalized trapezoidal density and its accumulated probability.

Exercise 2.5. Normalization of a piecewise constant density

IMG_8623.pdf, PDF page(s) 4.

Problem. The plotted density is $f_X(x) = k$ on $0 < x < 1$, $f_X(x) = 1/3$ on $1 < x < 2$, and zero elsewhere. Determine k .

Solution.

The area under the density is the sum of two rectangular areas:

$$1 = \int_0^1 k dx + \int_1^2 \frac{1}{3} dx = k + \frac{1}{3}.$$

Hence $k = 2/3$. Both density levels are nonnegative, so the normalized function is valid.

Exercise 2.6. An exponential component and a probability atom

IMG_8623.pdf, PDF page(s) 4, 5, 6.

Problem. A random voltage X has the generalized density

$$f_X(x) = k u(x + 7) e^{-1.72(x+7)} + 0.35\delta(x - 1).$$

Determine the normalizing constant, calculate $P(X \leq 0.5)$, $P(X > 3)$ and $P(0.5 \leq X \leq 3)$, and obtain and sketch the CDF.

Solution.

The Dirac impulse represents a probability atom $P(X = 1) = 0.35$. The remaining probability 0.65 is continuous. Normalize both parts:

$$1 = k \int_{-7}^{\infty} e^{-1.72(x+7)} dx + 0.35 = \frac{k}{1.72} + 0.35.$$

Therefore $k = 1.72(0.65) = 1.118$. The CDF is

$$F_X(x) = \begin{cases} 0, & x < -7, \\ 0.65[1 - e^{-1.72(x+7)}], & -7 \leq x < 1, \\ 0.65[1 - e^{-1.72(x+7)}] + 0.35, & x \geq 1. \end{cases}$$

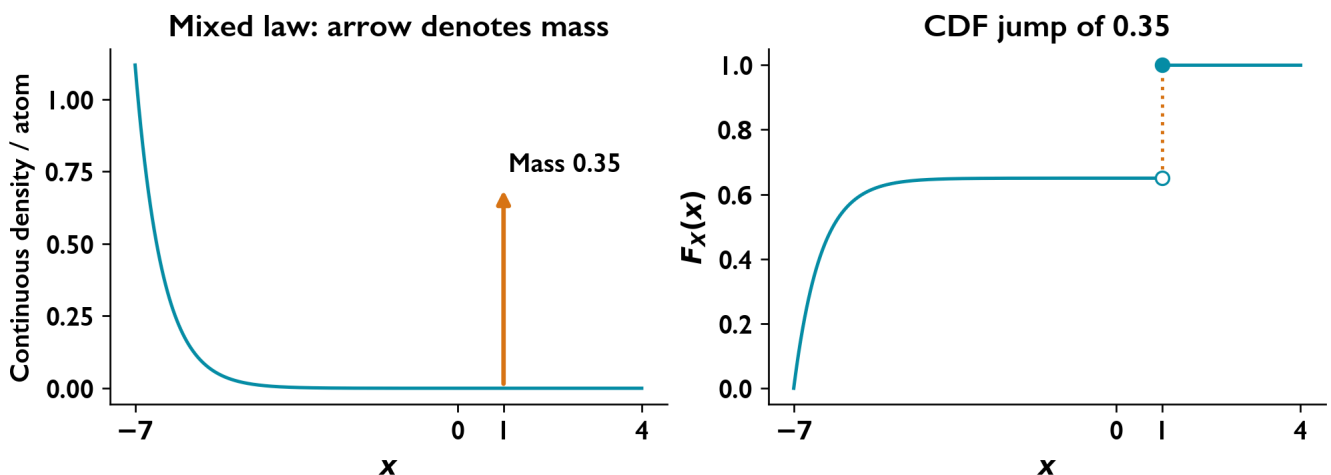
It has a jump of 0.35 at $x = 1$ and tends to one only as $x \rightarrow \infty$. Since there are no atoms at 0.5 or 3, strict and nonstrict inequalities agree at these endpoints. Consequently,

$$P(X \leq 0.5) = 0.65(1 - e^{-12.9}) \approx 0.6499983763,$$

$$P(X > 3) = 1 - F_X(3) = 0.65e^{-17.2} \approx 2.20317 \times 10^{-8},$$

$$P(0.5 \leq X \leq 3) = F_X(3) - F_X(0.5) = 0.35 + 0.65(e^{-12.9} - e^{-17.2}) \approx 0.3500016017.$$

The continuous tail is small but nonzero. It must remain in the exact answer; rounding the tail to zero would incorrectly imply finite support.



The arrow represents a probability mass of 0.35; it is not an ordinary finite density height.

Exercise 2.7. Gamma density, CDF and interval probabilities

IMG_8623.pdf, PDF pp. 6–7 (handwritten pp. 9–10)

Problem. A continuous random variable X has probability density

$$f_X(x) = xe^{-x}u(x).$$

Find its cumulative distribution function and evaluate $P(X \leq 1)$, $P(1 \leq X \leq 2)$, and $P(X > 2)$.

Solution.

The unit-step factor specifies the support $x \geq 0$. For $x < 0$, $F_X(x) = 0$. For $x \geq 0$, integration by parts gives

$$F_X(x) = \int_0^x te^{-t} dt = \left[-(t+1)e^{-t} \right]_0^x = 1 - (1+x)e^{-x}.$$

Hence

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - (1+x)e^{-x}, & x \geq 0. \end{cases}$$

The density is normalized because $\int_0^\infty xe^{-x} dx = 1$. Since X is continuous, the inclusion or exclusion of a single endpoint does not change the requested probabilities:

$$P(X \leq 1) = F_X(1) = 1 - \frac{2}{e} \approx 0.264241,$$

$$P(1 \leq X \leq 2) = F_X(2) - F_X(1) = \frac{2}{e} - \frac{3}{e^2} \approx 0.329753,$$

$$P(X > 2) = 1 - F_X(2) = \frac{3}{e^2} \approx 0.406006.$$

Exercise 2.8. Uniform density and cumulative probability

IMG_8623.pdf, PDF p. 7 (handwritten p. 10)

Problem. For $a < b$, a continuous random variable has density

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Determine and sketch its cumulative distribution function.

Solution.

The CDF is $F_X(x) = \int_{-\infty}^x f_X(t) dt$. Evaluate it on the three regions of the support:

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

For $a \leq x \leq b$, the probability accumulated over $[a, x]$ equals the width $x - a$ multiplied by the constant density $1/(b - a)$. The sketch is horizontal at zero for $x < a$, rises linearly from $(a, 0)$ to $(b, 1)$, and is horizontal at one for $x > b$. It is continuous, nondecreasing, and has the required limits zero and one.

Exercise 2.9. A sinusoidal CDF and its density

IMG_8623.pdf, PDF pp. 7–8 (handwritten pp. 10–11)

Problem. A random variable X has cumulative distribution

$$F_X(x) = \begin{cases} 0, & x < 0, \\ k \sin\left(\frac{\pi x}{40}\right), & 0 \leq x \leq 10, \\ k \sin\left(\frac{\pi}{4}\right), & x > 10. \end{cases}$$

Find k , determine $f_X(x)$, and calculate $P(X \leq 4)$.

Solution.

Every CDF satisfies $\lim_{x \rightarrow \infty} F_X(x) = 1$. Therefore

$$k \sin\left(\frac{\pi}{4}\right) = 1 \quad \Rightarrow \quad k = \sqrt{2}.$$

The CDF is continuous at both support boundaries, so there are no point masses at zero or ten. Differentiating its middle branch yields

$$f_X(x) = \begin{cases} \frac{\sqrt{2} \pi}{40} \cos\left(\frac{\pi x}{40}\right), & 0 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

The values assigned to the density at individual endpoints are immaterial. On $[0, 10]$, the cosine is nonnegative, and

$$\int_0^{10} f_X(x) dx = \sqrt{2} \sin\left(\frac{\pi}{4}\right) = 1,$$

which verifies that it is a valid density. Finally,

$$P(X \leq 4) = F_X(4) = \sqrt{2} \sin\left(\frac{\pi}{10}\right) \approx 0.437016.$$

Lesson 03. Expectation, Moments, and Transformations of Random Variables

Reference: Proakis and Salehi, Section 5.1.4, pp. 201–203; Problems 5.11–5.17; GJWHC moments, sample estimates, Laplace and sine/cosine transformations.

3.1 Expectation and statistical moments

The expectation or mean is a probability-weighted average. For a discrete variable and a density-bearing variable, respectively,

$$\mu_X = E[X] = \sum_k x_k p_X(x_k), \quad \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx.$$

A finite mean is guaranteed when $E[|X|] < \infty$. The definition also applies to mixed laws by adding the mass contributions and the density integral. Expectation is a deterministic number, even though the quantity being averaged is random.

For a measurable function g , the law of the unconscious statistician gives

$$E[g(X)] = \sum_k g(x_k) p_X(x_k) \quad \text{or} \quad E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx,$$

provided the expectation exists. The distribution of $g(X)$ need not be found first. The raw moment of order r is $E[X^r]$; the central moment is $E[(X - \mu_X)^r]$. The first raw moment is the mean, and the second raw moment is the mean square.

The variance and standard deviation are

$$\sigma_X^2 = \text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu_X^2, \quad \sigma_X = \sqrt{\sigma_X^2}.$$

Variance has squared units; standard deviation has the same units as X . The root-mean-square value is $\sqrt{E[X^2]}$. For a voltage across a resistance R , mean-square voltage becomes average power after division by R ; it is not automatically power in watts.

3.2 Affine rules and important moments

Linearity of expectation gives $E[aX + b] = a\mu_X + b$. Subtracting this mean shows that

$$\text{Var}(aX + b) = a^2 \sigma_X^2.$$

A constant shift changes the mean without changing the variance. Multiplication by a scales the standard deviation by $|a|$.

Probability law	Mean	Variance
Bernoulli with parameter p	p	$p(1-p)$
Binomial with parameters n, p	np	$np(1-p)$
Poisson with parameter λ	λ	λ
Uniform on $[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential with rate α	$1/\alpha$	$1/\alpha^2$
Gaussian $\mathcal{N}(\mu, \sigma^2)$	μ	σ^2

For a Bernoulli variable, $X^2 = X$, so $E[X^2] = p$ and the variance is $p - p^2$. The binomial moments follow by expressing the count as a sum of independent Bernoulli indicators. For a zero-mean Gaussian variable, symmetry makes every odd moment zero. Integration by parts gives

$$E[X^{2m}] = (2m-1)!! \sigma^{2m}, \quad m = 1, 2, \dots,$$

where $(2m-1)!! = 1 \cdot 3 \cdots (2m-1)$. In particular, $E[X^4] = 3\sigma^4$.

If X is Gaussian and $a \neq 0$, then

$$aX + b \sim \mathcal{N}(a\mu_X + b, a^2\sigma_X^2).$$

At $a = 0$, the result is the deterministic value b , represented by a point mass rather than a nondegenerate Gaussian density.

3.3 Estimation from observed samples

For N independent identically distributed observations with finite variance, the sample mean and unbiased sample variance are

$$\hat{\mu}_X = \frac{1}{N} \sum_{n=1}^N x_n,$$

$$\hat{\sigma}_X^2 = \frac{1}{N-1} \sum_{n=1}^N (x_n - \hat{\mu}_X)^2, \quad N > 1.$$

The factor $N - 1$ accounts for estimating the mean from the same data. Dividing by N instead has expectation $(N - 1) \sigma_X^2 / N$. These sample statistics estimate population quantities; their numerical values fluctuate from one finite record to another.

3.4 Obtaining a transformed distribution from its CDF

If $Y = g(X)$, the safest starting point is the event identity

$$F_Y(y) = P(g(X) \leq y).$$

Determine the values of X satisfying this inequality and sum or integrate their probabilities. This method handles increasing, decreasing, many-to-one, and saturating transformations.

For a strictly increasing transformation and density-bearing X ,

$$F_Y(y) = F_X(g^{-1}(y)).$$

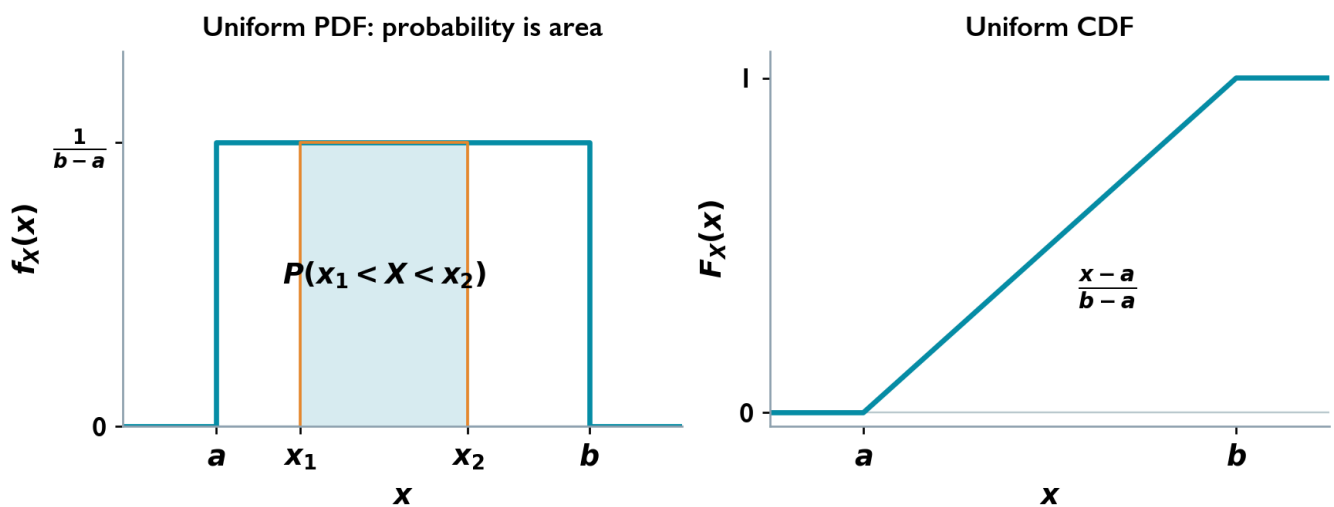
For a strictly decreasing transformation,

$$F_Y(y) = 1 - F_X(g^{-1}(y)).$$

If the inverse is differentiable, both cases lead to

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

The absolute value ensures a nonnegative density, and the support must be transformed along with the formula. For example, $X \sim \mathcal{U}[0, 1]$ and $Y = 2X + 1$ give $f_Y(y) = 1/2$ on $[1, 3]$ and zero elsewhere.



Uniform density and CDF: interval probabilities are areas under the density.

If $g(x) = y$ has several admissible inverse roots x_i with $g'(x_i) \neq 0$, each contributes probability:

$$f_Y(y) = \sum_i \frac{f_X(x_i)}{|g'(x_i)|}.$$

Only roots in the input support belong in this sum. Flat portions of g can create probability masses, and critical points require appropriate limits or direct CDF analysis. The density formula should not be applied blindly at such points.

3.5 Square-law and periodic transformations

Let $X \sim \mathcal{N}(0, \sigma^2)$ and $Y = X^2$. The output is nonnegative. For $y > 0$, its inverse roots are $\sqrt{y}$ and $-\sqrt{y}$, giving

$$f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{e^{-y/(2\sigma^2)}}{\sigma\sqrt{2\pi y}}.$$

The density is zero for $y < 0$. Its integrable singularity at zero is not a probability mass:

$$P(Y=0) = P(X=0) = 0.$$

For a uniformly distributed phase $\Theta \sim \mathcal{U}[0, 2\pi)$, the transformation $U = \sin \Theta$ also has two relevant branches for each $-1 < u < 1$. Since $|\cos \Theta| = \sqrt{1-u^2}$ at either root,

$$f_U(u) = \frac{1}{\pi\sqrt{1-u^2}}, \quad -1 < u < 1.$$

The density is zero outside this interval, and

$$F_U(u) = \frac{1}{2} + \frac{\arcsin u}{\pi}, \quad -1 \leq u \leq 1,$$

with values zero below -1 and one above 1 . The cosine transformation has the same marginal law. Nonuniform density of sine/cosine values therefore arises from a uniform angle through the geometry of the mapping.

For a half-wave rectifier $Y = \max(X, 0)$ with zero-mean Gaussian input, all negative input values collapse to zero. Hence $P(Y=0) = 1/2$, while the continuous density for $y > 0$ is $f_X(y)$. The result is mixed; retaining only the positive density would lose half the probability.

A symmetric clipping limiter $Y = \max(-b, \min(X, b))$, with $b > 0$ and $X \sim \mathcal{N}(0, \sigma^2)$, produces masses $P(Y=-b) = P(Y=b) = Q(b/\sigma)$ and retains density $f_X(y)$ for $-b < y < b$. Its continuous probability is $1 - 2Q(b/\sigma)$, so the masses and density together normalize to one. A hard limiter $Y = c \operatorname{sgn}(X)$, $c > 0$, instead gives a discrete law with

probability $1/2$ at each of $-c$ and c . The value assigned at $X = 0$ contributes no mass for this continuous input. Clipping, hard limiting, and quantization must therefore be analyzed using the resulting mixed or discrete laws rather than only a derivative formula.

3.6 The Laplace law and existence of moments

The symmetric double-exponential or Laplace density from the GJWHC material is

$$f_X(x) = \frac{b}{2} e^{-b|x|}, \quad b > 0, \quad x \in \mathbb{R}.$$

Normalization follows from $2 \int_0^\infty (b/2) e^{-bx} dx = 1$. Integration gives

$$F_X(x) = \begin{cases} \frac{1}{2} e^{bx}, & x < 0, \\ 1 - \frac{1}{2} e^{-bx}, & x \geq 0. \end{cases}$$

Symmetry gives $\mu_X = 0$. Using $\int_0^\infty x^2 e^{-bx} dx = 2/b^3$ yields $E[X^2] = 2/b^2$, which is also the variance.

Not every normalized density has finite moments. For instance, transforming a uniform angle on $(-\pi/2, \pi/2)$ by $Y = \tan \Theta$ gives the Cauchy density $1/[\pi(1+y^2)]$. Its mean and variance do not exist. Moment-based statements must therefore include their existence assumptions.

3.7 Probability bounds from moments

For $X \geq 0$ and $a > 0$, the pointwise inequality $a \mathbf{1}_{\{X \geq a\}} \leq X$ implies Markov's inequality:

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Apply it to $(X - \mu_X)^2$ to obtain Chebyshev's inequality:

$$P(|X - \mu_X| \geq \varepsilon) \leq \frac{\sigma_X^2}{\varepsilon^2}, \quad \varepsilon > 0.$$

These bounds require no particular distribution shape. Their generality can make them conservative, but they are useful when only moments are known and provide the foundation for sample-average convergence.

Solved Exercises for Lesson 03

Exercise 3.1. A transformation producing both a discrete mass and a continuous density

IMG_8587.pdf, PDF page(s) 2, 3.

Problem. The supplied graph relates $Y = g(X)$ to X , where $X \sim \mathcal{U}[-2, 2]$. The graph shows $Y = -1$ for $-2 \leq X < 0$, a straight line from $(0, 0)$ to $(3, 2)$, and $Y = 2$ beyond $X = 3$.

Obtain and sketch the CDF of X , and the PDF and CDF of Y .

Solution.

1. Write the input distribution.

$$f_X(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2, \\ 0, & \text{otherwise,} \end{cases} \quad F_X(x) = \begin{cases} 0, & x < -2, \\ \frac{x+2}{4}, & -2 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

The CDF rises linearly from $(-2, 0)$ to $(2, 1)$.

2. Restrict the transformation to the actual input support. The positive branch is $Y = (2/3)X$. Because $X \leq 2$, its output range is $0 \leq Y \leq 4/3$. The horizontal branch at $Y = 2$ is never reached and creates no probability mass.

The negative half of the input interval maps to the single value -1 :

$$\Pr(Y = -1) = \Pr(-2 \leq X < 0) = \frac{1}{2}.$$

For the positive branch, $x = 3y/2$, so

$$f_{Y,\text{cont}}(y) = f_X(3y/2) \left| \frac{dx}{dy} \right| = \frac{13}{42} = \frac{3}{8}, \quad 0 < y < \frac{4}{3}.$$

Thus Y has a mixed distribution. It has no ordinary PDF consisting solely of a finite-valued function. Its generalized density is

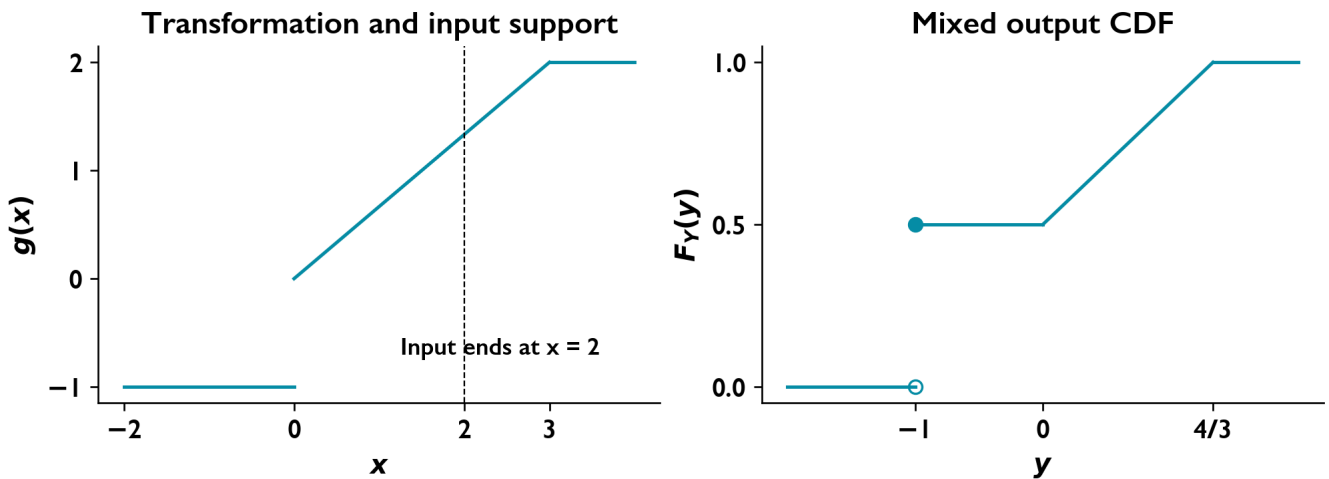
$$f_Y(y) = \frac{1}{2} \delta(y + 1) + \frac{3}{8} \mathbf{1}_{(0, 4/3)}(y).$$

The impulse represents the point mass; the continuous density integrates to the remaining probability $(3/8)(4/3) = 1/2$.

3. Obtain the output CDF.

$$F_Y(y) = \begin{cases} 0, & y < -1, \\ \frac{1}{2}, & -1 \leq y < 0, \\ \frac{1}{2} + \frac{3}{8}y, & 0 \leq y < \frac{4}{3}, \\ 1, & y \geq \frac{4}{3}. \end{cases}$$

The CDF jumps by $1/2$ at -1 , remains flat up to 0 , then rises linearly to 1 at $4/3$. This jump is essential; applying the continuous change-of-variable formula to the entire transformation would lose half of the probability.



The constant negative branch creates an atom at $y = -1$; the output never reaches the saturation branch at $y = 2$.

Exercise 3.2. Moments of a polynomial density

IMG_8623.pdf, PDF p. 10 (handwritten p. 13)

Problem. A continuous random variable X has density

$$f_X(x) = \begin{cases} 3 \times 10^{-3} x^2, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Find its mean μ_X , variance σ_X^2 , and second raw moment $E[X^2]$.

Solution.

First verify normalization:

$$\int_0^{10} 3 \times 10^{-3} x^2 dx = 3 \times 10^{-3} \frac{10^3}{3} = 1.$$

The mean is

$$\mu_X = E[X] = \int_0^{10} x f_X(x) dx = 3 \times 10^{-3} \left[\frac{x^4}{4} \right]_0^{10} = 7.5.$$

The second raw moment is

$$E[X^2] = \int_0^{10} x^2 f_X(x) dx = 3 \times 10^{-3} \left[\frac{x^5}{5} \right]_0^{10} = 60.$$

Use the relation between the second raw moment and the second central moment:

$$\sigma_X^2 = E[X^2] - \mu_X^2 = 60 - (7.5)^2 = 3.75.$$

Thus $\mu_X = 7.5$, $E[X^2] = 60$, and $\sigma_X^2 = 3.75$. The corresponding standard deviation is $\sigma_X = \sqrt{3.75} \approx 1.93649$.

Exercise 3.3. Mean and variance of an exponential variable

IMG_8623.pdf, PDF page(s) 13.

Problem. For the exponential density $f_X(x) = ae^{-ax}u(x)$ with $a > 0$, find the mean, second moment, variance and standard deviation.

Solution.

Using integration by parts, or the integral $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$ for $a > 0$, obtain

$$\mu_X = E[X] = a \int_0^\infty x e^{-ax} dx = \frac{1}{a},$$

$$E[X^2] = a \int_0^\infty x^2 e^{-ax} dx = \frac{2}{a^2}.$$

The variance is the second moment minus the square of the mean:

$$\sigma_X^2 = E[X^2] - \mu_X^2 = \frac{2}{a^2} - \frac{1}{a^2} = \frac{1}{a^2}.$$

Since $a > 0$, $\sigma_X = \sqrt{\sigma_X^2} = 1/a$. Thus an exponential variable has standard deviation equal to its mean.

Lesson 04. Joint Distributions, Conditional Moments, and Gaussian Vectors

Reference: Proakis and Salehi, Section 5.1.5, pp. 203–208; Problems 5.19–5.30; GJWHC joint laws, dependence and Rayleigh distribution.

4.1 Joint distributions and support

Two random variables X and Y defined on the same probability space are described jointly by

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

When an ordinary joint density exists,

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}, \quad f_{X,Y} \geq 0,$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1.$$

The support is the region on which the density is nonzero. For any region A ,

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy.$$

Integration limits must describe the intersection of A with the support. A triangular or curved support cannot be replaced by a rectangle merely because each variable individually lies in a fixed interval.

The marginal laws are obtained by integrating out the other variable:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Similarly, $F_X(x) = F_{X,Y}(x, \infty)$ and $F_Y(y) = F_{X,Y}(\infty, y)$. For discrete variables, integrals are replaced by sums of the joint PMF.

4.2 Conditional densities and conditional moments

For $f_Y(y) > 0$, define

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

At a fixed conditioning value y , this is a normalized density in x over the relevant section of the joint support. It does not use the ratio $P(X=x, Y=y)/P(Y=y)$, whose denominator is zero for ordinary continuous variables. At values where $f_Y(y) = 0$, the density ratio is undefined; any boundary limit must be identified as a limiting interpretation.

The conditional mean and conditional variance are

$$m_{X|Y}(y) = E[X|Y=y] = \int x f_{X|Y}(x|y) dx,$$

$$\text{Var}(X|Y=y) = \int x^2 f_{X|Y}(x|y) dx - m_{X|Y}(y)^2.$$

They are functions of y . Substituting the random variable Y gives the random conditional mean $E[X|Y]$. Averaging it gives the law of total expectation:

$$E[X] = E[E[X|Y]].$$

To derive the law of total variance, write

$$X - \mu_X = (X - E[X|Y]) + (E[X|Y] - \mu_X).$$

The cross term has zero expectation because $E[X - E[X|Y] | Y] = 0$. Therefore,

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]).$$

The two terms measure variation remaining within the conditional distributions and variation between their conditional means.

4.3 Independence, correlation, and covariance

Independence requires

$$P(X \in A, Y \in B) = P(X \in A) P(Y \in B)$$

for every pair of admissible sets. For joint density laws, this is equivalent to

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

almost everywhere. A conditional law then equals the corresponding marginal law. Checking the supports is part of checking this equality.

For an integrable function g ,

$$E[g(X, Y)] = \iint g(x, y) f_{X,Y}(x, y) dx dy.$$

The correlation of two scalar random variables is $E[XY]$. Their covariance is

$$\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y.$$

They are uncorrelated when the covariance is zero. For positive standard deviations, the correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{X,Y} \leq 1.$$

The bound follows from the Cauchy–Schwarz inequality. Equality indicates an affine relation almost surely: a positive slope for $\rho = 1$ and a negative slope for $\rho = -1$.

Independence implies zero covariance when the relevant moments exist, but the converse is false. For a uniform phase, $X = \sin \Theta$ and $Y = \cos \Theta$ have zero means and $E[XY] = 0$, yet $X^2 + Y^2 = 1$. Their joint law is supported on a circle and has no ordinary two-dimensional density.

For $Z = \sum_i a_i X_i$, linearity and expansion give

$$E[Z] = \sum_i a_i \mu_i,$$

$$\text{Var}(Z) = \sum_i a_i^2 \sigma_i^2 + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

Variance adds without cross terms for uncorrelated summands; independence is sufficient but stronger than necessary.

4.4 Joint transformations and the Jacobian

Let $Z = g(X, Y)$ and $W = h(X, Y)$. If the inverse transformation is differentiable and one-to-one in the region considered,

$$f_{Z,W}(z, w) = f_{X,Y}(x(z, w), y(z, w)) \left| \det \frac{\partial(x, y)}{\partial(z, w)} \right|.$$

Equivalently, divide the input density by the absolute determinant of the forward Jacobian. For multiple inverse branches, sum their contributions. The transformed support must also be specified. The determinant expresses the local area change; it cannot be omitted.

For polar coordinates,

$$X = R \cos \Theta, \quad Y = R \sin \Theta, \quad R \geq 0, \quad 0 \leq \Theta < 2\pi.$$

Use $\Theta = \text{atan2}(Y, X)$ modulo 2π to retain the correct quadrant. The inverse Jacobian determinant has magnitude r , so

$$f_{R,\Theta}(r, \theta) = r f_{X,Y}(r \cos \theta, r \sin \theta).$$

If X and Y are independent $\mathcal{N}(0, \sigma^2)$ variables, substitution yields

$$f_{R,\Theta}(r, \theta) = \frac{r}{2\pi\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r > 0, \quad 0 \leq \theta < 2\pi.$$

It factors into a Rayleigh density and a uniform phase density:

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0, \quad f_\Theta(\theta) = \frac{1}{2\pi}.$$

Thus magnitude and phase are independent. Integrating gives

$$F_R(r) = 1 - e^{-r^2/(2\sigma^2)}, \quad r \geq 0,$$

$$E[R] = \sigma \sqrt{\frac{\pi}{2}}, \quad E[R^2] = 2\sigma^2, \quad \text{Var}(R) = \left(2 - \frac{\pi}{2}\right) \sigma^2.$$

The density and CDF are zero for negative r . If the Cartesian pair instead has means $(A, 0)$, its magnitude follows the Rician law

$$f_R(r) = \frac{r}{\sigma^2} \exp\left[-\frac{r^2 + A^2}{2\sigma^2}\right] I_0\left(\frac{Ar}{\sigma^2}\right), \quad r \geq 0,$$

where $I_0(z) = (1/2\pi) \int_0^{2\pi} e^{z \cos \theta} d\theta$. The Rayleigh law is recovered at $A = 0$.

4.5 Jointly Gaussian variables and vectors

A vector is jointly Gaussian when every linear combination of its components is Gaussian, including possible deterministic combinations. Its law is completely specified by its mean vector and covariance matrix. For a nonsingular covariance matrix Σ , an n -dimensional Gaussian vector has density

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]}{(2\pi)^{n/2} \sqrt{\det \boldsymbol{\Sigma}}}.$$

Subvectors remain jointly Gaussian. An affine transformation $\mathbf{Z} = \mathbf{A}\mathbf{X} + \mathbf{b}$ has mean $\mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ and covariance $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$.

For two jointly Gaussian variables with $|\rho| < 1$, define $u = (x - \mu_X)/\sigma_X$ and $v = (y - \mu_Y)/\sigma_Y$. Their density is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)} \right].$$

Completing the square,

$$\frac{u^2 - 2\rho uv + v^2}{1 - \rho^2} = \frac{(u - \rho v)^2}{1 - \rho^2} + v^2,$$

shows that

$$X | Y = y \sim \mathcal{N} \left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y), \sigma_X^2 (1 - \rho^2) \right).$$

At $\rho = 0$, the joint density factors and the conditional law is unchanged. Therefore uncorrelated jointly Gaussian variables are independent. At $|\rho| = 1$, the distribution is concentrated on a line and has no ordinary two-dimensional density.

Gaussian marginal laws alone do not imply a joint Gaussian law. If $X \sim \mathcal{N}(0, 1)$ and an independent sign S is equally likely to be $+1$ or -1 , then $Y = SX$ is also standard Gaussian and $E[XY] = 0$. Nevertheless, $|Y| = |X|$, so the variables are dependent. The joint-Gaussian qualification is essential.

Orthogonal rotations can diagonalize a real covariance matrix. For centered, equal-variance Gaussian variables with covariance $\rho\sigma^2$, define $Z = X \cos \theta + Y \sin \theta$ and $W = -X \sin \theta + Y \cos \theta$. Direct expansion gives

$$\text{Cov}(Z, W) = \rho\sigma^2 \cos(2\theta).$$

When $\rho \neq 0$, the angles $\theta = \pi/4 + k\pi/2$ make the rotated variables uncorrelated and therefore independent. For a general Gaussian vector, choosing an orthogonal matrix of covariance eigenvectors gives a diagonal transformed covariance; components corresponding to positive

eigenvalues can additionally be scaled to unit variance. This decorrelation produces independence because of joint Gaussianity. The same rotation need not make non-Gaussian variables independent.

4.6 Independent sums and order statistics

For independent continuous variables, the density of their sum is the convolution

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

In particular, independent Gaussian sums are exactly Gaussian, with means and variances added. A central-limit approximation is unnecessary in that case.

For independent identically distributed variables $X_1, \dots, X_n$ with CDF F , the largest and smallest values satisfy

$$F_{\max}(x) = F(x)^n, \quad F_{\min}(x) = 1 - [1 - F(x)]^n.$$

The first identity states that all observations must be no greater than x ; the second uses the complementary event that all exceed x . If F has density f , differentiation gives

$$f_{\max}(x) = nF(x)^{n-1}f(x), \quad f_{\min}(x) = n[1 - F(x)]^{n-1}f(x).$$

These results illustrate how joint independence converts a multivariable probability into products of one-variable probabilities.

Solved Exercises for Lesson 04

Exercise 4.1. Normalization and marginals on a curved support

IMG_8587.pdf, PDF page(s) 1.

Problem. The joint density of X and Y is

$$f_{XY}(x, y) = \begin{cases} k(1 + 2xy), & 0 < x \leq 1, \quad 0 < x^2 < y \leq x + 0.5 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the normalization constant k . (b) Determine $f_X(x)$. (c) Determine $f_Y(y)$.

Solution.

I. Identify the support. The printed condition $x + 0.5 \leq 1$ further restricts x to $0 < x \leq 1/2$. Hence

$$\mathcal{D} = \{ (x, y) : 0 < x \leq \frac{1}{2}, x^2 < y \leq x + \frac{1}{2} \}.$$

The support must be used before either marginal is calculated.

2. Normalize the joint density.

$$1 = k \int_0^{1/2} \int_{x^2}^{x+1/2} (1 + 2xy) dy dx.$$

The inner integral is

$$\left[y + xy^2 \right]_{x^2}^{x+1/2} = \frac{1}{2} + \frac{5}{4}x + x^3 - x^5.$$

Therefore

$$1 = k \left[\frac{1}{2}x + \frac{5}{8}x^2 + \frac{1}{4}x^4 - \frac{1}{6}x^6 \right]_0^{1/2} = k \frac{161}{384}, \quad k = \frac{384}{161}.$$

3. Obtain the marginal of X .

$$f_X(x) = \begin{cases} \frac{384}{161} \left(\frac{1}{2} + \frac{5}{4}x + x^3 - x^5 \right), & 0 < x < \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

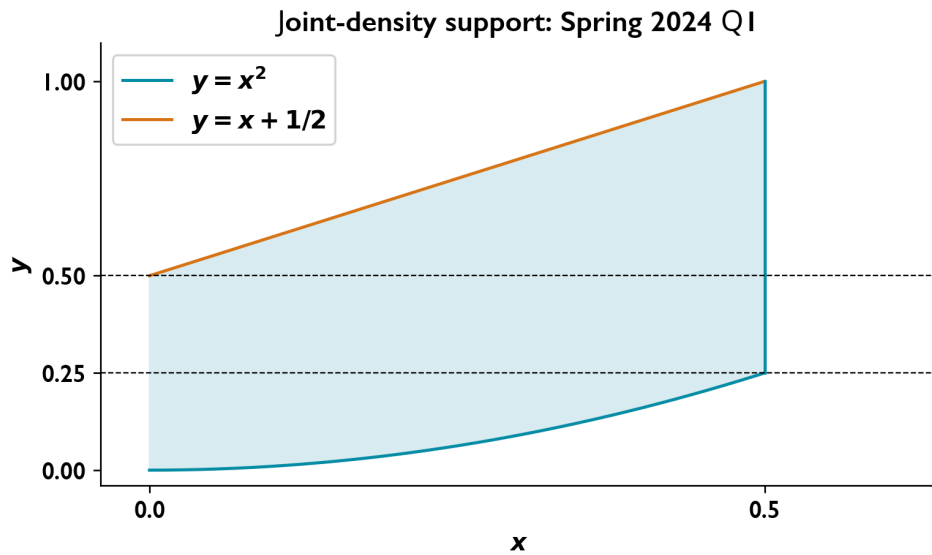
4. Reverse the integration limits for the marginal of Y . For fixed y , the allowed interval is

$$\max \left(0, y - \frac{1}{2} \right) < x < \min \left(\frac{1}{2}, \sqrt{y} \right).$$

The upper boundary changes at $y = 1/4$; the lower boundary changes at $y = 1/2$. Consequently,

$$f_Y(y) = \frac{384}{161} \begin{cases} \sqrt{y} + y^2, & 0 < y < \frac{1}{4}, \\ \frac{1}{2} + \frac{1}{4}y, & \frac{1}{4} \leq y < \frac{1}{2}, \\ 1 - y + y^2 - y^3, & \frac{1}{2} \leq y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

This follows by evaluating $k[x + yx^2]$ at the corresponding endpoints. Values at isolated endpoints do not affect a continuous probability distribution.



The curved support fixes the marginal integration limits. Horizontal boundary changes occur at $y = 1/4$ and $y = 1/2$.

Exercise 4.2. A joint density on a triangular strip

IMG_8587.pdf, PDF page(s) 6.

Problem. The joint density is

$$f_{XY}(x, y) = kx, \quad 1 \leq x + y \leq 2, \quad x \geq 0, \quad y \geq 0,$$

and zero elsewhere. (a) Find k . (b) Obtain the marginal densities of X and Y .

Solution.

1. Normalize over the strip. Use $s = x + y$; its Jacobian with respect to (x, s) is 1. For $1 \leq s \leq 2, 0 \leq x \leq s$. Thus

$$1 = k \int_1^2 \int_0^s x \, dx \, ds = \frac{k}{2} \int_1^2 s^2 \, ds = \frac{7k}{6}, \quad k = \frac{6}{7}.$$

2. Integrate along a vertical line. If $0 \leq x \leq 1$, the allowed interval is $1 - x \leq y \leq 2 - x$, of length 1. If $1 < x \leq 2$, it is $0 \leq y \leq 2 - x$. Therefore

$$f_X(x) = \begin{cases} \frac{6}{7}x, & 0 \leq x \leq 1, \\ \frac{6}{7}x(2-x), & 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

3. Integrate along a horizontal line. If $0 \leq y \leq 1$, $1 - y \leq x \leq 2 - y$; if $1 < y \leq 2$, $0 \leq x \leq 2 - y$. Hence

$$f_Y(y) = \begin{cases} \frac{3}{7}(3-2y), & 0 \leq y \leq 1, \\ \frac{3}{7}(2-y)^2, & 1 < y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Each marginal integrates to 1. Although the support is symmetric in x, y , the density kx is not; therefore the two marginal densities need not coincide.

Exercise 4.3. An ordered exponential joint law

IMG_8623.pdf, PDF pp. 10–11 (handwritten pp. 13–14)

Problem. Two continuous random variables X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} ke^{-(x+y)}, & x > y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find k , the marginal densities $f_X(x)$ and $f_Y(y)$, determine whether X and Y are independent, and obtain $f_{X|Y}(x|y)$.

Solution.

The support is the wedge $x > y > 0$. Its geometry determines the integration limits. Normalize by integrating first over x :

$$1 = \int_0^\infty \int_y^\infty ke^{-(x+y)} dx dy = k \int_0^\infty e^{-2y} dy = \frac{k}{2}.$$

Therefore $k = 2$. For a fixed $x > 0$, the permitted values of y lie in $(0, x)$, so

$$f_X(x) = \int_0^x 2e^{-x-y} dy = 2e^{-x}(1 - e^{-x}), \quad x > 0.$$

For a fixed $y > 0$, the permitted values of x lie in (y, ∞) , giving

$$f_Y(y) = \int_y^\infty 2e^{-x-y} dx = 2e^{-2y}, \quad y > 0.$$

Both marginal densities are zero outside their stated supports. Their product is positive for every $x > 0, y > 0$, whereas the joint density vanishes whenever $0 < x < y$. Hence the factorization $f_{X,Y} = f_X f_Y$ fails, and the variables are dependent. For a specified $y > 0$,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} e^{-(x-y)}, & x > y, \\ 0, & \text{otherwise.} \end{cases}$$

It integrates to one over (y, ∞) . Thus, conditional on $Y = y$, $X - y$ has an exponential density with rate one.

Exercise 4.4. Dependence through an exponential rate

IMG_8623.pdf, PDF p. 12 (handwritten p. 15)

Problem. Let

$$f_{X,Y}(x, y) = ye^{-y(x+1)} u(x) u(y).$$

Find $f_Y(y)$, show that X and Y are not statistically independent, and determine $f_{X|Y}(x|y)$.

Solution.

For $y > 0$, integrate over $x \geq 0$:

$$f_Y(y) = \int_0^\infty ye^{-y(x+1)} dx = ye^{-y} \int_0^\infty e^{-yx} dx = e^{-y}.$$

Thus $f_Y(y) = e^{-y}u(y)$, up to the arbitrary value assigned at the single endpoint. This integrates to one and confirms normalization of the joint density. To test independence, also calculate

$$f_X(x) = \int_0^\infty ye^{-(x+1)y} dy = \frac{1}{(x+1)^2}, \quad x \geq 0,$$

where $\int_0^\infty ye^{-ay} dy = 1/a^2$ for $a > 0$. Hence

$$f_X(x) f_Y(y) = \frac{e^{-y}}{(x+1)^2} \neq ye^{-y(x+1)}$$

in general, so X and Y are dependent. Finally, for $y > 0$,

$$f_{X|Y}(x|y) = \frac{ye^{-y(x+1)}}{e^{-y}} = \begin{cases} ye^{-yx}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

This conditional density is exponential with rate y , and $\int_0^\infty ye^{-yx} dx = 1$. Its explicit dependence on the conditioning value y supplies a second explanation of the dependence between the variables.

Exercise 4.5. Joint density and conditional moments

University of Tripoli, EE416, Spring 2025 midterm, Q1.

Problem. Let X and Y have joint density

$$f_{X,Y}(x, y) = \begin{cases} 6xy, & 0 \leq x \leq 1, 0 \leq y \leq \sqrt{x}, \\ 0, & \text{otherwise.} \end{cases}$$

Find (a) both marginal densities; (b) whether X and Y are independent; (c) $f_{X|Y}(x|y)$; (d) $E[X|Y=y]$; and (e) $\text{Var}(X|Y=y)$. The final printed request refers to “autocorrelation R_{xy} in the (x, y) plane”; interpret the available joint-variable data carefully and show the support.

Solution.

1. Support and normalization

The support can be written in either order:

$$0 \leq x \leq 1, \quad 0 \leq y \leq \sqrt{x}, \quad \text{or} \quad 0 \leq y \leq 1, \quad y^2 \leq x \leq 1.$$

Its total probability is $\int_0^1 \int_0^{\sqrt{x}} 6xy \, dy \, dx = \int_0^1 3x^2 \, dx = 1$.

2. Marginal densities and independence

$$f_X(x) = 3x^2, \quad 0 \leq x \leq 1,$$

$$f_Y(y) = \int_{y^2}^1 6xy \, dx = 3y(1 - y^4), \quad 0 \leq y \leq 1.$$

Both densities vanish outside these intervals. The joint support is not the Cartesian product of the marginal supports, so the variables are dependent. For example, $x < y^2$ has zero joint density while the product of the marginals is positive at interior points there.

3. Conditional density and moments

For $0 < y < 1$,

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{1 - y^4}, & y^2 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Consequently,

$$E[X|Y=y] = \frac{2(1 - y^6)}{3(1 - y^4)} = \frac{2(1 + y^2 + y^4)}{3(1 + y^2)},$$

$$E[X^2|Y=y] = \frac{1 - y^8}{2(1 - y^4)} = \frac{1 + y^4}{2},$$

$$\text{Var}(X|Y=y) = \frac{1 + y^4}{2} - \left[\frac{2(1 + y^2 + y^4)}{3(1 + y^2)} \right]^2.$$

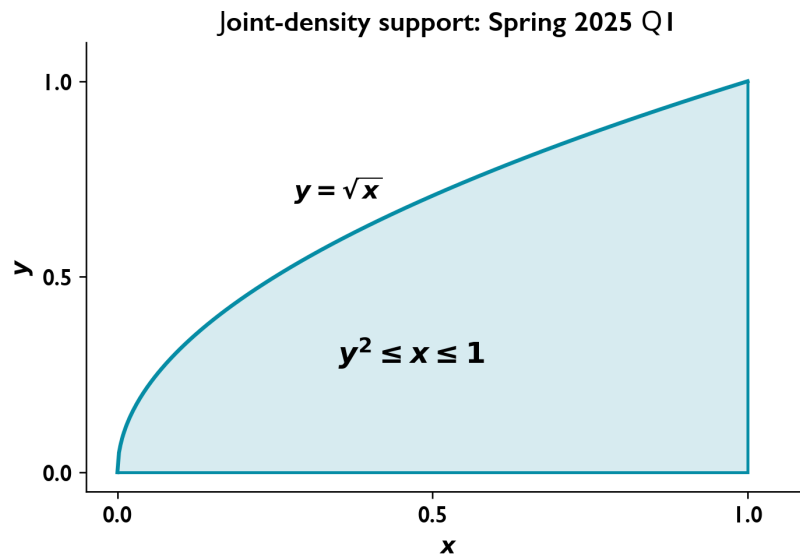
At $y = 0$ and $y = 1$, the marginal density is zero and the density-ratio definition does not determine a unique conditional law. The continuous limiting versions give mean $2/3$ and variance $1/18$ as $y \rightarrow 0$, and mean 1 and variance 0 as $y \rightarrow 1$.

4. Meaning of the final request

A temporal autocorrelation cannot be recovered from a joint density of two scalar variables without a process or lag definition. The corresponding scalar cross moment is

$$E[XY] = \int_0^1 \int_0^{\sqrt{x}} 6x^2 y^2 dy dx = \frac{4}{9}.$$

Also $E[X] = 3/4$, $E[Y] = 4/7$, and $\text{Cov}(X, Y) = 4/9 - (3/4)(4/7) = 1/63$. The support diagram shows the curved region $y^2 \leq x \leq 1$; it is not a graph of a temporal correlation.



The joint support is bounded by $y = \sqrt{x}$, $y = 0$ and $x = 1$.

Lesson 05. Sums of Random Variables, the Law of Large Numbers, and the Central Limit Theorem

Reference: Proakis and Salehi, Chapter 5, §5.1.6, pp. 208–209; Problems 5.20–5.21 and 5.31; GJWHC main pp. 13–14.

5.1 Distribution of an independent sum

Communication waveforms often combine many contributions: noise sources add, measurements are averaged, and a detector may integrate many samples. The distribution of a sum depends on the joint law of its summands. Independence permits a particularly useful convolution rule.

Let $Z = X + Y$, where X and Y are independent continuous random variables. The transformation $(x, y) \mapsto (x, z = x + y)$ has unit Jacobian. Integrating its joint density over the unobserved variable gives

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z-x) dx = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

Thus $f_Z = f_X * f_Y$. The integral must respect the support of both densities. For independent discrete variables, replace the integral by a sum:

$$p_Z(z) = \sum_x p_X(x) p_Y(z-x).$$

For a dependent pair, the first joint-density integral remains valid, but replacing the joint density by a product is generally incorrect. Repeating convolution determines the law of a sum of several independent variables. Equivalently, their characteristic functions multiply; this offers a useful alternative when direct convolution is cumbersome.

The mean of any integrable sum is the sum of the means, regardless of independence. For finite-variance variables,

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mu_i,$$

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \sigma_i^2 + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Independence, or merely zero pairwise covariance, removes the covariance terms. Independence is still required for the convolution factorization above.

5.2 Exponential sums and the Erlang distribution

For independent exponential variables with rates λ_1 and λ_2 , the support condition restricts the convolution to $0 < x < z$. When the rates differ,

$$f_Z(z) = \lambda_1 \lambda_2 e^{-\lambda_2 z} \int_0^z e^{-(\lambda_1 - \lambda_2)x} dx = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} (e^{-\lambda_1 z} - e^{-\lambda_2 z}), \quad z \geq 0.$$

The density is zero for negative z . For equal rates λ , take the limit, or integrate the constant exponential factor directly, to obtain $f_Z(z) = \lambda^2 z e^{-\lambda z} u(z)$.

More generally, the sum of n independent exponential variables with common rate $\lambda > 0$ has the Erlang density

$$f_{S_n}(s) = \frac{\lambda^n s^{n-1} e^{-\lambda s}}{(n-1)!} u(s), \quad E[S_n] = \frac{n}{\lambda}, \quad \text{Var}(S_n) = \frac{n}{\lambda^2}.$$

Repeated convolution proves this formula by induction: integrating x^{n-1} over $[0, s]$ supplies the next power and factorial. The changing shape illustrates why a sum need not retain the original distribution family, except in special closed families such as Gaussian laws.

5.3 Sample averages and the weak law of large numbers

Suppose $X_1, X_2, \dots$ are independent and identically distributed, with finite mean μ and variance σ^2 . Define the sample average

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Linearity and the variance formula give

$$E[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The average is an unbiased estimate of the population mean; its standard deviation decreases as $1/\sqrt{n}$. Pairwise uncorrelated variables with the same finite mean and variance also satisfy these two moment identities, even without full independence.

Chebyshev's inequality converts variance reduction into a precise probability statement. For every fixed $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

Therefore $\bar{X}_n$ converges in probability to μ . This is the weak law of large numbers under the stated assumptions. For a desired deviation probability no greater than δ , the sufficient bound is $n \geq \sigma^2 / (\delta \varepsilon^2)$. It is a guarantee, often conservative, rather than a distributional approximation.

The strong law uses almost-sure convergence instead. It concerns the eventual behavior of individual infinite sample sequences and should not be identified with the weaker convergence proved by Chebyshev. Neither law means that every finite average is exactly equal to the mean. Persistent dependence can prevent variance reduction: repeated copies of one random variable have a sample average equal to that variable for every n .

The voltage-averaging example in GJWHC is exact rather than asymptotic. If $X \sim \mathcal{N}(0, 100)$ in squared-voltage units and $Y = X + 5$, then $Y \sim \mathcal{N}(5, 100)$ and $P(Y > 0) = 1 - Q(0.5) \approx 0.69146$. Averaging two independent readings gives $M = (Y_1 + Y_2)/2 \sim \mathcal{N}(5, 50)$ and

$$P(M > 0) = 1 - Q\left(\frac{5}{\sqrt{50}}\right) \approx 0.76025.$$

The offset is unchanged while independent averaging reduces variance. If the readings were correlated, the covariance term would have to be included before drawing this conclusion.

5.4 The central limit theorem

The central limit theorem describes the fluctuations around the mean. For i.i.d. variables with finite mean and finite positive variance,

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0, 1).$$

The arrow denotes convergence in distribution. For sufficiently large n , this motivates

$$S_n \approx \mathcal{N}(n\mu, n\sigma^2), \quad \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

The unnormalized sum does not converge to one fixed Gaussian distribution: its mean and variance generally change with n . An independent sum of Gaussian variables is exactly Gaussian at every n ; the CLT is an asymptotic approximation for more general summands. It supplies no universal finite sample size at which every approximation becomes accurate. Highly skewed laws or extreme-tail probabilities can require much larger samples.

For a binomial count $K \sim \text{Bin}(n, p)$, use mean np and variance $np(1-p)$. Because counts occupy integer values, a continuity correction matches each integer to an interval of width one:

$$P(a \leq K \leq b) \approx \Phi\left(\frac{b + 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - 1/2 - np}{\sqrt{np(1-p)}}\right).$$

The approximation is useful when both expected successes and failures are sufficiently numerous. Exact binomial calculation remains preferable when accuracy is important and feasible. Thermal noise is approximately Gaussian because many microscopic contributions

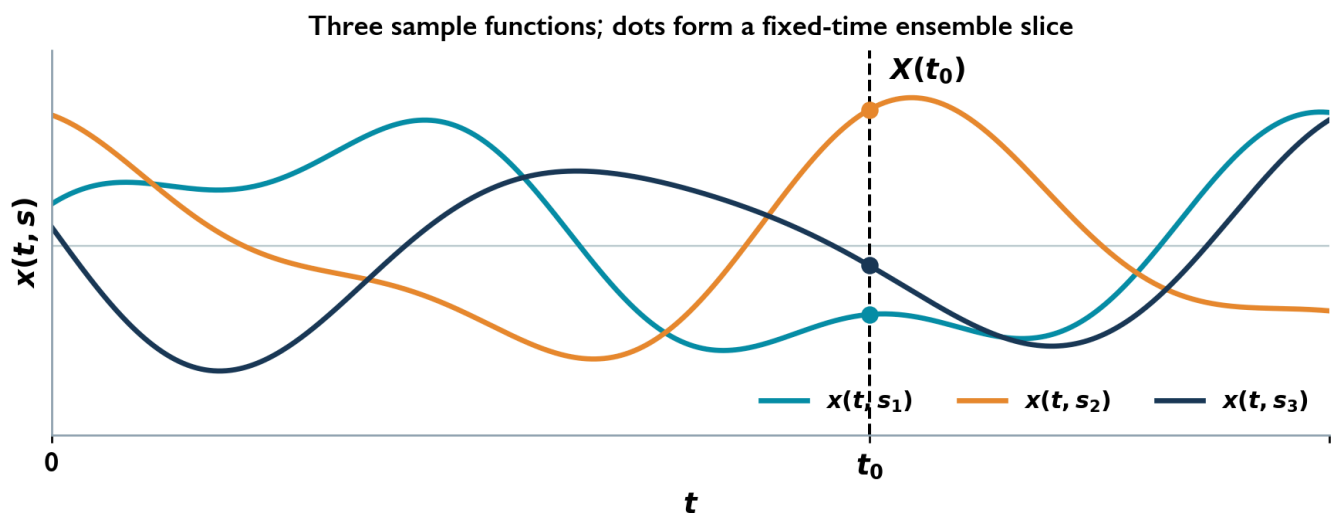
combine; independence and finite-variance assumptions explain this model, while its frequency spectrum requires the separate process analysis developed below.

Lesson 06. Random Processes, Stationarity and Ergodicity

Reference: Proakis and Salehi, Fundamentals of Communication Systems, 2nd ed., Chapter 5 §§5.2–5.2.3, printed pp. 209–218; GJWHC random-process, stationarity, and ergodicity material.

6.1 The process, the realization, and the ensemble

A random process assigns a waveform to every outcome of a random experiment. Write $X(t, s)$, where t denotes time and s denotes an outcome in the sample space. Fixing $s = s_j$ gives the deterministic sample function $x_j(t) = X(t, s_j)$. The collection of possible sample functions is the ensemble. Fixing $t = t_0$ instead gives the random variable $X(t_0)$, whose probability law describes the amplitudes observed across the ensemble at that instant.



Ensemble of sample paths and fixed-time slice.

For any finite time set $t_1, \dots, t_n$, the vector $(X(t_1), \dots, X(t_n))$ has a joint distribution. These finite-dimensional distributions describe temporal dependence; a collection of individual amplitude distributions does not determine that dependence. Continuous-time processes are indexed by real t , whereas discrete-time processes $X[n]$ use integer indices. A process can have continuous time and discrete amplitudes, or discrete time and continuous amplitudes.

A simple random walk illustrates this distinction. Let $X[0] = 0$ and $X[n] = \sum_{k=1}^n U_k$, where independent increments equal $+1$ or -1 with equal probability. Then $E[X[n]] = 0$, but $\text{Var}(X[n]) = n$. Its spreading amplitude distribution makes it nonstationary. For nonnegative integer indices, $R_X[m, n] = \min(m, n)$ because only shared increments survive the expected product. The dependence on absolute indices also violates WSS.

A binary switching process connects a random transition count to temporal correlation. Let $Z(t)$ take values zero and one with equal marginal probabilities, and let every transition toggle its state. If the transition count over duration $\tau > 0$ is independent of the starting state and has the reference exercise law

$$P(N_\tau = n) = \frac{(a\tau)^n}{(1 + a\tau)^{n+1}}, \quad n = 0, 1, \dots, \quad a > 0,$$

then the state is unchanged exactly when the count is even. Summing its geometric even terms gives $P(N_\tau \text{ even}) = (1 + a\tau) / (1 + 2a\tau)$. Consequently,

$$\mu_Z = \frac{1}{2}, \quad R_Z(\tau) = \frac{1}{4} \left(1 + \frac{1}{1 + 2a|\tau|} \right), \quad C_Z(\tau) = \frac{1}{4(1 + 2a|\tau|)}.$$

With stationary transition-count statistics, this is a WSS binary process. The example illustrates how correlations can be derived from switching probabilities even when amplitude values are discrete.

6.2 Ensemble statistical averages

For a real process with finite second moments, define

$$\mu_X(t) = E[X(t)], \quad R_X(t_1, t_2) = E[X(t_1)X(t_2)],$$

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2).$$

The mean is an ensemble average, the autocorrelation measures the mean product at two times, and the autocovariance removes the contribution of the means. These functions are deterministic even though their underlying signals are random. At equal times,

$$R_X(t, t) = E[X^2(t)], \quad C_X(t, t) = \text{Var}(X(t)).$$

Thus mean square and variance coincide only when the mean is zero. If a density exists at a fixed time, the mean can be evaluated as $\mu_X(t) = \int_{-\infty}^{\infty} x f_{X(t)}(x) dx$. A recorded waveform supplies one realization; averaging its amplitudes is a different operation whose agreement with ensemble averaging requires justification.

6.3 Three stationarity conditions

First-order stationarity requires the one-time distribution to be invariant under every time shift:

$$F_{X(t+\tau)}(x) = F_{X(t)}(x).$$

It implies a constant mean when that mean exists. A constant mean alone does not establish first-order stationarity.

Strict-sense stationarity (SSS) requires every finite-dimensional joint distribution to remain unchanged by a common shift of all its time arguments. This condition preserves the entire probability structure, including dependencies between arbitrarily many samples.

Wide-sense stationarity (WSS) requires finite second moments, a constant mean, and autocorrelation depending only on the separation between its arguments. Throughout this course, the lag convention is

$$R_X(\tau) = E[X(t+\tau)X(t)], \quad \mu_X(t) = \mu_X.$$

SSS with finite second moments implies WSS. WSS alone generally does not imply SSS or first-order stationarity. For a Gaussian process, however, the mean and covariance determine every finite-dimensional law, so Gaussian WSS implies SSS. For real WSS processes,

$$R_X(-\tau) = R_X(\tau), \quad |R_X(\tau)| \leq R_X(0), \quad R_X(0) = \sigma_X^2 + \mu_X^2.$$

6.4 Random phase and nonstationary sinusoidal models

Consider $X(t) = A \cos(2\pi f_c t + \Theta)$, where A and f_c are deterministic and Θ is uniform on $[0, 2\pi)$. A time shift replaces Θ by $(\Theta + 2\pi f_c \tau) \bmod 2\pi$, which is still uniform. The whole vector of shifted samples therefore has the original joint law: the process is SSS. Averaging over phase gives $\mu_X = 0$. The product identity yields

$$X(t+\tau)X(t) = \frac{A^2}{2} [\cos(2\pi f_c \tau) + \cos(4\pi f_c t + 2\pi f_c \tau + 2\Theta)].$$

The second term averages to zero, giving

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau), \quad E[X^2(t)] = \frac{A^2}{2}.$$

A uniform phase on only half a period does not have the same shift invariance. If $\Theta \sim \mathcal{U}[0, \pi]$, then $E[A \cos(2\pi f_c t + \Theta)] = -(2A/\pi) \sin(2\pi f_c t)$, which varies with time for $A f_c \neq 0$. Uniformity alone is insufficient; the phase support must match the full-period invariance used in the stationary model.

Random amplitude does not provide the same invariance. If $X(t) = A \cos(2\pi f_c t)$ with $f_c \neq 0$ and $A \sim \mathcal{U}[0, 1]$, then $\mu_X(t) = \frac{1}{2} \cos(2\pi f_c t)$ varies with time. Even zero-mean amplitude is insufficient: if $E[A] = 0$ and $\text{Var}(A) = \sigma_A^2 > 0$, then $\text{Var}(X(t)) = \sigma_A^2 \cos^2(2\pi f_c t)$ varies, preventing WSS.

Random frequency also requires care. For $X(t) = \sin(2\pi Wt)$ with $W \sim \mathcal{U}[0, B]$, $B > 0$,

$$\mu_X(t) = \frac{1 - \cos(2\pi Bt)}{2\pi Bt} \quad (t \neq 0), \quad \mu_X(0) = 0.$$

The changing mean establishes nonstationarity. The value at zero follows directly from $X(0) = 0$ and agrees with the continuous limit.

For an affine-time process $X(t) = A + Bt$, with independent $A, B \sim \mathcal{U}[-1, 1]$,

$$\mu_X(t) = 0, \quad R_X(t_1, t_2) = \frac{1}{3}(1 + t_1 t_2), \quad \text{Var}(X(t)) = \frac{1}{3}(1 + t^2).$$

The constant mean therefore does not rescue a time-varying variance. Starting an integrator at zero gives another GJWHC example. If $X(t) = A \cos(2\pi f_c t)$ with $A \sim \mathcal{N}(0, \sigma_A^2)$ and $f_c \neq 0$, then

$$Y(t) = \int_0^t X(u) du = \frac{A}{2\pi f_c} \sin(2\pi f_c t).$$

At a fixed time, $Y(t)$ is Gaussian with variance $\sigma_A^2 \sin^2(2\pi f_c t) / (2\pi f_c)^2$ when the sine is nonzero. At its zeros it is a deterministic point mass at zero. Its varying variance rules out WSS; the fixed integration origin is part of this model.

A Gaussian quadrature sinusoid provides a useful comparison. Let

$X(t) = A \cos(\omega_c t) + B \sin(\omega_c t)$, where A, B are independent zero-mean Gaussian variables with variances σ_A^2, σ_B^2 and $\omega_c \neq 0$. Then

$$R_X(t_1, t_2) = \frac{\sigma_A^2 + \sigma_B^2}{2} \cos[\omega_c(t_1 - t_2)] + \frac{\sigma_A^2 - \sigma_B^2}{2} \cos[\omega_c(t_1 + t_2)].$$

Equal variances eliminate the time-sum term, giving WSS and, by joint Gaussianity, SSS.

Unequal variances leave an absolute-time dependence. Thus two random Gaussian coefficients can produce stationarity under a condition that a single random coefficient does not satisfy.

6.5 Ergodicity and time averages

Ergodicity concerns whether one sufficiently long realization reproduces a specified ensemble statistic. Mean ergodicity, in the almost-sure sense used here, requires

$$\lim_{L \rightarrow \infty} \frac{1}{L} \int_{-L/2}^{L/2} X(t) dt = \mu_X$$

for almost every realization. Correlation ergodicity similarly requires the time average of $X(t + \tau) X(t)$ to approach $R_X(\tau)$ for each specified lag. Other convergence senses may be used and must be stated.

Stationarity does not guarantee ergodicity. For $X(t) = A$, where A is a nondegenerate random variable with finite second moment, the process is SSS and WSS, with $\mu_X = E[A]$ and $R_X(\tau) = E[A^2]$. Every path is constant, so its time mean is A , rather than the common ensemble mean. In contrast, the fixed-amplitude random-phase sinusoid with $f_c \neq 0$ has time mean zero and time autocorrelation $A^2 \cos(2\pi f_c \tau)/2$; it is ergodic for these two statistics.

6.6 Cyclostationarity

A second-order cyclostationary process has a mean and two-time autocorrelation periodic under a common time shift T_0 :

$$\mu_X(t + T_0) = \mu_X(t), \quad R_X(t + \tau + T_0, t + T_0) = R_X(t + \tau, t).$$

For zero-mean WSS $X(t)$ and $f_c > 0$, deterministic modulation $Y(t) = X(t) \cos(2\pi f_c t)$ gives

$$R_Y(t + \tau, t) = \frac{R_X(\tau)}{2} [\cos(2\pi f_c \tau) + \cos(4\pi f_c t + 2\pi f_c \tau)].$$

This generally depends periodically on t , so Y is cyclostationary, with a valid second-order period $T_0 = 1/(2f_c)$. A stationary mean need not prevent periodically varying second-order statistics. Averaging over a period removes the second cosine:

$$\bar{R}_Y(\tau) = \frac{1}{T_0} \int_0^{T_0} R_Y(t + \tau, t) dt = \frac{1}{2} R_X(\tau) \cos(2\pi f_c \tau).$$

6.7 Multiple processes

Processes are independent when every finite collection of samples from one is independent of every finite collection from the other. They are uncorrelated when their cross covariance vanishes at every time pair. Independence implies this second-order condition when moments exist; its converse generally fails.

Joint WSS requires both processes to be WSS and their cross-correlation to depend only on lag. Define

$$R_{XY}(\tau) = E[X(t + \tau) Y(t)], \quad R_{XY}(\tau) = R_{YX}(-\tau).$$

The sum of jointly WSS processes is itself WSS. For $Z = X + Y$, expand the product to obtain

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau).$$

Both cross terms must be retained unless their disappearance is justified. Zero cross covariance gives $R_{XY} = \mu_X \mu_Y$, so uncorrelated processes with nonzero means can still contribute cross-correlation.

Lesson 07. Correlation Functions and Deterministic Energy Signals

Reference: Proakis and Salehi, Chapter 5, §§5.2.1–5.2.3; Problems 5.33–5.38 and 5.48; GJWHC main pp. 16–18 and supplemental pp. 11–15.

7.1 Ensemble correlation and covariance

For a real process with finite second moments, the two-time autocorrelation is

$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$. A WSS process has constant mean μ_X and a correlation depending only on the lag. Throughout this lesson the lag convention is

$$R_X(\tau) = E[X(t+\tau)X(t)], \quad C_X(\tau) = R_X(\tau) - \mu_X^2.$$

Correlation includes the mean contribution; covariance describes fluctuations after the mean is removed. The zero-lag values are

$$R_X(0) = E[X^2(t)] = \mu_X^2 + \sigma_X^2, \quad C_X(0) = \sigma_X^2.$$

The normalized covariance $C_X(\tau)/\sigma_X^2$, when $\sigma_X^2 > 0$, compares statistical dependence without mixing in the DC level. A correlation need not approach zero at large lag. A nonzero mean, a random constant component, or a persistent sinusoid can prevent such decay.

7.2 Necessary conditions and positive definiteness

For real WSS processes, $R_X(-\tau) = R_X(\tau)$. Cauchy–Schwarz gives

$$|R_X(\tau)| = |E[X(t+\tau)X(t)]| \leq \sqrt{E[X^2(t+\tau)]E[X^2(t)]} = R_X(0).$$

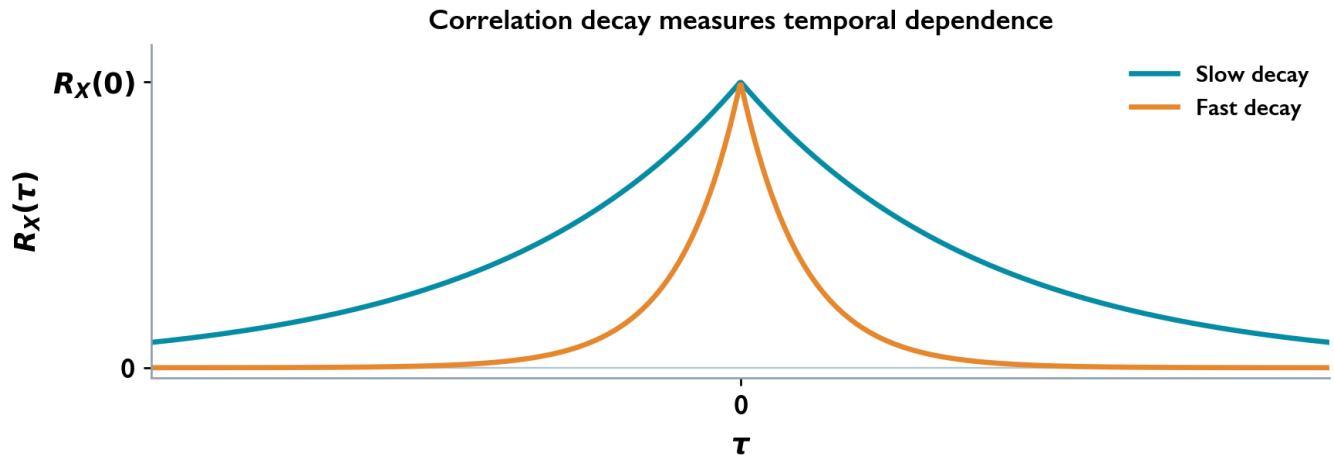
Thus the magnitude is bounded by the zero-lag value. Negative correlation values are permitted: they describe a tendency for separated samples to have opposite signs relative to the relevant reference. If $R_X(0) = 0$, then the process is zero in mean square and every correlation value must vanish.

Evenness and the zero-lag bound are necessary but insufficient. For every finite set of times $t_1, \dots, t_m$ and real coefficients $a_1, \dots, a_m$, a valid autocorrelation must satisfy

$$\sum_{i=1}^m \sum_{j=1}^m a_i a_j R_X(t_i - t_j) = E \left[\left(\sum_{i=1}^m a_i X(t_i) \right)^2 \right] \geq 0.$$

This positive-semidefinite condition states that every linear combination has nonnegative mean-square value. The same property holds for the covariance function. For continuous covariance models, a nonnegative Fourier spectral measure provides the equivalent spectral characterization. Checking a few lags or a small correlation matrix can expose an invalid candidate but cannot prove all required inequalities.

For example, $\cos(2\pi f_0 \tau)$ is valid because a random-phase sinusoid realizes it, with suitable amplitude. In contrast, τ^2 has zero value at the origin and positive values elsewhere, so it violates the bound immediately. An odd nonzero sine cannot be the autocorrelation of a real WSS process. The triangular function discussed below is valid; extending its exterior upward would destroy the zero-lag bound.



Two illustrative autocorrelation widths on the same lag scale.

A short correlation width usually indicates rapid fluctuations and a broad spectral distribution; a wider correlation indicates slower fluctuations. This interpretation concerns decorrelating processes. A periodic correlation has repeated peaks and should not be assigned a single decay time from the central peak alone.

7.3 Crosscorrelation and jointly stationary processes

Define

$$R_{XY}(\tau) = E[X(t + \tau)Y(t)].$$

The order of X and Y determines the lag orientation. For real jointly WSS processes,

$$R_{YX}(\tau) = R_{XY}(-\tau), \quad C_{XY}(\tau) = R_{XY}(\tau) - \mu_X \mu_Y.$$

Crosscorrelation is generally not even. Its bounds are

$$|R_{XY}(\tau)| \leq \sqrt{R_X(0)R_Y(0)}, \quad |C_{XY}(\tau)| \leq \sigma_X \sigma_Y.$$

The arithmetic-geometric mean inequality also gives

$\sqrt{R_X(0)R_Y(0)} \leq [R_X(0) + R_Y(0)]/2$. Both individual processes must be WSS, and their crosscorrelation must depend only on lag, to call them jointly WSS. Individual stationarity alone does not establish the required joint condition.

Processes are uncorrelated when their crosscovariance vanishes at every pair of times, rather than merely at equal times. Independence requires factorization of every finite collection from the two processes. Independence implies zero crosscovariance when moments exist. The

converse requires additional structure, such as joint Gaussianity. A zero correlation at one chosen lag says much less than process independence.

A lag-only cross-correlation does not by itself establish individual stationarity. For independent zero-mean, positive-variance variables A, B , the processes $X(t) = tA$ and $Y(t) = tB$ have $R_{XY}(t_1, t_2) = 0$ for every time pair, yet each has variance proportional to t^2 . Both fail WSS even though their cross-correlation is constant.

For $Z = X + Y$, expand the product before simplifying:

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau).$$

If the processes are uncorrelated, both crosscorrelations equal $\mu_X \mu_Y$. They vanish when at least one mean is zero. Omitting cross terms solely because two processes are described as uncorrelated can therefore lose a nonzero DC contribution.

7.4 Time averages and finite-energy correlations

An ensemble expectation averages across realizations. For a deterministic power waveform, a different operation is the time-average correlation

$$\bar{R}_x(\tau) = \lim_{L \rightarrow \infty} \frac{1}{L} \int_{-L/2}^{L/2} x(t + \tau) x(t) dt,$$

when the limit exists. A correlation-ergodic process permits its ensemble correlation to be recovered from this limit on almost every realization. WSS alone provides no such guarantee. Finite-record estimates fluctuate and may be affected by truncation; they are not exact ensemble quantities.

For a finite-energy deterministic signal, use the unnormalized energy correlation

$$R_x^{(E)}(\tau) = \int_{-\infty}^{\infty} x(t + \tau) x(t) dt, \quad R_x^{(E)}(0) = E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt.$$

For complex signals conjugate the second factor. This correlation is Hermitian; it is even for real signals, and its magnitude is bounded by E_x . Its transform is the energy spectral density $|X(f)|^2$, not a stationary-process PSD.

A pulse of amplitude A and duration T has overlap length $T - |\tau|$ for $|\tau| \leq T$. Hence

$$R_x^{(E)}(\tau) = A^2 (T - |\tau|) u(T - |\tau|), \quad E_x = A^2 T.$$

Its Fourier transform is $X(f) = ATe^{-j\pi fT} \text{sinc}(fT)$, so its energy spectral density is $A^2 T^2 \text{sinc}^2(fT)$. A single pulse has zero infinite-time average power. Randomly shifting that one pulse over a finite interval does not generally produce stationarity, because the probability of pulse occupancy changes with absolute time. A stationary sequence of repeated random symbols, developed in the next lesson, instead produces a power autocorrelation with zero-lag value A^2 and a PSD proportional to T , rather than T^2 .

Solved Exercises for Lesson 07

Exercise 7.1. Autocorrelation of a rectangular energy pulse

IMG_8623.pdf, PDF page(s) 15, 16.

Problem. A deterministic finite-energy pulse is $x(t) = A$ for $0 \leq t \leq 1$ and zero elsewhere. Find and sketch its energy autocorrelation, using $R_x(\tau) = \int_{-\infty}^{\infty} x(t) x(t + \tau) dt$.

Solution.

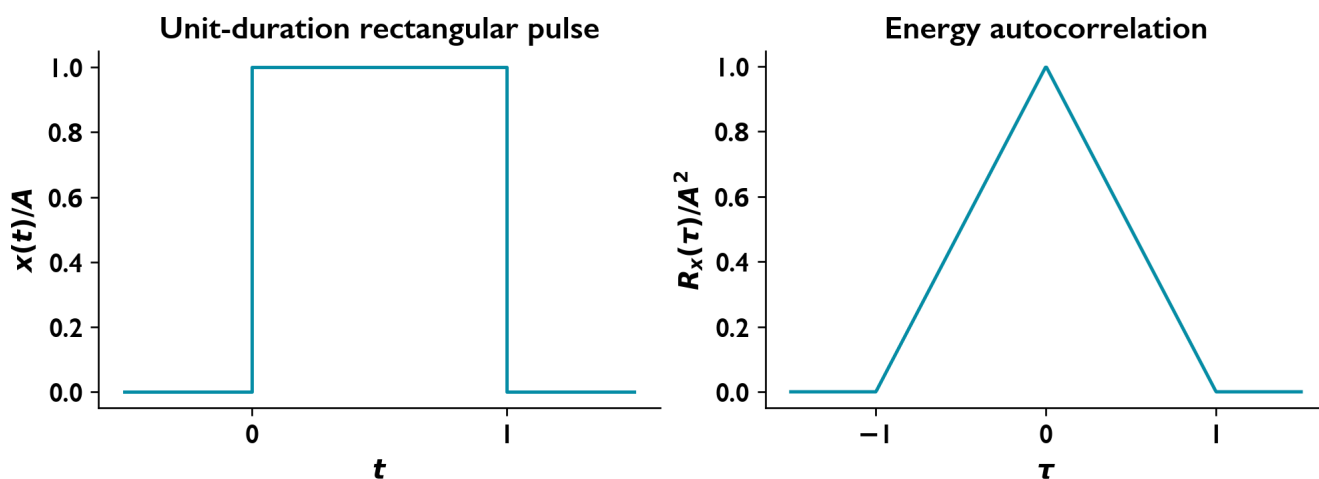
The integrand equals A^2 only where the two intervals $[0, 1]$ and $[-\tau, 1 - \tau]$ overlap. Their overlap length is $1 - |\tau|$ when $|\tau| \leq 1$ and zero otherwise. Therefore

$$R_x(\tau) = \begin{cases} A^2 (1 - |\tau|), & |\tau| \leq 1, \\ 0, & |\tau| > 1. \end{cases}$$

For $0 \leq \tau \leq 1$, the explicit integral is $\int_0^{1-\tau} A^2 dt = A^2 (1 - \tau)$. For $-1 \leq \tau \leq 0$, it is $\int_{-\tau}^1 A^2 dt = A^2 (1 + \tau)$. The graph is an even triangular function with peak A^2 at zero lag and zeros at $\tau = \pm 1$. Check the zero-lag value:

$$R_x(0) = \int_0^1 A^2 dt = A^2 = E_x.$$

This is an energy correlation of a deterministic pulse. It is distinct from the ensemble autocorrelation of a WSS random process. The pulse has finite energy and zero infinite-time average power.



Overlap of a unit-duration pulse produces its triangular energy autocorrelation.

Exercise 7.2. Crosscorrelation of a rectangle and a ramp

IMG_8623.pdf, PDF page(s) 17, 18. Also: IMG_8641.pdf, PDF page 5 (solution continuation).

Problem. Let $x_1(t) = 2$ for $-1 \leq t \leq 1$ and zero elsewhere, and let $x_2(t) = 1 - t$ for $0 \leq t \leq 1$ and zero elsewhere. Find and sketch the energy crosscorrelations $R_{12}(\tau) = \int x_1(t + \tau) x_2(t) dt$ and $R_{21}(\tau) = \int x_2(t + \tau) x_1(t) dt$.

Solution.

It is convenient to evaluate $R_{21}(\tau) = \int x_1(t) x_2(t + \tau) dt$ first, then reflect the result.

The shifted ramp is nonzero on $[-\tau, 1 - \tau]$. Intersect this interval with the rectangular pulse interval $[-1, 1]$ and integrate $2(1 - t - \tau)$ over the intersection. There is no overlap for $\tau < -1$ or $\tau > 2$.

For $-1 \leq \tau \leq 0$, the overlap is $[-\tau, 1]$:

$$R_{21}(\tau) = 2 \int_{-\tau}^1 (1 - t - \tau) dt = 1 - \tau^2.$$

For $0 \leq \tau \leq 1$, the complete ramp lies within the rectangle:

$$R_{21}(\tau) = 2 \int_{-\tau}^{1-\tau} (1 - t - \tau) dt = 1.$$

For $1 \leq \tau \leq 2$, the overlap is $[-1, 1 - \tau]$:

$$R_{21}(\tau) = 2 \int_{-1}^{1-\tau} (1 - t - \tau) dt = (2 - \tau)^2.$$

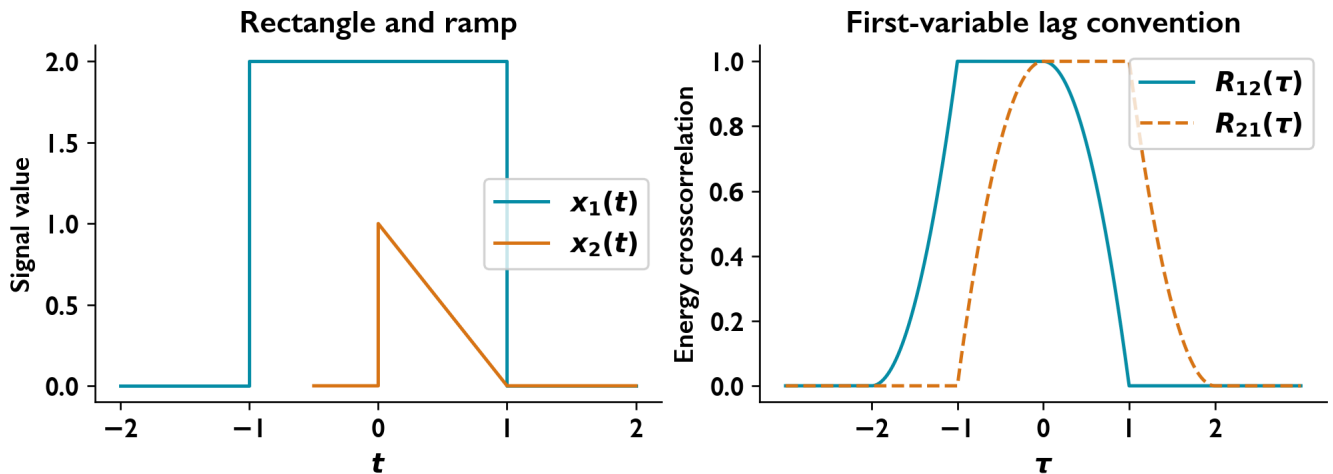
Thus

$$R_{21}(\tau) = \begin{cases} 0, & \tau < -1, \\ 1 - \tau^2, & -1 \leq \tau < 0, \\ 1, & 0 \leq \tau \leq 1, \\ (2 - \tau)^2, & 1 < \tau \leq 2, \\ 0, & \tau > 2. \end{cases}$$

For real signals $R_{12}(\tau) = R_{21}(-\tau)$, so

$$R_{12}(\tau) = \begin{cases} (\tau + 2)^2, & -2 \leq \tau < -1, \\ 1, & -1 \leq \tau \leq 0, \\ 1 - \tau^2, & 0 < \tau \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The crosscorrelation is generally not even. Stating the order and lag convention is essential: reversing the order reflects the graph about zero lag. All adjacent pieces agree at their boundaries; the maximum value is one.



Crosscorrelation order is defined by shifting the first signal. Reversal reflects the lag axis.

Exercise 7.3. Admissibility of autocorrelation candidates

IMG_8623.pdf, PDF page(s) 19. Also: IMG_8641.pdf, PDF page 1 (same three candidates).

Problem. Determine whether each of the following functions can be the autocorrelation of a real WSS process with finite second moment: (1) $r_1(\tau) = \sin(2\pi f_0 \tau)$, where $f_0 \neq 0$; (2) $r_2(\tau) = \tau^2$; (3) $r_3(\tau) = 1 - |\tau|$ for $|\tau| \leq 1$, and $r_3(\tau) = 1 + |\tau|$ for $|\tau| > 1$.

Solution.

A real WSS autocorrelation is even and satisfies $|R_x(\tau)| \leq R_x(0)$ for every lag. These are necessary conditions; passing them alone would not prove validity because positive semidefiniteness is also required.

1. For $f_0 \neq 0$, the sine is odd rather than even. Moreover $r_1(0) = 0$, whereas the function is nonzero at some lags. Either violation excludes it. If $f_0 = 0$, the identically zero function would be valid for the zero process.
2. Although τ^2 is even, $r_2(0) = 0$ and $r_2(\tau) > 0$ for all nonzero lags. It violates the zero-lag bound and is not an autocorrelation.
3. This function is even and has $r_3(0) = 1$, but at $\tau = 2$ it equals $3 > 1$. It violates $|r_3(\tau)| \leq 1$ and is not an autocorrelation.

Therefore none of the three nontrivial candidates is valid. The triangular function $(1 - |\tau|)$ on $|\tau| \leq 1$, extended by zero outside, would be valid; replacing the exterior by $1 + |\tau|$ destroys that validity.

Lesson 08. Power Spectral Density, Cross Spectra, and Random Symbol Sequences

Reference: Proakis and Salehi, Chapter 5, §§5.2.5–5.2.6; Problems 5.49–5.54; GJWHC main pp. 18–20 and binary-sequence supplement pp. 15–17.

8.1 The Wiener–Khinchin relation

The power spectral density distributes the mean-square value of a WSS process over frequency. Using frequency f in hertz and the transform kernel $e^{-j2\pi f\tau}$,

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau, \quad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df.$$

This Wiener–Khinchin pair applies as an ordinary Fourier transform when the required integrability holds, and as a spectral measure when lines or other generalized components occur. At zero lag,

$$P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f) df.$$

For voltage across a normalized one-ohm load, PSD units are V^2/Hz , equivalently W/Hz . Physical power across resistance R_L requires division of the voltage mean square by R_L . Delta arrows represent integrated line weights, not ordinary density heights.

The finite-record interpretation makes the power normalization explicit. With

$$X_L(f) = \int_{-L/2}^{L/2} X(t) e^{-j2\pi ft} dt, \text{ the expected periodogram is } E[|X_L(f)|^2]/L; \text{ its}$$

appropriate long-record limit yields the PSD or spectral measure. Omitting $1/L$ changes the dimensions and confuses power with energy.

For a real process, S_X is real, even and nonnegative. A constant mean contributes

$$S_X(f) = S_{X-\mu_X}(f) + \mu_X^2 \delta(f).$$

Consequently the continuous fluctuation-spectrum area gives variance, while the full spectrum gives mean square. A DC line can also arise from a random constant component. Its presence alone does not establish the deterministic mean of the process.

8.2 Useful transform pairs and spectral lines

Use $\text{sinc}(u) = \sin(\pi u)/(\pi u)$ with $\text{sinc}(0) = 1$, and $\Lambda(u) = \max(1 - |u|, 0)$.

Three important pairs are

$$R_X(\tau) = P\Lambda(\tau/T) \quad \leftrightarrow \quad S_X(f) = PT \text{sinc}^2(fT),$$

$$S_X(f) = S_0\Lambda(f/B) \quad \leftrightarrow \quad R_X(\tau) = S_0B \text{sinc}^2(B\tau),$$

$$S_X(f) = C \mathbf{1}_{\{|f| \leq B\}} \leftrightarrow R_X(\tau) = 2BC \operatorname{sinc}(2B\tau).$$

Their zero-lag values equal their spectral areas. Zero covariance at selected sample spacings establishes uncorrelated centered samples; independence additionally requires a suitable joint law, such as joint Gaussianity.

The GJWHC triangle-plus-line example makes centering concrete. Suppose

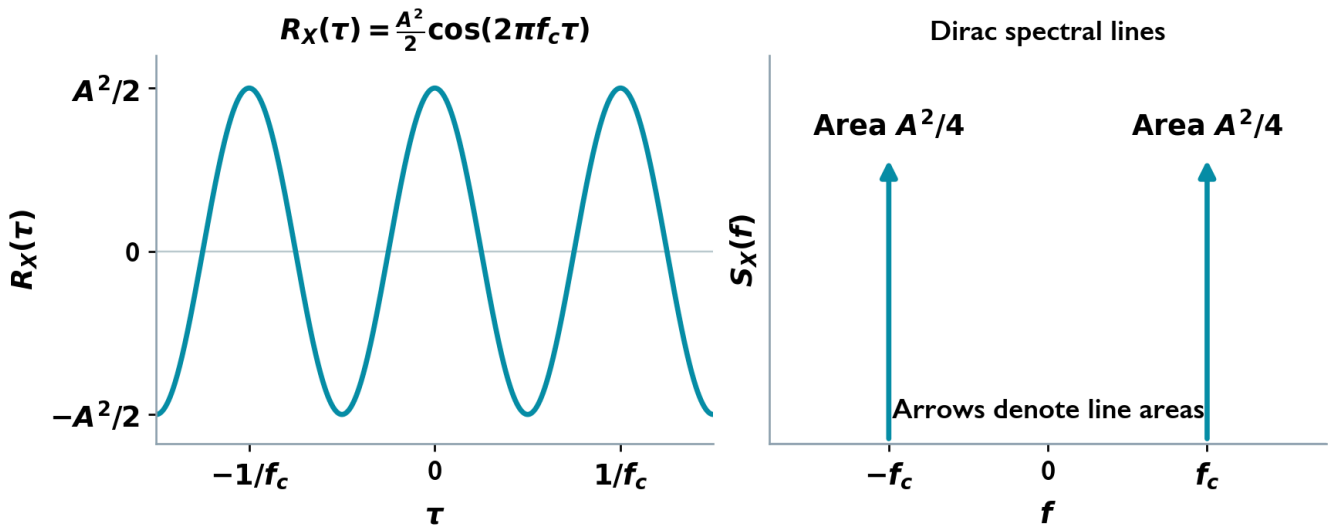
$$S_X(f) = \Lambda(f/f_0) + \delta(f), \quad f_0 > 0.$$

Then $R_X(\tau) = f_0 \operatorname{sinc}^2(f_0\tau) + 1$ and the total mean square is $f_0 + 1$. If the line represents a deterministic unit mean, the covariance is $C_X(\tau) = f_0 \operatorname{sinc}^2(f_0\tau)$, and centered samples spaced by $T_s = m/f_0$, for positive integer m , are uncorrelated. Their autocorrelation at nonzero grid lags is one, rather than zero. If the same DC line instead represents a zero-mean random constant, this inference about covariance and sample independence is unavailable. A spectral line identifies persistent second-order power, not its mean interpretation without additional information.

For a stationary random-phase sinusoid with amplitude A and nonzero carrier f_c ,

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau), \quad S_X(f) = \frac{A^2}{4} [\delta(f - f_c) + \delta(f + f_c)].$$

The two line weights sum to $A^2/2$.



Random-phase sinusoid autocorrelation and two-sided line spectrum; arrows represent line areas.

8.3 Cross spectra and spectra of sums

For jointly WSS processes define $S_{XY}(f) = \mathcal{F}\{R_{XY}(\tau)\}$ using

$R_{XY}(\tau) = E[X(t+\tau)Y(t)]$. For real processes,

$$S_{YX}(f) = S_{XY}^*(f), \quad S_{XY}(-f) = S_{XY}^*(f).$$

A cross spectrum is generally complex, and need not be nonnegative. Its phase records relative timing. For example, if $Y(t) = X(t - d)$, then $S_{XY}(f) = e^{j2\pi fd} S_X(f)$. For ordinary spectral densities, positive semidefiniteness of the spectral matrix implies $|S_{XY}(f)|^2 \leq S_X(f) S_Y(f)$.

The PSD of $Z = X + Y$ is

$$S_Z = S_X + S_Y + S_{XY} + S_{YX} = S_X + S_Y + 2 \operatorname{Re} S_{XY}.$$

Spectra add directly when the real cross contribution vanishes. Zero crosscovariance with two nonzero means leaves $S_{XY} = \mu_X \mu_Y \delta(f)$, producing the extra term $2\mu_X \mu_Y \delta(f)$. Centering removes this mean contribution.

8.4 Carrier modulation and cyclostationary spectra

Let X be WSS and $Y(t) = X(t) \cos(2\pi f_c t)$, with $f_c \neq 0$. Its correlation depends on the time origin:

$$R_Y(t + \tau, t) = \frac{R_X(\tau)}{2} [\cos(2\pi f_c \tau) + \cos(4\pi f_c t + 2\pi f_c \tau)].$$

This is generally cyclostationary. Averaging over a carrier period removes the second term and yields

$$\bar{R}_Y(\tau) = \frac{1}{2} R_X(\tau) \cos(2\pi f_c \tau), \quad \bar{S}_Y(f) = \frac{1}{4} [S_X(f - f_c) + S_X(f + f_c)].$$

The bars distinguish a period-averaged spectrum from an ordinary WSS PSD. An independent uniform random carrier phase can make the modulated process WSS and give the same correlation and spectrum.

8.5 Stationary symbol sequences and pulse shaping

Let $\{a_n\}$ be a WSS discrete sequence with $R_a[k] = E[a_{n+k} a_n^*]$, symbol period T , and pulse $g(t)$ with transform $G(f)$. The pulse train $X(t) = \sum_n a_n g(t - nT)$ usually has a preferred deterministic clock origin and is cyclostationary. A timing offset $U \sim U[0, T)$, independent of the sequence, gives $X(t) = \sum_n a_n g(t - nT - U)$ with WSS second-order statistics, provided the series and moments exist.

Averaging over the symbol clock and grouping pairs separated by k symbols gives

$$\bar{R}_X(\tau) = \frac{1}{T} \sum_k R_a[k] \int_{-\infty}^{\infty} g(v + \tau - kT) g^*(v) dv,$$

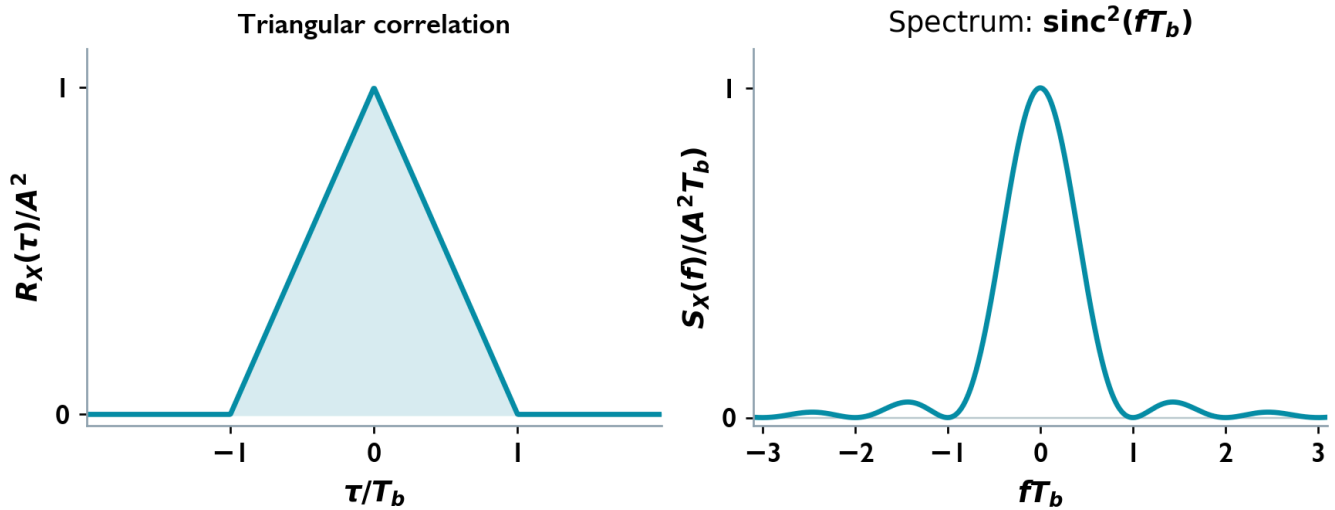
$$\bar{S}_X(f) = \frac{|G(f)|^2}{T} \sum_k R_a[k] e^{-j2\pi f k T}.$$

For nonzero symbol mean μ_a , separate covariance and mean. The mean part produces lines

$$S_{\text{lines}}(f) = \frac{|\mu_a|^2}{T^2} \sum_m |G(m/T)|^2 \delta(f - m/T).$$

For zero-mean independent symbols of variance σ_a^2 , $R_a[k] = \sigma_a^2 \delta[k]$ and $\bar{S}_X = \sigma_a^2 |G|^2 / T$. Correlation introduced by discrete filtering $b_n = \sum_\ell c_\ell a_{n-\ell}$ multiplies the sequence spectrum by $|\sum_\ell c_\ell e^{-j2\pi f \ell T}|^2$, creating frequency-dependent reinforcement and possible nulls.

Equiprobable rectangular bipolar symbols of amplitude A have stationary correlation $A^2 \Lambda(\tau/T)$ and PSD $A^2 T \text{sinc}^2(fT)$. Samples less than one symbol apart share a symbol with probability $1 - |\tau|/T$; different independent symbols contribute zero expected product. The first zeros occur at $\pm 1/T$, giving null-to-null main-lobe width $2/T$. Sidelobes extend indefinitely, so this width is not a strict support bandwidth.



Normalized triangular correlation and sinc-squared PSD of rectangular independent bipolar symbols.

Solved Exercises for Lesson 08

Exercise 8.1. Spectra of linear combinations of processes

IMG_8641.pdf, PDF page(s) 7.

Problem. Let $X(t)$ and $Y(t)$ be zero-mean stationary random processes, and let $Z(t) = X(t) + Y(t)$. Determine the PSD of $Z(t)$, and state the simplification when $X(t)$ and $Y(t)$ are independent.

Solution.

To obtain a time-independent PSD for the sum, assume the two processes are jointly WSS. Define $R_{XY}(\tau) = E[X(t + \tau)Y(t)]$. Expanding the product yields

$$\begin{aligned} R_Z(\tau) &= E\{[X(t) + Y(t)][X(t + \tau) + Y(t + \tau)]\} \\ &= R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau). \end{aligned}$$

Taking Fourier transforms,

$$S_Z(f) = S_X(f) + S_Y(f) + S_{XY}(f) + S_{YX}(f) = S_X(f) + S_Y(f) + 2 \operatorname{Re} S_{XY}(f).$$

For independent zero-mean processes, $R_{XY}(\tau) = 0$ at every lag, and hence

$$S_Z(f) = S_X(f) + S_Y(f).$$

Independence is sufficient; zero cross-correlation at every lag is also sufficient. Individual WSS alone does not ensure joint WSS, so the original premise needs this additional condition before a WSS PSD for the sum can be asserted.

Exercise 8.2. A spectral null in a correlated symbol sequence

University of Tripoli, EE416, Spring 2025 midterm, Q2.

Problem. An i.i.d. sequence $\{\alpha_n\}$ takes values -1 and $+1$ with equal probability. Define $\beta_n = \alpha_n + K\alpha_{n-3}$, with real K , and

$$X(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT).$$

If the power spectrum has a null at $f = 1/(3T)$, determine K .

Solution.

1. Symbol autocorrelation

Since the symbols have zero mean and unit variance,

$$R_\alpha[k] = \delta[k], \quad R_\beta[k] = (1 + K^2)\delta[k] + K\delta[k-3] + K\delta[k+3].$$

Here $\delta[k]$ is the Kronecker delta. The discrete-time spectral factor is

$$\sum_k R_\beta[k] e^{-j2\pi f k T} = 1 + K^2 + 2K \cos(6\pi f T).$$

2. Continuous-time spectrum and the null

For the period-averaged spectrum of the fixed-clock pulse train, or its WSS version with an independent uniform timing offset,

$$\bar{S}_X(f) = \frac{|G(f)|^2}{T} [1 + K^2 + 2K \cos(6\pi f T)].$$

At $f = 1/(3T)$, $\cos(6\pi f T) = 1$. Provided $G(1/(3T)) \neq 0$, the null condition gives $(1 + K)^2 = 0$, hence

$$K = -1.$$

The sequence filter is then $1 - z^{-3}$, which creates zeros at integer multiples of $1/(3T)$. If the pulse already satisfies $G(1/(3T)) = 0$, the specified null alone does not determine K .

Lesson 09. LTI Filtering and Gaussian Processes

Reference: Proakis and Salehi, Chapter 5, §§5.2.4–5.2.6 and §5.3.1; GJWHC filtering and cascade examples; notation follows the supplied Chapter 2.

9.1 Filtering a random process

An LTI system acts on every realization of its input through the same convolution. If its impulse response is $h(t)$ and its transfer function is $H(f)$, the output process is

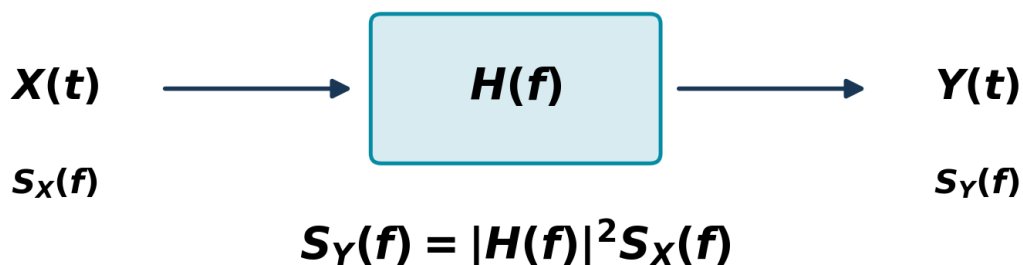
$$Y(t) = \int_{-\infty}^{\infty} h(u) X(t-u) du, \quad H(f) = \int_{-\infty}^{\infty} h(u) e^{-j2\pi fu} du.$$

Assume that $h(t)$ is real and $X(t)$ is real and WSS, with finite second moments, and that the output integral exists in mean square. Absolute integrability of h is a useful sufficient condition; more generally, the spectral integral for output power must be finite. Under these conditions, $Y(t)$ is WSS and the input and output are jointly WSS. A transient response produced by an imposed initial condition must be separated from this stationary convolution model.

Expectation can be interchanged with the integral under the stated moment conditions. Since the input mean is constant,

$$\mu_Y = \int h(u) E\{X(t-u)\} du = \mu_X \int h(u) du = \mu_X H(0).$$

Thus an LTI system scales a constant mean by its DC gain. A zero-mean input remains zero mean. Stationarity is inherited because shifting both observation times leaves every input second-order quantity unchanged inside the convolution integrals.



An LTI system maps input correlation and PSD to output correlation and PSD.

9.2 Output correlation and input–output orientation

Use the Chapter 2 cross-correlation convention

$$R_{XY}(\tau) = E\{X(t + \tau) Y(t)\}.$$

For a real WSS process, $R_X(\tau) = E\{X(t) X(t + \tau)\}$ is even. Correlating the input with the convolution output gives

$$R_{XY}(\tau) = \int h(u) R_X(\tau + u) du = R_X(\tau) * h(-\tau).$$

Reversing the variables reverses the lag:

$$R_{YX}(\tau) = R_{XY}(-\tau) = R_X(\tau) * h(\tau).$$

The two orders are generally different. This distinction is especially visible for a causal filter driven by white noise: the output–input correlation follows the causal impulse response, whereas the input–output correlation follows its time reversal.

For the output autocorrelation, expand both output integrals:

$$R_Y(\tau) = \iint h(u) h(v) R_X(\tau - u + v) du dv = R_X(\tau) * h(\tau) * h(-\tau).$$

The covariance obeys the same convolution formula with C_X in place of R_X . This centered form separates fluctuations from a deterministic mean.

9.3 Spectral description and cascaded systems

Taking Fourier transforms converts these convolutions into products:

$$S_Y(f) = |H(f)|^2 S_X(f), \quad S_{YX}(f) = H(f) S_X(f), \quad S_{XY}(f) = H^*(f) S_X(f).$$

The output PSD depends on the magnitude squared of the transfer function. The cross spectrum retains its phase and therefore identifies how input and output align in time. Always identify the variable order before substituting a cross-spectrum formula.

The output mean-square power and variance are

$$E\{Y^2(t)\} = \int S_Y(f) df, \quad \sigma_Y^2 = \int S_Y(f) df - \mu_Y^2.$$

For a nonzero mean, the full PSD contains the line $\mu_Y^2 \delta(f)$. If S_X instead denotes the centered fluctuation spectrum, its filtered area directly gives variance; the deterministic DC line must then be stated separately. For a cascade, $H_{\text{eq}} = H_1 H_2$, so $S_Y = |H_1 H_2|^2 S_X$. A valid intermediate output can be reused rather than analyzed as a new independent input.

9.4 Delays, differences, differentiation and quadrature

A pure delay, $Y(t) = X(t - d)$, has $H(f) = e^{-j2\pi fd}$. It preserves the mean, autocorrelation and PSD, but changes the input-output correlations to

$$R_{YX}(\tau) = R_X(\tau - d), \quad R_{XY}(\tau) = R_X(\tau + d).$$

A shifted difference, $Y(t) = X(t + a) - X(t - a)$, has zero mean and

$$R_Y(\tau) = 2R_X(\tau) - R_X(\tau - 2a) - R_X(\tau + 2a),$$

$$H(f) = 2j \sin(2\pi fa), \quad S_Y(f) = 4 \sin^2(2\pi fa) S_X(f).$$

An ideal differentiator has $H(f) = j2\pi f$. Its WSS output exists in mean square when

$$\int_{-\infty}^{\infty} (2\pi f)^2 S_X(f) df < \infty.$$

When the required derivatives exist,

$$R_Y(\tau) = -R_X''(\tau), \quad R_{YX}(\tau) = R_X'(\tau), \quad R_{XY}(\tau) = -R_X'(\tau).$$

Evenness of R_X gives zero same-time covariance between $X(t)$ and its derivative. Their unequal-time cross-correlation generally remains nonzero, so the processes are not generally independent. Their cross spectrum is imaginary; this is why the PSD of their real-valued sum can add without a real cross term.

A Hilbert or quadrature filter has $H(f) = -j \operatorname{sgn}(f)$. Away from DC its magnitude is one, so it preserves the fluctuation PSD and power while shifting spectral phase by a quarter cycle. Exclude a DC component when asserting complete power preservation. These examples demonstrate why unchanged output power does not imply unchanged input-output correlation.

More explicitly, the Hilbert impulse response is $h(t) = \text{PV} [1/(\pi t)]$, with the integral interpreted in the principal-value sense. When the input has no DC spectral mass, write $\widehat{R}_X$ for the Hilbert transform of R_X . The corresponding relations are

$$\mu_Y = 0, \quad R_Y(\tau) = R_X(\tau), \quad R_{YX}(\tau) = \widehat{R}_X(\tau), \quad R_{XY}(\tau) = -\widehat{R}_X(\tau).$$

A finite-time moving average is another useful filter:

$$Y(t) = \frac{1}{2T_a} \int_{t-T_a}^{t+T_a} X(u) du, \quad \text{quad}h(t) = \frac{1}{2T_a} \mathbf{1}_{\{|t| \leq T_a\}}, \quad T_a > 0.$$

It has $H(f) = \text{sinc}(2T_a f)$, preserves the mean, and gives

$$S_Y(f) = \text{sinc}^2(2T_a f) S_X(f), \quad R_Y = R_X * \left[\frac{1}{2T_a} \Lambda\left(\frac{\tau}{2T_a}\right) \right].$$

The cross-correlation is $R_{XY} = R_X * h$, because this averaging kernel is even. The triangular kernel in R_Y results from the overlap of two averaging windows.

The stationarity-through-LTI result has no general converse. For example, let

$X(t) = B(t) + \cos(2\pi t/T)$ with zero-mean WSS $B(t)$. The input has a time-varying mean, but $Y(t) = X(t) - X(t-T) = B(t) - B(t-T)$ is WSS. The filter has removed the nonstationary component, so a stationary output does not prove that its input was stationary.

9.5 Gaussian processes and linear closure

A Gaussian process is defined by every finite collection of time samples: for any $t_1, \dots, t_m$, the vector $[X(t_1), \dots, X(t_m)]^T$ is jointly Gaussian, including possible degenerate cases. Individual Gaussian marginals alone are insufficient. The mean function and covariance function determine all finite-dimensional Gaussian laws. Consequently, a Gaussian WSS process is also strictly stationary.

An LTI output formed from a Gaussian input is Gaussian because each finite approximation to the convolution is a linear combination of jointly Gaussian samples, and a convergent mean-square limit retains the Gaussian law. The input and output are jointly Gaussian as well. This result requires existence of the output; an ideal differentiator applied to unfiltered ideal white noise does not have finite ordinary output variance.

For jointly Gaussian processes, zero cross covariance at every pair of times implies process independence. Zero covariance only at one common time gives independence of that particular pair of random variables. The distinction concerns entire sample collections, not terminology alone.

9.6 Conditional Gaussian prediction

A Gaussian observation makes conditional prediction explicit. For jointly Gaussian variables with positive variances and $|\rho| < 1$,

$$X | Y = y \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y), \sigma_X^2 (1 - \rho^2)\right).$$

The conditional mean is linear in the observed deviation, and the conditional variance decreases as correlation strengthens. At $|\rho| = 1$, the conditional law is deterministic on a line rather than an ordinary positive-variance density.

For a jointly Gaussian target vector $\mathbf{U}$ and observation vector $\mathbf{V}$, with invertible observation covariance $\mathbf{C}_V$,

$$E\{\mathbf{U} \mid \mathbf{V} = \mathbf{v}\} = \boldsymbol{\mu}_U + \mathbf{C}_{UV}\mathbf{C}_V^{-1}(\mathbf{v} - \boldsymbol{\mu}_V),$$

$$\mathbf{C}_{U|V} = \mathbf{C}_{UU} - \mathbf{C}_{UV}\mathbf{C}_V^{-1}\mathbf{C}_{VU}.$$

Time samples of a Gaussian process can be substituted directly into these vector formulas. Their entries are obtained from the covariance function, linking statistical prediction to the filtering and correlation framework developed above.

Solved Exercises for Lesson 09

Exercise 9.1. Independent Gaussian noise branches and an estimation-error network

IMG_8587.pdf, PDF page(s) 4, 5.

Problem. Two independent Gaussian white-noise processes $W_1(t)$ and $W_2(t)$ have

$$R_{W_i}(\tau) = \frac{\eta}{2}\delta(\tau).$$

Each passes through the same LTI filter $h(t)$, producing $N_1(t)$ and $N_2(t)$. Their difference $N(t) = N_1(t) - N_2(t)$ passes through a third filter $h_o(t)$. The final subtractor forms $\varepsilon(t) = N_1(t) - (h_o * N)(t)$.

Find the density of $N(t)$, the joint density of $N_1(t)$ and $N_2(t)$, $\Pr[N(t) > -1]$, and the simplest expression for $\varepsilon(t)$.

Solution.

1. Determine the branch statistics. Let

$$E_h = \int_{-\infty}^{\infty} |h(u)|^2 du, \quad \sigma_b^2 = \frac{\eta}{2}E_h.$$

Assume real filters, $\eta > 0$, and $0 < E_h < \infty$ for the ordinary Gaussian densities below. Finite E_h is necessary for finite instantaneous output variance. LTI filtering preserves Gaussianity. Since the input processes are independent and the branches do not share an input, N_1 and N_2 remain independent, each with mean zero and variance σ_b^2 . Their PSDs are

$$S_{N_1}(f) = S_{N_2}(f) = \frac{\eta}{2}|H(f)|^2.$$

2. Form the difference. At any fixed time,

$$N = N_1 - N_2 \sim \mathcal{N}(0, 2\sigma_b^2) = \mathcal{N}(0, \eta E_h).$$

Consequently

$$f_N(n) = \frac{1}{\sqrt{2\pi\eta E_h}} \exp\left(-\frac{n^2}{2\eta E_h}\right).$$

Its PSD is $S_N(f) = \eta |H(f)|^2$. Here the second parameter in $\mathcal{N}(\mu, \sigma^2)$ denotes variance.

3. Use independence for the joint density.

$$f_{N_1 N_2}(n_1, n_2) = \frac{1}{2\pi\sigma_b^2} \exp\left[-\frac{n_1^2 + n_2^2}{2\sigma_b^2}\right].$$

4. Standardize the requested probability. With Φ the standard-normal CDF,

$$\Pr(N > -1) = \Phi\left(\frac{1}{\sqrt{\eta E_h}}\right) = 1 - Q\left(\frac{1}{\sqrt{\eta E_h}}\right).$$

A numerical value requires η and $h(t)$; the supplied screenshots do not provide them.

5. Simplify the network algebra. Without assuming a specific third filter,

$$\varepsilon = (\delta - h_o) * N_1 + h_o * N_2,$$

and independence gives

$$S_\varepsilon(f) = \frac{\eta}{2} |H(f)|^2 \left(|1 - H_o(f)|^2 + |H_o(f)|^2 \right).$$

If the third filter is intended to be the minimum mean-square estimator of N_1 from $N = N_1 - N_2$, then

$$H_o(f) = \frac{S_{N_1 N}(f)}{S_N(f)} = \frac{1}{2}, \quad h_o(t) = \frac{1}{2} \delta(t).$$

In that optimum case,

$$\varepsilon(t) = \frac{N_1(t) + N_2(t)}{2}, \quad \text{Var}(\varepsilon) = \frac{\sigma_b^2}{2} = \frac{\eta E_h}{4}.$$

This optimum is stated separately because the screenshots do not specify a numerical third-filter response or explicitly impose an optimality condition.

Degenerate case. If $\eta E_h = 0$, both branch outputs and their difference vanish in mean square. Their joint law is a point mass at the origin, and $\Pr[N(t) > -1] = 1$. The positive-variance density formulas do not apply in this case.

Exercise 9.2. Output autocorrelation of a rectangular LTI filter

IMG_8587.pdf, PDF page(s) 6.

Problem. A WSS input process has

$$R_X(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The impulse response is $h(t) = u(t) - u(t-1)$. Use convolution to determine the output autocorrelation $R_Y(\tau)$.

Solution.

1. Use the LTI autocorrelation relation. For a real impulse response,

$$R_Y = R_X * h * \tilde{h}, \quad \tilde{h}(t) = h(-t).$$

The convolution of a unit rectangular pulse with its time reversal is the triangular function:

$$(h * \tilde{h})(\tau) = \Lambda(\tau), \quad \Lambda(\tau) = \max(1 - |\tau|, 0).$$

Since $R_X = \Lambda$, the required function is $R_Y = \Lambda * \Lambda$.

2. Evaluate the overlap integral.

$$R_Y(\tau) = \int_{-1}^1 (1 - |v|) \max(1 - |\tau - v|, 0) dv.$$

Both triangles are even, so the result is even. Let $a = |\tau|$. Splitting the overlap where either absolute value changes sign gives

$$R_Y(\tau) = \begin{cases} \frac{2}{3} - a^2 + \frac{1}{2}a^3, & 0 \leq a \leq 1, \\ \frac{1}{6}(2 - a)^3, & 1 < a \leq 2, \\ 0, & a > 2, \end{cases} \quad a = |\tau|.$$

The branches agree at $a = 1$, where both equal $1/6$. The output average power is

$$R_Y(0) = 2/3.$$

As an independent frequency-domain check, $S_X(f) = \text{sinc}^2(f)$, $|H(f)|^2 = \text{sinc}^2(f)$, and $S_Y(f) = \text{sinc}^4(f)$, where $\text{sinc}(z) = \sin(\pi z)/(\pi z)$.

Exercise 9.3. A delay-and-difference filter

IMG_8641.pdf, PDF page(s) 14.

Problem. A stationary real random process $X(t)$ is transformed according to

$$Y(t) = X(t + a) - X(t - a).$$

Find (a) the autocorrelation of $Y(t)$ and (b) its PSD in terms of those of $X(t)$.

Solution.

(a) Autocorrelation. Define $R_X(\tau) = E[X(t+\tau)X(t)]$. Expanding the four products and using stationarity gives

$$\begin{aligned} R_Y(\tau) &= E\{[X(t+a) - X(t-a)][X(t+\tau+a) - X(t+\tau-a)]\} \\ &= 2R_X(\tau) - R_X(\tau-2a) - R_X(\tau+2a). \end{aligned}$$

Its mean is zero because the two shifted input means cancel. The output power is therefore $R_Y(0) = 2[R_X(0) - R_X(2a)]$ for a real WSS input. **(b) PSD.** The transform of the shifted difference has transfer function

$$H(f) = e^{j2\pi fa} - e^{-j2\pi fa} = 2j \sin(2\pi fa).$$

Consequently,

$$S_Y(f) = |H(f)|^2 S_X(f) = 4 \sin^2(2\pi fa) S_X(f).$$

The same result follows from Fourier transforming the autocorrelation and using $2 - 2 \cos(4\pi fa) = 4 \sin^2(2\pi fa)$. For $a > 0$, the displayed operator includes an advance and is noncausal; adding a common delay leaves the PSD unchanged.

Lesson 10. White Noise, Thermal Noise and Equivalent Bandwidth

Reference: Proakis and Salehi, Chapter 5, §§5.3.2–5.3.3; GJWHC ideal low-pass and RC filtering; supplied Chapter 2 §§2.6–2.7 and §2.9.

10.1 Whiteness and the continuous-time ideal model

Whiteness describes second-order spectral structure; Gaussianity describes probability distributions. A process may be Gaussian with a nonflat PSD, or white without being Gaussian. These properties should therefore be specified separately.

The standard ideal zero-mean white-noise model has the two-sided PSD

$$S_n(f) = \frac{N_0}{2}, \quad -\infty < f < \infty,$$

and the distributional autocorrelation

$$R_n(\tau) = \frac{N_0}{2} \delta(\tau).$$

Here N_0 is a spectral-density parameter, not the total noise power. If power is measured in watts, N_0 has units watts per hertz. The corresponding one-sided density for positive frequencies is N_0 , because the negative-frequency contribution has been combined with the positive-frequency contribution.

The total ideal power diverges: $\int_{-\infty}^{\infty} (N_0/2) df = \infty$. Accordingly, unfiltered continuous-time white noise is treated as a generalized random process. It does not possess an ordinary finite-variance point-sample Gaussian density. The delta correlation expresses a limit of ever narrower correlation functions as bandwidth increases.

A precise operational interpretation uses an integrated or filtered measurement. For a square-integrable weighting function $q(t)$,

$$Z_q = \int n(t) q(t) dt, \quad \text{Var}(Z_q) = \frac{N_0}{2} \int |q(t)|^2 dt.$$

For Gaussian white noise, these measurements are jointly Gaussian. Their covariance is proportional to the inner product of the weighting functions, so orthogonal measurements are independent. This supplies a finite-variance interpretation of white-noise independence without treating ideal point values as ordinary random variables.

10.2 Thermal noise and its engineering approximation

Thermal agitation produces random electrical motion even when no information signal is applied. A large number of microscopic contributions supports the Gaussian approximation through the central limit theorem. Over a fixed temperature and an ordinary communication band, thermal noise is commonly modeled as stationary, zero mean, Gaussian and approximately white.

For the available-noise-power convention used here,

$$N_0 = kT, \quad S_n(f) \approx \frac{kT}{2},$$

where k is Boltzmann's constant, approximately 1.380649×10^{-23} J/K, and T is absolute temperature in kelvin. Temperature in Celsius must be converted using $T = T_C + 273.15$. The density grows in direct proportion to absolute temperature. A voltage-noise PSD across a specified resistance involves the circuit's resistance and loading; it must not be equated to available power density without that conversion.

The physically relevant spectrum eventually decreases at sufficiently high frequencies. An even, two-sided form of the thermal spectrum in the reference model is

$$S_{th}(f) = \frac{h|f|}{2[\exp(h|f|/(kT)) - 1]},$$

with the value at zero defined by its limit. Here h is Planck's constant. If $h|f| \ll kT$, the expansion $e^x - 1 \approx x$ gives $S_{th}(f) \approx kT/2$. This condition explains why the white approximation is useful over an engineering band while infinite-bandwidth whiteness remains an idealization. No abrupt physical cutoff is implied.

10.3 Finite noise power through a filter

If white noise drives an LTI filter,

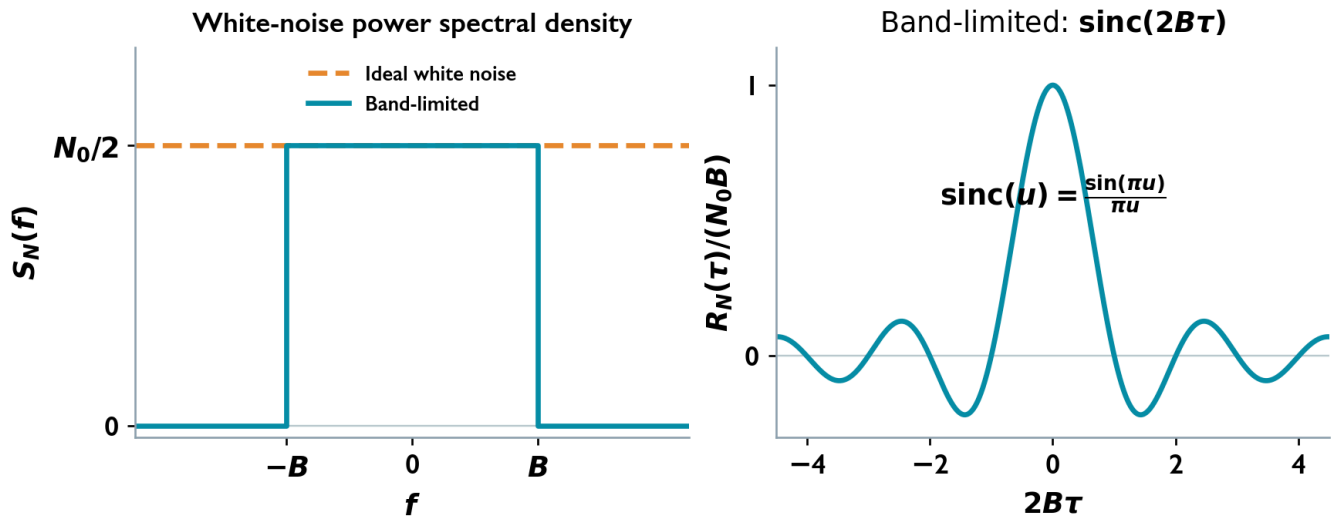
$$S_{N_o}(f) = \frac{N_0}{2} |H(f)|^2, \quad P_{N_o} = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df.$$

A finite integral of squared gain is the condition for finite output noise power. An amplitude response must be squared before power integration. For a real filter, $|H(-f)|^2 = |H(f)|^2$, so the total power is also $N_0 \int_0^{\infty} |H(f)|^2 df$. The output remains Gaussian when the input is Gaussian, but its spectrum generally becomes colored.

For a unity-gain ideal low-pass filter with edges $-B$ and B ,

$$S_{N_o}(f) = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right), \quad R_{N_o}(\tau) = N_0 B \text{sinc}(2B\tau),$$

where $\text{sinc}(x) = \sin(\pi x) / (\pi x)$. The output variance is $N_0 B$. At distinct samples spaced by an integer multiple of $1/(2B)$, this autocorrelation is zero. If the filtered noise is Gaussian, any finite collection on such a sampling grid consists of independent Gaussian variables. Other sample separations generally give correlated values.



Ideal white density and its finite-bandwidth low-pass output; the output correlation is a sinc.

10.4 RC filtering and correlation time

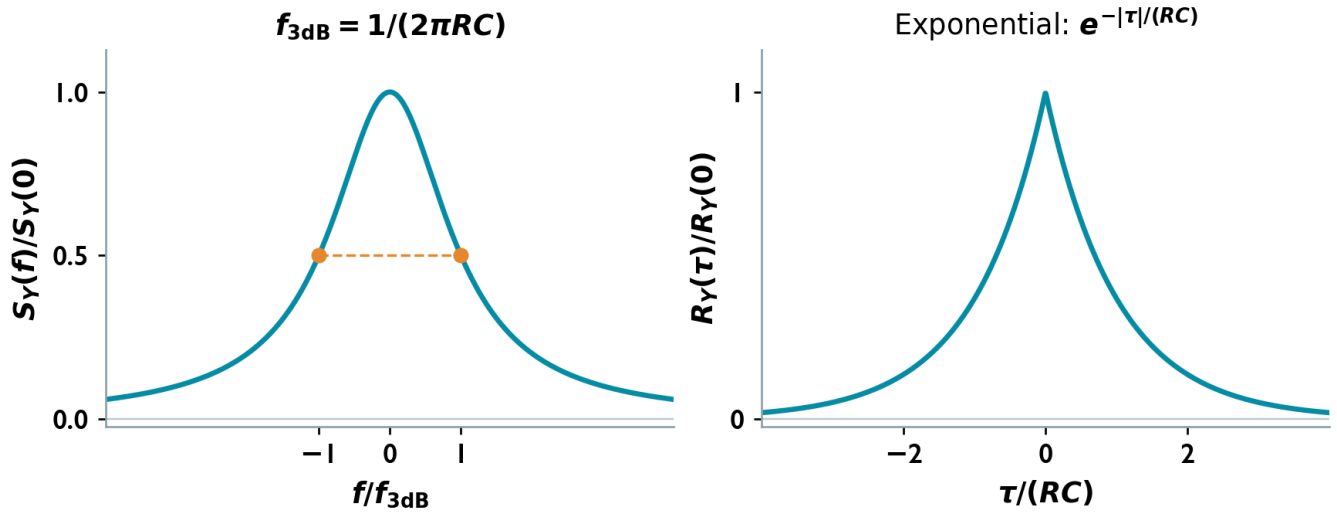
For an RC low-pass filter, define $\tau_c = RC > 0$. Its frequency response and output PSD are

$$H(f) = \frac{1}{1 + j2\pi f\tau_c}, \quad S_{N_o}(f) = \frac{N_0/2}{1 + (2\pi f\tau_c)^2}.$$

The Fourier pair for an exponential gives

$$R_{N_o}(\tau) = \frac{N_0}{4\tau_c} e^{-|\tau|/\tau_c}, \quad \sigma_{N_o}^2 = \frac{N_0}{4RC}.$$

Unlike the ideal low-pass sinc, this exponential correlation has no finite nonzero-lag zeros. The correlation time RC controls how quickly samples lose correlation. Increasing RC narrows the output spectrum and reduces its noise power. The same output variance can be computed from the PSD area or from zero-lag autocorrelation; agreement is a useful check on factors of two and frequency units.



RC-filtered white noise has a Lorentzian PSD and an exponential autocorrelation.

10.5 Noise-equivalent bandwidth

A practical filter passes a weighted range of frequencies rather than a perfect rectangle. Let

$$g = \max_f |H(f)|^2 > 0.$$

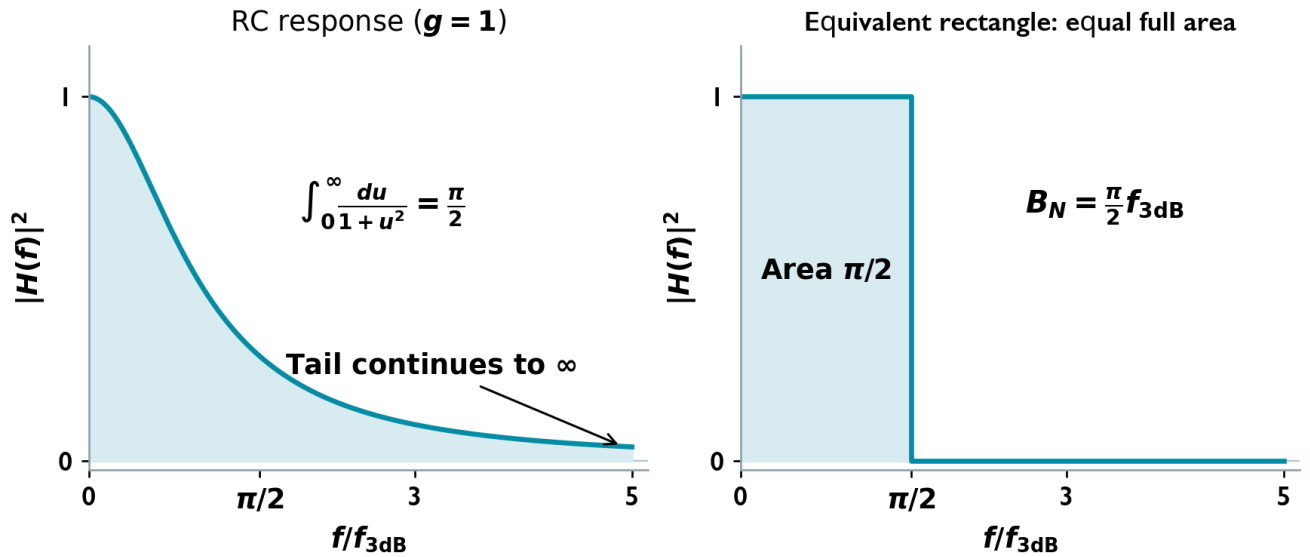
Define its one-sided noise-equivalent bandwidth by

$$B_N = \frac{1}{2g} \int_{-\infty}^{\infty} |H(f)|^2 df = \frac{1}{g} \int_0^{\infty} |H(f)|^2 df$$

for a real filter. The output power is then

$$P_{N_o} = g N_0 B_N.$$

This bandwidth is the width of an ideal positive-frequency rectangle of height g with the same area as the actual squared-gain response. For a band-pass filter, the rectangle is located in the passband; the definition still includes its negative-frequency mirror through the factor two. Ordinary passband width and equivalent bandwidth agree for a rectangular unity-gain filter, but need not agree for a practical response.



The equivalent rectangle and the actual squared-gain curve have equal positive-frequency area.

For the RC filter, $g = 1$. Substituting $u = 2\pi RCf$ gives

$$\begin{aligned}
 B_N &= \int_0^\infty \frac{df}{1 + (2\pi RCf)^2} \\
 &= \frac{1}{2\pi RC} \int_0^\infty \frac{du}{1 + u^2} \\
 &= \frac{1}{2\pi RC} \left[\tan^{-1} u \right]_0^\infty = \frac{1}{4RC}.
 \end{aligned}$$

Since $f_{3dB} = 1 / (2\pi RC)$, the equivalent bandwidth is $B_N = (\pi/2) f_{3dB}$. The 3-dB cutoff alone therefore understates the total white noise passed by this filter.

10.6 Recovering a Gaussian output density

After establishing Gaussianity and a finite output variance, the marginal distribution follows immediately. For zero-mean filtered Gaussian noise with $\sigma_o^2 > 0$,

$$f_{N_o(t)}(x) = \frac{1}{\sqrt{2\pi\sigma_o^2}} \exp\left(-\frac{x^2}{2\sigma_o^2}\right), \quad \sigma_o^2 = \int S_{N_o}(f) df.$$

Thus the ideal low-pass output uses $\sigma_o^2 = N_0 B$, whereas the RC output uses $N_0 / (4RC)$. A two-sided threshold probability is $P(|N_o(t)| > a) = 2Q(a/\sigma_o)$ for $a \geq 0$. The PSD area supplies the variance, Gaussian closure supplies the law, and the Q function supplies the probability. These are separate logical steps; the same PSD area for a non-Gaussian process would not establish a Gaussian PDF.

Solved Exercises for Lesson 10

Exercise 10.1. White noise through a first-order lowpass filter

IMG_864I.pdf, PDF page(s) 4.

Problem. Zero-mean white noise $N_i(t)$, with $R_{N_i}(\tau) = \frac{N_0}{2}\delta(\tau)$, is applied to the first-order low-pass filter

$$H(\omega) = \frac{1}{1 + j\omega/\omega_0}, \quad \omega_0 > 0.$$

Find (a) the impulse response, (b) the output-noise autocorrelation, (c) the input-output cross-correlation, and (d) the output noise power.

Solution.

(a) **Impulse response.** Rewrite the transfer function as $H(\omega) = \omega_0 / (\omega_0 + j\omega)$. The Fourier transform pair $e^{-at}u(t) \leftrightarrow 1/(a + j\omega)$, for $a > 0$, gives

$$h(t) = \omega_0 e^{-\omega_0 t} u(t).$$

(b) **Output autocorrelation.** In hertz, $H(f) = \omega_0 / (\omega_0 + j2\pi f)$, so

$$S_{N_o}(f) = \frac{N_0}{2} \frac{\omega_0^2}{\omega_0^2 + (2\pi f)^2}.$$

Since $e^{-a|\tau|} \leftrightarrow 2a / [a^2 + (2\pi f)^2]$,

$$R_{N_o}(\tau) = \frac{N_0 \omega_0}{4} e^{-\omega_0 |\tau|}.$$

(c) **Cross-correlation.** Define $R_{N_i N_o}(\tau) = E[N_i(t + \tau) N_o(t)]$. Using the convolution expression for the output,

$$R_{N_i N_o}(\tau) = \int h(v) R_{N_i}(\tau + v) dv = \frac{N_0}{2} h(-\tau) = \frac{N_0 \omega_0}{2} e^{\omega_0 \tau} u(-\tau).$$

The reverse-order correlation is $R_{N_o N_i}(\tau) = R_{N_i N_o}(-\tau) = \frac{N_0}{2} h(\tau)$. Its cross spectrum is $S_{N_o N_i}(f) = H(f) S_{N_i}(f)$; in the original order, $S_{N_i N_o}(f) = H^*(f) S_{N_i}(f)$. (d) **Output power.** The output has zero mean, hence

$$P_{N_o} = R_{N_o}(0) = \frac{N_0 \omega_0}{4}.$$

Equivalently, integrating the Lorentzian output PSD gives the same result. Ideal input white noise has infinite total power, although the stable filter produces finite output power.

Exercise 10.2. Ideal lowpass filtering of white Gaussian noise

IMG_864I.pdf, PDF page(s) 8.

Problem. White noise with two-sided PSD $N_0/2$ is passed through an ideal low-pass filter

$$H(f) = \begin{cases} 1, & |f| \leq B, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the output-noise autocorrelation.

Solution.

The output PSD is

$$S_{N_o}(f) = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right).$$

By the inverse Fourier transform,

$$\begin{aligned} R_{N_o}(\tau) &= \frac{N_0}{2} \int_{-B}^B e^{j2\pi f\tau} df \\ &= N_0 B \text{sinc}(2B\tau), \end{aligned}$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$, with $\text{sinc}(0) = 1$. The noise power is $R_{N_o}(0) = N_0 B$, and the first nonzero-lag zeros occur at $\tau = \pm 1/(2B)$. Filtering changes the white-noise impulse correlation into a sinc correlation.

Exercise 10.3. Triangular noise through an ideal bandpass filter

IMG_8641.pdf, PDF page(s) 10.

Problem. A zero-mean noise process has the two-sided PSD

$$S_{N_i}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right), & |f| < 10^8 \text{ Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

The noise passes through a unity-gain ideal band-pass filter with positive-frequency passband width 2 MHz, centered at 50 MHz, and the corresponding negative-frequency passband. Find (a) the output PSD and (b) the input and output noise powers.

Solution.

(a) Output PSD. The positive-frequency passband extends from 49 to 51 MHz. Therefore,

$$S_{N_o}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right), & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Noise powers. The total input power is the area of the two-sided triangle:

$$P_{N_i} = \frac{1}{2} (2 \times 10^8) (10^{-8}) = 1.$$

The output PSD is even, so both passbands must be included:

$$\begin{aligned} P_{N_o} &= 2 \left(10^{-8} \right) \int_{49 \times 10^6}^{51 \times 10^6} \left(1 - \frac{f}{10^8} \right) df \\ &= 2 \left(10^{-8} \right) \left[f - \frac{f^2}{2 \times 10^8} \right]_{49 \times 10^6}^{51 \times 10^6} \\ &= 0.02. \end{aligned}$$

Thus the filter passes $100P_{N_o}/P_{N_i} = 2\%$ of the input noise power. The numerical powers are in watts if the stated PSD is in watts per hertz; otherwise they use the PSD's integrated power units.

Exercise 10.4. A nonideal bandpass filter with a ramp gain

IMG_8641.pdf, PDF page(s) 9.

Problem. Repeat the preceding triangular-noise example for the nonideal filter whose magnitude squared is

$$|H(f)|^2 = \begin{cases} |f| - 49 \times 10^6, & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise,} \end{cases}$$

with f expressed in hertz. Determine the output PSD, output noise power, and output/input power ratio.

Solution.

Multiply the input spectrum by the stated magnitude squared:

$$S_{N_o}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8} \right) (|f| - 49 \times 10^6), & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

To integrate, set $a = 49 \times 10^6$, $L = 2 \times 10^6$, and $u = f - a$. The positive-band integral is

$$\int_0^L \left(0.51 - \frac{u}{10^8} \right) u du = 0.51 \frac{L^2}{2} - \frac{L^3}{3 \times 10^8}.$$

Hence

$$\begin{aligned} P_{N_o} &= 2 \times 10^{-8} \left[0.51 \frac{(2 \times 10^6)^2}{2} - \frac{(2 \times 10^6)^3}{3 \times 10^8} \right] \\ &= \frac{59600}{3} \approx 1.98667 \times 10^4. \end{aligned}$$

Since $P_{N_i} = 1$, the power ratio is $59600/3$, and its percentage is approximately $1.98667 \times 10^6\%$. This large result follows directly from the specified filter: its peak $|H|^2$ is 2×10^6 . No unit-gain normalization is stated. If a normalized ramp were intended, the filter specification would have

to include a denominator; that would be a different problem.

Exercise 10.5. Rectangular bandpass noise and its quadrature components

IMG_8641.pdf, PDF page(s) 11.

Problem. White noise with two-sided PSD $N_0/2$ is passed through an ideal band-pass filter with magnitude K on each interval $|f - f_c| \leq B$ and $|f + f_c| \leq B$, and zero elsewhere. Assume $f_c > B$. Determine the output-noise autocorrelation.

Solution.

The output PSD is the sum of two shifted rectangles:

$$S_{N_o}(f) = \frac{K^2 N_0}{2} \left[\text{rect} \left(\frac{f - f_c}{2B} \right) + \text{rect} \left(\frac{f + f_c}{2B} \right) \right].$$

Each rectangle of width $2B$ transforms to $2B \text{sinc}(2B\tau)$. Frequency translation adds a complex sinusoidal factor. Thus

$$\begin{aligned} R_{N_o}(\tau) &= K^2 N_0 B \text{sinc}(2B\tau) (e^{j2\pi f_c \tau} + e^{-j2\pi f_c \tau}) \\ &= 2K^2 N_0 B \text{sinc}(2B\tau) \cos(2\pi f_c \tau). \end{aligned}$$

Here $\text{sinc}(x) = \sin(\pi x) / (\pi x)$. The output power is $2K^2 N_0 B$. The correlation oscillates at f_c inside a sinc envelope; the bandwidth of each passband is $2B$, which explains the factor of two relative to a low-pass filter with limits $\pm B$.

Exercise 10.6. Numerical specialization of the first-order filter

IMG_8641.pdf, PDF page(s) 14 (question), 13 (solution continuation).

Problem. White noise with $R_X(\tau) = \frac{N_0}{2} \delta(\tau)$ is applied to a low-pass filter with

$$H(\omega) = \frac{1}{1 + j\omega/10}.$$

Find the impulse response and the output-noise autocorrelation.

Solution.

Rewrite $H(\omega) = 10 / (10 + j\omega)$. Therefore,

$$h(t) = 10e^{-10t}u(t).$$

The output PSD in hertz is

$$S_Y(f) = \frac{N_0}{2} \frac{100}{100 + (2\pi f)^2}.$$

Using $e^{-a|\tau|} \leftrightarrow 2a / [a^2 + (2\pi f)^2]$ with $a = 10$,

$$R_Y(\tau) = \frac{5N_0}{2} e^{-10|\tau|}.$$

The output noise power is $R_Y(0) = 5N_0/2$. The parameter 10 has units of radians per second, so the 3-dB cutoff in hertz is $10 / (2\pi)$. This numerical example is the $\omega_0 = 10$ case of the first-order LPF exercise above.

Exercise 10.7. White noise and an exponential impulse response

University of Tripoli, EE416, Spring 2025 midterm, Q3.

Problem. A zero-mean white Gaussian input has $S_X(f) = \eta/2$ and drives an LTI filter with $h(t) = e^{-t}u(t)$. Find (a) $S_Y(f)$, (b) $R_Y(\tau)$, (c) $E[Y(t)^2]$, (d) input and output noise powers in a bandwidth of 10^3 Hz, and (e) the percentage of output noise power.

Solution.

1. Spectrum, autocorrelation and total output power

Using frequency in hertz,

$$H(f) = \int_0^\infty e^{-t} e^{-j2\pi ft} dt = \frac{1}{1 + j2\pi f},$$

$$S_Y(f) = \frac{\eta}{2 [1 + (2\pi f)^2]}.$$

The Fourier pair for an exponential gives

$$R_Y(\tau) = \frac{\eta}{4} e^{-|\tau|}, \quad E[Y(t)^2] = R_Y(0) = \frac{\eta}{4}.$$

The output remains zero-mean Gaussian. Ideal white input has infinite total power, but the filter output has finite power.

2. Powers within a stated frequency interval

Use the one-sided lowpass bandwidth convention $|f| \leq B$, with $B = 10^3$ Hz. Then

$$P_{X,B} = \int_{-B}^B \frac{\eta}{2} df = \eta B = 1000\eta,$$

$$P_{Y,B} = \frac{\eta}{2\pi} \tan^{-1}(2\pi B).$$

3. Two distinct percentages

The wording “percentage of output power” does not specify a denominator. The fraction of the total output contained in this band is

$$100 \frac{P_{Y,B}}{P_{Y,\text{total}}} = 100 \frac{2}{\pi} \tan^{-1} (2\pi B) \approx 99.9899\%.$$

The output-to-input power percentage within the same band is instead

$$100 \frac{P_{Y,B}}{P_{X,B}} = 100 \frac{\tan^{-1} (2\pi B)}{2\pi B} \approx 0.0249975\%.$$

These ratios answer different questions. If “bandwidth 10^3 Hz” means the full two-sided span, set $B = 500$ Hz in these formulas.

Lesson 11. Narrowband Noise, I/Q Representation and Envelope Statistics

Reference: Proakis and Salehi, Chapter 5, §5.3.3 and Problems 5.30 and 5.57–5.62; GJWHC narrowband examples; supplied Chapter 2 §§2.8–2.9.

11.1 Why a baseband description is useful

A real narrowband noise process has power concentrated in two separated spectral lobes near $+f_c$ and $-f_c$. Its waveform oscillates rapidly at the carrier frequency, while its amplitude and phase change more slowly. Describing those slower changes simplifies receiver analysis.

For the standard quadrature convention used in Chapter 2,

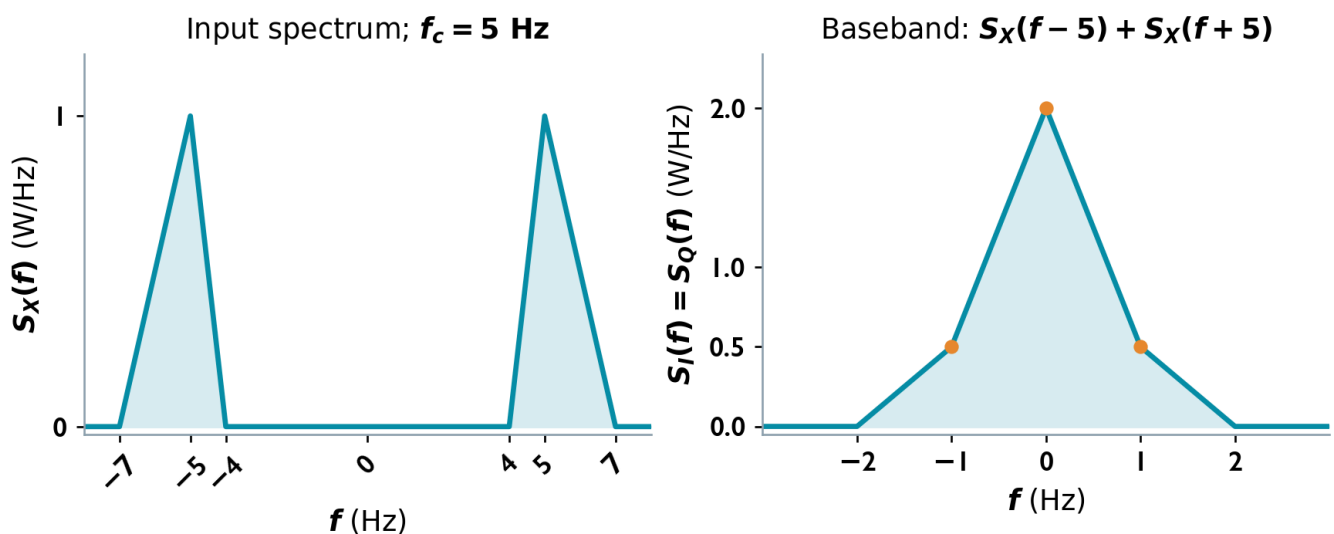
$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

The processes n_c and n_s are the in-phase and quadrature components. They can be obtained by coherent downconversion:

$$n_c(t) = \text{LPF} \{2n(t) \cos(2\pi f_c t)\}, \quad n_s(t) = \text{LPF} \{-2n(t) \sin(2\pi f_c t)\}.$$

The low-pass filters must preserve the shifted baseband support and reject the images near $\pm 2f_c$. This requires separated original lobes and a suitable carrier choice; the representation is not a claim about arbitrary wideband processes.

For zero-mean stationary Gaussian bandpass noise, these components are zero-mean, low-pass, jointly WSS and jointly Gaussian. Their spectral shapes depend on the chosen reference carrier even though reconstructing them returns the same physical bandpass noise.



Two bandpass lobes translate to baseband and combine to form the I/Q PSD.

11.2 Component spectra for a general bandpass shape

Let $S_n(f)$ be the even PSD of the real noise. Translation of the positive lobe by $-f_c$ and the negative lobe by $+f_c$ gives

$$S_{n_c}(f) = S_{n_s}(f) = S_n(f + f_c) + S_n(f - f_c),$$

within the retained low-pass support, and zero outside it. Both shifted terms must be included wherever their supports overlap. For example, an asymmetric triangular lobe produces a common component PSD with changes of slope; it need not produce a single triangle.

Using the explicit lag convention

$$R_{n_c n_s}(\tau) = E\{n_c(t + \tau) n_s(t)\},$$

the cross spectrum is

$$S_{n_c n_s}(f) = j[S_n(f + f_c) - S_n(f - f_c)], \quad S_{n_s n_c}(f) = S_{n_c n_s}^*(f).$$

For a real noise process, the bracket is real and odd. Consequently, the cross spectrum is imaginary and odd, and

$$C_{n_c n_s}(0) = \int S_{n_c n_s}(f) df = 0.$$

Integrating either component PSD gives

$$\sigma_{n_c}^2 = \sigma_{n_s}^2 = \int S_n(f) df = P_n.$$

These are the individual component powers, each equal to the bandpass-noise power. They are not half that power. Reconstruction multiplies each component by a sinusoid, and the resulting half-power factors make the total equal to P_n .

For the asymmetric GJWHC triangle in the figure, the positive lobe has support $[4, 7]$ Hz and height one at 5 Hz; the negative lobe is its reflection. With reference carrier $f_c = 5$ Hz, direct shifted-spectrum addition gives

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} 2 - \frac{3}{2}|f|, & |f| \leq 1, \\ 1 - \frac{1}{2}|f|, & 1 < |f| \leq 2, \\ 0, & |f| > 2. \end{cases}$$

The baseband peak is two, with changes of slope at $f = \pm 1$ Hz. Each original lobe has area $3/2$, so $P_n = 3$, and integrating either component PSD also gives three. For Gaussian noise the same-time component pair is therefore independent $\mathcal{N}(0, 3)$, with joint density $\exp[-(u^2 + v^2)/6] / (6\pi)$. The asymmetric shape can create unequal-time cross-correlation without changing this same-time independence.

11.3 Symmetric and asymmetric rectangular examples

Suppose the positive-frequency passband has edges $f_c - W$ and $f_c + W$, with a mirrored negative-frequency passband, and $S_n = N_0/2$ on both lobes. Each lobe has width $2W$, so

$$P_n = 2N_0W, \quad S_{n_c}(f) = S_{n_s}(f) = N_0, \quad |f| \leq W.$$

The equal shifted spectra make $S_{n_cn_s} = 0$ at every baseband frequency. The bandpass autocorrelation is

$$R_n(\tau) = 2N_0W \operatorname{sinc}(2W\tau) \cos(2\pi f_c \tau).$$

Now choose the asymmetric positive lobe $f_c \leq f \leq f_c + W$, with its negative-frequency mirror. Each lobe has width W , so $P_n = N_0W$. The translated supports meet at zero rather than fully overlap; the common component PSD is $N_0/2$ on $|f| \leq W$. The cross spectrum is generally nonzero, even though its oddness still makes the same-time cross covariance zero. Define the lobe width before computing power: a half-width W and a complete positive-frequency width W describe different filters.

11.4 Same-time independence and process independence

At any fixed time, $n_c(t)$ and $n_s(t)$ are jointly Gaussian, have equal variance $\sigma_n^2 = P_n$, and have zero covariance. Therefore they are independent random variables at that time. Spectral asymmetry about the chosen carrier does not invalidate this same-time result.

Independence of the two entire processes is stronger: it requires zero cross covariance at every lag. For jointly Gaussian components, symmetry of the original positive lobe about f_c makes the two translated spectra equal, sets the cross spectrum to zero everywhere, and gives process independence. Without that symmetry, component values at different times may be correlated. Merely observing zero covariance at a common instant cannot settle the all-lag question.

For real constants a, b , define $Z(t) = an_c(t) + bn_s(t)$. Its PSD is

$$S_Z(f) = a^2 S_{n_c}(f) + b^2 S_{n_s}(f) + 2ab \operatorname{Re} S_{n_cn_s}(f).$$

Since the cross spectrum is imaginary and the auto spectra are equal,

$$S_Z(f) = (a^2 + b^2) S_{n_c}(f).$$

This particular same-time linear combination has the stated PSD even for an asymmetric spectrum. Delayed combinations need their cross terms reevaluated because a delay introduces spectral phase. Taking $a = \cos \alpha$ and $b = -\sin \alpha$ preserves the common component PSD, representing a rotation of the quadrature axes.

11.5 Envelope and phase of Gaussian noise

Define the envelope and phase by

$$r(t) = \sqrt{n_c^2(t) + n_s^2(t)}, \quad \theta(t) = \text{atan2}(n_s(t), n_c(t)).$$

Then $n(t) = r(t) \cos[2\pi f_c t + \theta(t)]$. The two-argument arctangent identifies all quadrants. At zero envelope, phase can be assigned arbitrarily; that point has probability zero in the positive-variance continuous Gaussian model.

At a fixed time, let $U = n_c(t)$ and $V = n_s(t)$. Their joint density is

$$f_{U,V}(u, v) = \frac{1}{2\pi\sigma_n^2} \exp\left[-\frac{u^2 + v^2}{2\sigma_n^2}\right].$$

Use $u = r \cos \theta$, $v = r \sin \theta$, with $r \geq 0$ and $0 \leq \theta < 2\pi$. The inverse-transform Jacobian has magnitude r , giving

$$f_{r,\theta}(r, \theta) = \frac{r}{2\pi\sigma_n^2} \exp\left(-\frac{r^2}{2\sigma_n^2}\right).$$

This factors into a Rayleigh envelope density and a uniform phase density:

$$f_r(r) = \frac{r}{\sigma_n^2} e^{-r^2/(2\sigma_n^2)}, \quad r \geq 0, \quad f_\theta(\theta) = \frac{1}{2\pi}.$$

Thus envelope and phase are independent at a fixed time. This does not assert temporal independence of either process. Useful envelope results are

$$F_r(r) = 1 - e^{-r^2/(2\sigma_n^2)}, \quad r \geq 0, \quad E\{r\} = \sigma_n \sqrt{\frac{\pi}{2}},$$

$$E\{r^2\} = 2\sigma_n^2, \quad \text{Var}(r) = \left(2 - \frac{\pi}{2}\right) \sigma_n^2.$$

The mean-square envelope is twice the average real bandpass power. This is consistent with the sinusoidal reconstruction, not a contradiction of $\sigma_n^2 = P_n$.

11.6 A carrier plus noise: the Rician envelope

Adding a deterministic carrier $A \cos(2\pi f_c t)$ shifts the in-phase Gaussian component. The received envelope becomes

$$r_A(t) = \sqrt{[A + n_c(t)]^2 + n_s^2(t)}.$$

The Cartesian density is centered at $(A, 0)$ rather than at the origin. Polar transformation and integration over phase give the Rician density

$$f_{r_A}(r) = \frac{r}{\sigma_n^2} \exp\left[-\frac{r^2 + A^2}{2\sigma_n^2}\right] I_0\left(\frac{Ar}{\sigma_n^2}\right), \quad r \geq 0,$$

where $I_0(z) = (1/2\pi) \int_0^{2\pi} e^{z \cos \theta} d\theta$ is the modified Bessel function of order zero. The phase integral is exactly the step that produces I_0 . Setting $A = 0$ recovers Rayleigh noise. With a nonzero carrier, envelope and phase are generally not independent, and the phase is not uniform. The parameter $K = A^2 / (2\sigma_n^2)$ compares the carrier power $A^2/2$ with the bandpass-noise power. This connects the probability model to envelope-based receiver measurements.

11.7 Reference-carrier changes and bandpass pulse trains

Let the complex envelope at carrier f_c be $z_c(t) = n_c(t) + jn_s(t)$. Referencing the same real bandpass waveform to another suitable carrier f_r gives

$$z_r(t) = z_c(t) e^{j2\pi(f_c - f_r)t}.$$

This rotates the quadrature pair with time and changes its baseband spectral location. Its magnitude is unchanged, so the physical envelope $|z_r(t)| = |z_c(t)|$ is independent of the reference carrier. The phase changes by the corresponding linear-in-time term. The low-pass supports must still be retained correctly when implementing either representation.

For a bandpass pulse $p(t) = p_c(t) \cos(2\pi f_c t) - p_s(t) \sin(2\pi f_c t)$, define $p_b(t) = p_c(t) + jp_s(t)$. For real symbol coefficients A_n , the pulse train $X(t) = \sum_n A_n p(t - nT)$ has complex envelope

$$X_c(t) + jX_s(t) = \sum_n A_n p_b(t - nT) e^{-j2\pi f_c nT}.$$

Real and imaginary parts give its in-phase and quadrature components. The factor $e^{-j2\pi f_c nT}$ accounts for the carrier phase shift of each delayed pulse; omitting it would generally reconstruct a different bandpass signal. This representation connects pulse shaping to the I/Q process framework without assuming that the deterministic symbol clock already gives ordinary stationarity.

Solved Exercises for Lesson 11

Exercise 11.1. Envelope and phase of a bandpass Gaussian process

IMG_864I.pdf, PDF page(s) 6.

Problem. A sinusoidal carrier $x(t) = A \cos(2\pi f_c t)$ is added to narrowband noise

$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

Express $y(t) = x(t) + n(t)$ in envelope-phase form.

Solution.

Collect the in-phase and quadrature terms:

$$y(t) = [A + n_c(t)] \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

Write $y(t) = \rho(t) \cos[2\pi f_c t + \theta(t)]$. Comparing coefficients gives

$$\rho(t) \cos \theta(t) = A + n_c(t), \quad \rho(t) \sin \theta(t) = n_s(t).$$

Consequently,

$$\rho(t) = \sqrt{[A + n_c(t)]^2 + n_s^2(t)}, \quad \theta(t) = \text{atan2}(n_s(t), A + n_c(t)).$$

The two-argument arctangent preserves the correct quadrant. If the transmitted envelope is time varying, replace A by the specified envelope, such as $A_c[1 + m(t)]$. The phase is undefined when both Cartesian components vanish, but the waveform is then zero at that instant.

Exercise 11.2. Asymmetric triangular noise and I/Q spectra

IMG_864I.pdf, PDF page(s) 13–12 (question and continuation).

Problem. The PSD of narrowband noise $n(t)$ consists of two mirrored triangular lobes. The positive-frequency lobe is zero outside $4 \leq f \leq 7$ Hz, rises linearly from zero at 4 Hz to height 1 at 5 Hz, and decreases linearly to zero at 7 Hz. The negative-frequency lobe is its mirror. Use carrier frequency $f_c = 5$ Hz. Find (a) the PSDs of the in-phase and quadrature components and (b) their cross-spectral densities.

Solution.

Use

$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t),$$

and define $R_{ab}(\tau) = E[a(t + \tau)b(t)]$. Write the positive lobe as

$$T(f) = \begin{cases} f - 4, & 4 \leq f \leq 5, \\ (7 - f)/2, & 5 \leq f \leq 7, \\ 0, & \text{otherwise.} \end{cases}$$

The full PSD is $S_n(f) = T(f) + T(-f)$. Translation to baseband gives

$$S_{n_c}(f) = S_{n_s}(f) = S_n(f + 5) + S_n(f - 5).$$

For $r = |f|$, the common result is

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} 2 - \frac{3}{2}r, & 0 \leq r \leq 1, \\ 1 - \frac{1}{2}r, & 1 \leq r \leq 2, \\ 0, & r > 2. \end{cases}$$

The slopes change at $f = \pm 1$ Hz because the original lobes are asymmetric about the selected carrier. The cross spectra are

$$S_{n_c n_s}(f) = j[S_n(f + 5) - S_n(f - 5)], \quad S_{n_s n_c}(f) = S_{n_c n_s}^*(f).$$

Equivalently,

$$S_{n_c n_s}(f) = +j \operatorname{sgn}(f) q(|f|), \quad q(r) = \begin{cases} r/2, & 0 \leq r \leq 1, \\ 1 - r/2, & 1 \leq r \leq 2, \\ 0, & r > 2. \end{cases}$$

The cross spectrum is imaginary and odd, while each auto spectrum is real, even, and nonnegative. The in-phase and quadrature components have zero covariance at zero lag, because the integral of this odd cross spectrum is zero. They are not uncorrelated at every lag, because the cross spectrum is not identically zero. Each component has power 3, equal to the total passband-noise power. Reversing the cross-correlation argument order conjugates the cross spectra.

Exercise 11.3. A stepped bandpass filter and carrier selection

IMG_8641.pdf, PDF page(s) 12, 16, 15 (original lecture pages 34–36).

Problem. Zero-mean white noise with two-sided PSD $N_0/2$ is applied to a band-pass filter having

$$|H(f)|^2 = \begin{cases} 1/2, & 10 \leq |f|/10^3 < 12, \\ 1, & 12 \leq |f|/10^3 \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

Find the output noise PSD and total power. Then obtain the in-phase and quadrature PSDs and cross spectra for carrier choices $f_c = 10, 12, 14$ kHz.

Solution.

The output-noise PSD is

$$S_n(f) = \begin{cases} N_0/4, & 10 \leq |f|/10^3 < 12, \\ N_0/2, & 12 \leq |f|/10^3 \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

Integrate both mirrored lobes, converting the widths from kilohertz to hertz:

$$P_n = 2 \left[\frac{N_0}{4} (2000) + \frac{N_0}{2} (2000) \right] = 3000N_0.$$

For each carrier, set $r = |f|/10^3$ and use $S_{n_c} = S_{n_s} = S_n(f + f_c) + S_n(f - f_c)$. Carrier 10 kHz.

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} N_0/4, & 0 < r < 2, \\ N_0/2, & 2 < r < 4, \\ 0, & r > 4. \end{cases}$$

Carrier 12 kHz.

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} 3N_0/4, & r < 2, \\ 0, & r > 2. \end{cases}$$

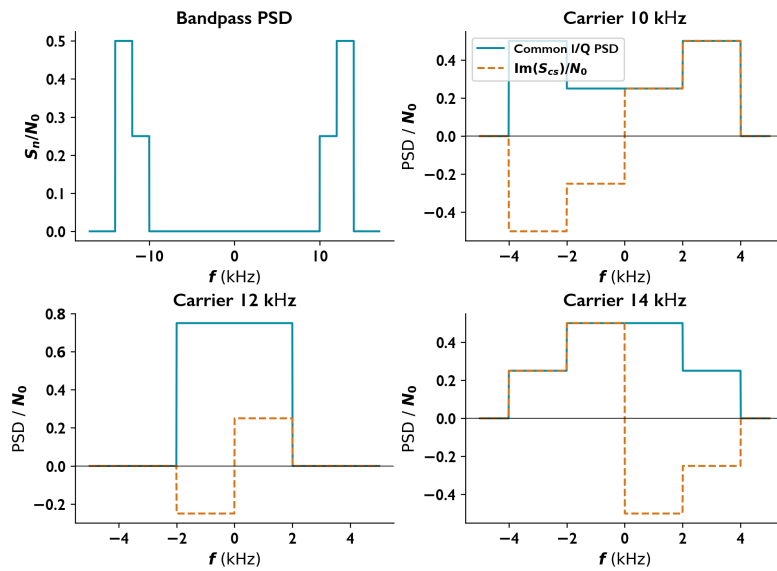
Carrier 14 kHz.

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} N_0/2, & 0 < r < 2, \\ N_0/4, & 2 < r < 4, \\ 0, & r > 4. \end{cases}$$

Values exactly at breakpoints do not affect any integrated power. Each common PSD integrates to $3000N_0$. Using $R_{ab}(\tau) = E[a(t + \tau)b(t)]$, the cross spectrum is

$$S_{n_cn_s}(f) = j[S_n(f + f_c) - S_n(f - f_c)], \quad S_{n_sn_c} = S_{n_cn_s}^*.$$

For 10 kHz, $S_{n_cn_s} = +j \operatorname{sgn}(f) S_{n_c}(f)$. For 12 kHz, $S_{n_cn_s} = +j \operatorname{sgn}(f) N_0/4$ on $0 < r < 2$, and zero elsewhere. For 14 kHz, $S_{n_cn_s} = -j \operatorname{sgn}(f) S_{n_c}(f)$. These signs follow the stated cross-correlation order. The factor 10^3 in the total power is essential because the graphs use kilohertz.



Frequency axes are in kHz. The dashed orange curve is the imaginary cross spectrum under the stated first-variable lag convention.

Lesson 12. Noise in Analog Modulation and Coherent Reception

Reference: Proakis and Salehi, Chapter 6, Sections 6.1.1–6.1.4 and 6.2; Chapter 5, Section 5.3.3.

12.1 Bandwidth and noise conventions

Noise performance follows from three separate operations: filtering the received spectrum, demodulating the retained signal, and integrating the output-noise spectrum. A formula is valid only when its bandwidth and noise-density conventions agree with those operations.

Let the real, zero-mean message $m(t)$ have power P_m and spectrum confined to $|f| \leq B$. Let additive white Gaussian noise have two-sided PSD $N_0/2$. An ideal baseband receiver passing $[-B, B]$ admits noise power

$$P_{n, \text{BB}} = \int_{-B}^B \frac{N_0}{2} df = N_0 B, \quad \text{SNR}_{\text{BB}} = \frac{P_R}{N_0 B}.$$

Here P_R is received signal power. For double-sideband reception, a real unity-gain BPF passes $[f_c - B, f_c + B]$ and its negative-frequency mirror. Each lobe has full width $2B$, and the total admitted noise power is $2N_0 B$. Thus a specification of positive-lobe width W means $B = W/2$. A message tone outside the corresponding baseband interval is rejected before demodulation; its power must not be included in the retained signal power.

Write the filtered noise as

$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

For the symmetric BPF just defined, $S_{n_c}(f) = S_{n_s}(f) = N_0$ on $[-B, B]$. Each component has variance $2N_0 B$. The factor of two follows from translating and adding the two passband-noise lobes; treating a component PSD as $N_0/2$ would underestimate coherent-detector noise.

12.2 Double-sideband suppressed-carrier reception

For $s(t) = A_c m(t) \cos(2\pi f_c t)$, with nonoverlapping shifted message spectra, the passband signal power is $P_R = A_c^2 P_m / 2$. Multiplication by $2 \cos(2\pi f_c t)$ followed by a unity-gain LPF passing $[-B, B]$ gives

$$y(t) = A_c m(t) + n_c(t).$$

The multiplier's doubled-frequency terms fall outside the LPF. Consequently,

$$\text{SNR}_{\text{out}} = \frac{A_c^2 P_m}{2N_0 B} = \frac{P_R}{N_0 B}, \quad \text{SNR}_{\text{in}} = \frac{P_R}{2N_0 B}.$$

The output is twice the actual BPF-input SNR, yet equals the baseband benchmark at the same P_R and B . These statements describe different denominators and are consistent. Using a local oscillator of amplitude one divides both output signal and noise powers by four and leaves SNR unchanged.

A phase error ϕ gives $y = A_c m \cos \phi + n_c \cos \phi + n_s \sin \phi$. Since symmetric Gaussian quadrature noise has equal powers and zero cross covariance, output noise power remains $2N_0 B$ while desired power is multiplied by $\cos^2 \phi$.

12.3 Single-sideband reception

An SSB waveform can be written

$$s(t) = A_c [m(t) \cos(2\pi f_c t) \mp \widehat{m}(t) \sin(2\pi f_c t)],$$

where $\widehat{m}$ is the Hilbert transform. With no DC spectral line, it has the same power as m . The selected positive-frequency sideband has width B , its negative mirror also has width B , and $P_R = A_c^2 P_m$. The BPF therefore admits noise power $N_0 B$. Its component PSD is $N_0/2$ on $[-B, B]$, so coherent reception yields

$$y = A_c m + n_c, \quad \text{SNR}_{\text{out}} = \frac{A_c^2 P_m}{N_0 B} = \frac{P_R}{N_0 B}.$$

SSB and DSB-SC have the same output SNR at equal received power and message bandwidth, while SSB uses half the positive-frequency transmission bandwidth. The quadrature-noise processes for the asymmetric SSB filter need not be independent at every lag.

12.4 Conventional AM and envelope detection

For a normalized zero-mean message and modulation index a , write

$s = A_c [1 + am] \cos(2\pi f_c t)$. Coherent demodulation followed by DC removal gives $y = A_c am + n_c$. Hence

$$P_R = \frac{A_c^2}{2} (1 + a^2 P_m), \quad \eta = \frac{a^2 P_m}{1 + a^2 P_m}, \quad \text{SNR}_{\text{out}} = \eta \frac{P_R}{N_0 B}.$$

The carrier contributes received power but no recovered message power. For

$m = \cos(2\pi f_m t)$, $P_m = 1/2$ and $\eta = a^2 / (2 + a^2)$; at $a = 1$, efficiency is $1/3$.

An envelope detector observes $\sqrt{[A_c (1 + am) + n_c]^2 + n_s^2}$. When the instantaneous signal envelope is positive and dominates noise, expansion gives approximately $A_c (1 + am) + n_c$, leading to the same high-SNR result after DC removal. Near an envelope zero, or at low received SNR, this approximation fails and signal and noise interact nonlinearly.

12.5 FM noise and the threshold limitation

For $s = A_c \cos \left[2\pi f_c t + 2\pi k_f \int^t m(u) du \right]$, k_f is frequency sensitivity in hertz per message unit. At high carrier-to-noise ratio, the small phase perturbation is approximately $-n_s/A_c$. An ideal normalized discriminator differentiates that phase, giving

$$S_{n,\text{out}}(f) = \frac{N_0 f^2}{A_c^2 k_f^2}, \quad |f| \leq B; \quad P_{n,\text{out}} = \frac{2N_0 B^3}{3A_c^2 k_f^2}.$$

The rising f^2 spectrum explains high-frequency noise enhancement and the use of preemphasis and deemphasis. Increasing deviation improves the high-SNR output while increasing transmission bandwidth. Carson's estimate is $W_{\text{FM}} \approx 2(\Delta f + B)$ for each positive-frequency passband, where $\Delta f = k_f \max |m|$.

The approximation also requires adequate predetection carrier-to-noise ratio over that larger band. Below the receiver's threshold, large phase excursions produce impulsive errors and the high-SNR equation ceases to predict performance. A stated threshold condition must therefore be checked alongside the channel-bandwidth constraint; its numerical value is a receiver assumption, not a universal constant.

Solved Exercises for Lesson 12

Exercise 12.1. AM signal and noise through a nonuniform bandpass filter

IMG_8587.pdf, PDF page(s) 6.

Problem. A carrier of amplitude $A_c = 10$ mV and frequency $f_c = 90$ kHz is 50% amplitude modulated by a sinusoid of frequency $f_m = 750$ Hz. The two-sided thermal-noise PSD is $\eta/2 = 10^{-3}$ W/Hz. The magnitude response is even and, on its positive-frequency band, is

$$|H(f)| = \begin{cases} 1, & 89.5 \leq f \leq 90.5 \text{ kHz}, \\ 0.5, & 89 \leq f < 89.5 \text{ kHz or } 90.5 < f \leq 91 \text{ kHz}, \\ 0, & \text{otherwise.} \end{cases}$$

The filtered signal is coherently multiplied by $2 \cos(2\pi f_c t)$, then passed through an ideal LPF of bandwidth 1 kHz.

- Determine SNR at the demodulator input.
- Determine SNR at its output.

Solution.

1. Resolve the AM signal into spectral lines. With $\mu = 0.5$,

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} [\cos(2\pi(f_c + f_m)t) + \cos(2\pi(f_c - f_m)t)].$$

The carrier has filter gain 1. The sidebands at 89.25 and 90.75 kHz each have gain 0.5. Using a 1-ohm power normalization,

$$P_{s,i} = \frac{A_c^2}{2} + 2 \left[\frac{1}{2} \left(\frac{A_c \mu}{4} \right)^2 \right] = 5.15625 \times 10^{-5} \text{ W}.$$

2. Integrate the filtered two-sided noise PSD. The noise-equivalent integral is

$$\int_{-\infty}^{\infty} |H(f)|^2 df = 2 [1000 + 2(500)(0.5)^2] = 2500 \text{ Hz}.$$

Thus $P_{n,i} = 10^{-3} (2500) = 2.5 \text{ W}$, and

$$\text{SNR}_i = \frac{5.15625 \times 10^{-5}}{2.5} = 2.0625 \times 10^{-5} \approx -46.86 \text{ dB}.$$

3. Calculate the recovered message. Coherent mixing and lowpass filtering give

$$s_o(t) = A_c + \frac{A_c \mu}{2} \cos(2\pi f_m t).$$

The recovered-message amplitude is $A_c \mu / 2 = 2.5 \text{ mV}$, giving

$$P_{m,o} = \frac{1}{2} \left(\frac{A_c \mu}{2} \right)^2 = 3.125 \times 10^{-6} \text{ W}.$$

The DC term comes from the carrier and is normally removed when message SNR is specified.

4. Translate both passbands into baseband. For the multiplier amplitude 2,

$$S_{n,o}(f) = \frac{\eta}{2} \left(|H(f - f_c)|^2 + |H(f + f_c)|^2 \right), \quad |f| \leq 1000.$$

It equals 2×10^{-3} for $|f| \leq 500$, and 0.5×10^{-3} for $500 < |f| \leq 1000$. Therefore

$$P_{n,o} = (2 \times 10^{-3})(1000) + (0.5 \times 10^{-3})(1000) = 2.5 \text{ W}.$$

The recovered-message SNR is

$$\text{SNR}_o = \frac{3.125 \times 10^{-6}}{2.5} = 1.25 \times 10^{-6} \approx -59.03 \text{ dB}.$$

If the DC carrier component is explicitly counted as signal power, the output ratio instead becomes $(A_c^2 + P_{m,o}) / P_{n,o} = 4.125 \times 10^{-5}$. That ratio is not the usual recovered-message SNR.

Exercise 12.2. FM power, Carson bandwidth, and threshold limitation

IMG_8587.pdf, PDF page(s) 6.

Problem. An audio signal has bandwidth $W = 15$ kHz, average power $P_m = 1/6$, and peak magnitude 1. It is transmitted using FM through a channel with 60 dB power loss and two-sided AWGN PSD $\eta/2 = 10^{-12}$ W/Hz. The destination SNR must be at least 40 dB.

- a. Find the required bandwidth and average transmitted power for deviation ratio $\beta = 5$. (b) If threshold occurs when carrier power is 10 times the channel noise power, determine the largest usable deviation ratio and the corresponding bandwidth.

Solution.

1. Use the above-threshold FM SNR relation. With $N_0 = \eta = 2 \times 10^{-12}$ and $|m(t)|_{\max} = 1$, the peak frequency deviation is $\Delta f = \beta W$. Ideal FM demodulation without pre-emphasis/de-emphasis gives

$$\text{SNR}_o = \frac{3(\Delta f)^2 P_m P_r}{N_0 W^3} = \frac{3\beta^2 P_m P_r}{N_0 W},$$

where P_r is received carrier power.

2. Solve part (a). Carson's rule gives

$$B_T \simeq 2(\beta + 1)W = 180 \text{ kHz}.$$

For $\text{SNR}_o = 10^4$,

$$P_r = \frac{10^4 N_0 W}{3\beta^2 P_m} = \frac{10^4 (2 \times 10^{-12}) (15000)}{3(25)(1/6)} = 2.4 \times 10^{-5} \text{ W}.$$

The 60 dB power loss is a factor 10^6 , so

$$P_T = 10^6 P_r = 24 \text{ W}.$$

The predetection channel noise power is $N_0 B_T = 3.6 \times 10^{-7}$ W; the carrier-to-noise ratio is 66.67, comfortably above the threshold value 10.

3. Solve part (b) at the transmitted power obtained in part (a). Keeping $P_r = 2.4 \times 10^{-5}$ W, the threshold condition requires

$$\frac{P_r}{N_0 B_T} \geq 10 \quad \Rightarrow \quad B_T \leq \frac{P_r}{10 N_0} = 1.2 \times 10^6 \text{ Hz}.$$

Hence, using Carson's rule,

$$\beta_{\max} = \frac{1.2 \times 10^6}{2(15000)} - 1 = 39, \quad B_{T, \max} = 1.2 \text{ MHz}.$$

Equality is the threshold boundary; a practical design would retain margin.

Interpretation of the power assumption. If part (b) instead intends to reduce transmitter power anew for every β while meeting exactly 40 dB output SNR, the two conditions become

$$P_r = \frac{6 \times 10^{-4}}{\beta^2}, \quad P_r \geq 6 \times 10^{-7} (\beta + 1).$$

Their boundary is $\beta^2 (\beta + 1) = 1000$, whose positive solution is $\beta = 9.67753$, with $B_T = 320.326$ kHz. This is a different optimization problem. The primary result above retains the part-(a) transmitter power, as normally understood in a two-part design question.

Exercise 12.3. Minimum FM transmitter power under a channel bandwidth limit

IMG_8587.pdf, PDF page(s) 7.

Problem. A normalized message has bandwidth $W = 5$ kHz and average power $P_m = 0.5$. The channel has available bandwidth 60 kHz, 40 dB power attenuation, and two-sided white-noise PSD 10^{-9} W/Hz. FM without pre-emphasis/de-emphasis is proposed. Find the minimum transmitter power and corresponding modulation index for at least 40 dB output SNR.

Solution.

1. Use the bandwidth allowance to maximize deviation. For a normalized peak message of 1, the deviation ratio is $\beta = \Delta f / W$. Carson's rule gives

$$2 (\beta + 1) W \leq 60000 \Rightarrow \beta \leq 5.$$

The required received power decreases as $1/\beta^2$, so the minimum-power choice is $\beta = 5$, with $\Delta f = 25$ kHz.

2. Calculate the minimum received and transmitted powers. Since the two-sided PSD is $N_0/2 = 10^{-9}$, $N_0 = 2 \times 10^{-9}$. The above-threshold FM result is

$$\text{SNR}_o = \frac{3\beta^2 P_m P_r}{N_0 W}.$$

Setting $\text{SNR}_o = 10^4$ yields

$$P_{r, \min} = \frac{10^4 (2 \times 10^{-9}) (5000)}{3 (25) (0.5)} = 2.66667 \times 10^{-3} \text{ W}.$$

The 40 dB power loss is 10^4 , hence

$$P_{T, \min} = 26.667 \text{ W}, \quad \beta = 5.$$

3. Check consistency with FM threshold. The predetection noise power over the 60 kHz channel is $N_0 B = 1.2 \times 10^{-4}$ W, giving $P_r / (N_0 B) = 22.22$. This exceeds a commonly used 10-to-1 threshold criterion, so the assumed above-threshold formula is consistent under that criterion.

Exercise 12.4. DSB-SC sideband rejection and coherent SNR

University of Tripoli, EE416, Spring 2025 midterm, Q4.

Problem. The modulated signal is

$$s(t) = 10 \left[3 \cos(2\pi 10^2 t) + 2 \cos(2\pi 10^4 t) \right] \cos(\pi 10^8 t).$$

White noise has two-sided PSD $N_0/2 = 2 \times 10^{-6}$ W/Hz. An ideal bandpass filter is centered at the carrier and has bandwidth 10 kHz. Determine (a) the modulation type; (b) SNR at the filter output; (c) the PSD and average power of the in-phase noise $n_c(t)$; and (d) SNR after coherent detection.

Solution.

1. Modulation and passed signal

This is double-sideband suppressed-carrier modulation, with $f_c = 5 \times 10^7$ Hz. Define the stated BPF bandwidth as the full width of its positive-frequency passband. Its edges are $f_c \pm B$, where $B = 5$ kHz, with a reflected negative-frequency passband. The 100 Hz message tone passes, whereas the 10 kHz tone is rejected. Thus the filtered signal is

$s_o(t) = 30 \cos(2\pi 100t) \cos(2\pi f_c t)$. Its average power on a unit resistance is

$$P_s = 30^2 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 225 \text{ W}.$$

2. Predetection noise and SNR

Both passbands contribute noise:

$$P_n = \frac{N_0}{2} (4B) = 2N_0 B = 0.04 \text{ W}.$$

Therefore

$$\text{SNR}_{\text{BPF}} = \frac{225}{0.04} = 5625 \approx 37.50 \text{ dB}.$$

3. In-phase component

For $n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t)$,

$$S_{n_c}(f) = \begin{cases} N_0 = 4 \times 10^{-6}, & |f| \leq 5000 \text{ Hz}, \\ 0, & \text{otherwise,} \end{cases}$$

$$E[n_c(t)^2] = \int S_{n_c}(f) df = 0.04 \text{ W}.$$

The quadrature component has the same PSD and power.

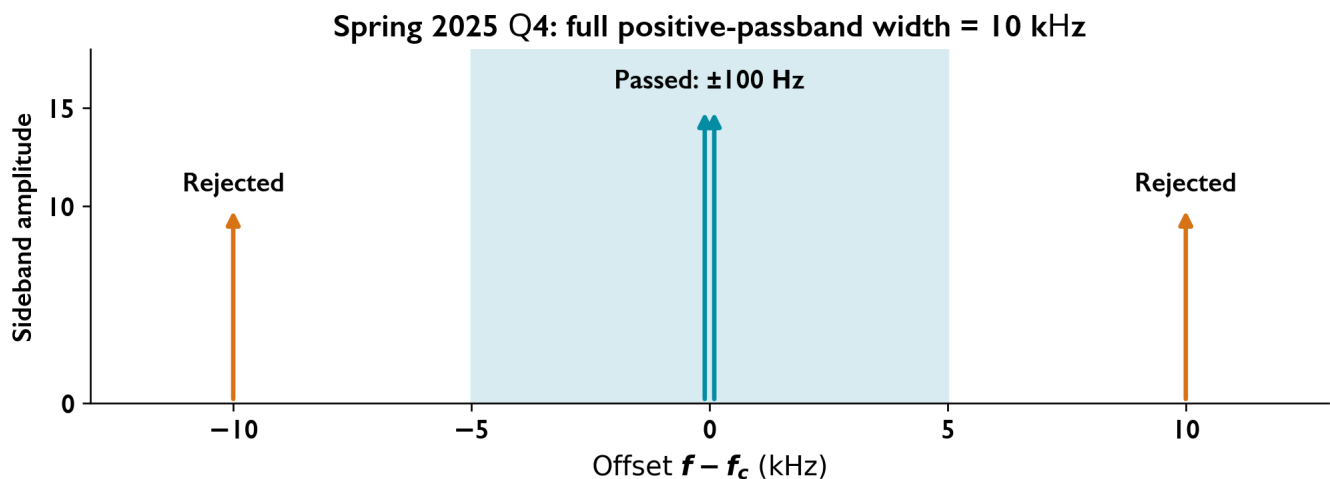
4. Coherent detector

Multiply by $2 \cos(2\pi f_c t)$ and lowpass-filter. With a detector lowpass edge $B = 5 \text{ kHz}$, the output is $30 \cos(2\pi 100t) + n_c(t)$, with signal power 450 W . Hence

$$\text{SNR}_{\text{det}} = \frac{450}{0.04} = 11250 \approx 40.51 \text{ dB}.$$

The 3 dB increase follows from retaining the in-phase noise component. Multiplication by $\cos$ rather than $2 \cos$ rescales signal and noise equally and leaves SNR unchanged.

The detector LPF bandwidth was not specified in the photograph. For a different edge B_d with $100 \leq B_d \leq 5000 \text{ Hz}$, the output noise power is $2N_0 B_d$, and $\text{SNR}_{\text{det}} = 450 / (2N_0 B_d)$. A narrower LPF therefore gives a different SNR. If a course instead uses "10 kHz bandwidth" to mean the half-width B , both message tones pass and all power calculations must be recomputed with $B = 10 \text{ kHz}$.



Positive-frequency sidebands relative to the carrier. The reflected negative-frequency band also contributes noise power.

Lesson 13. Sampling, Quantization, and Linear Prediction

Reference: Proakis and Salehi, Chapter 5, Sections 5.2–5.3; Chapter 7, Sections 7.1, 7.2, and 7.5.2; Chapter 9, pulse-shaping and bandwidth conventions.

13.1 What sampling specifies

Sampling a random process produces a sequence of random variables, rather than predetermined numerical amplitudes. For sampling interval $T_s = 1/f_s$, define $X[n] = X(nT_s)$. A specified mean and PSD determine statistical information about this sequence. Actual values require a realization, measured data, or a simulation rule.

For a real WSS process,

$$\mu_{X[n]} = \mu_X, \quad R_d[k] = R_X(kT_s), \quad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df.$$

If the process is Gaussian, every sampled vector is jointly Gaussian. For zero mean, the vector $(X[1], \dots, X[M])^T$ has covariance entries $K_{ij} = R_X((i-j)T_s)$. These entries determine its joint distribution, including possible degeneracy; they do not select one sample vector from that distribution.

For example, the spectral family

$$S_X(f) = \frac{2\sigma_X^2 a}{a^2 + (2\pi f)^2}, \quad a > 0,$$

has $R_X(\tau) = \sigma_X^2 e^{-a|\tau|}$. Its sampled covariance is $R_d[k] = \sigma_X^2 \rho^{|k|}$ with $\rho = e^{-aT_s}$. Nearby samples are correlated, and Gaussianity makes their dependence completely describable through this covariance matrix.

13.2 Sampling bandwidth and aliasing

For a signal confined to $|f| \leq B$, ideal reconstruction requires $f_s \geq 2B$ under the standard ideal bandlimited model; practical designs include a transition band and sample faster. The sampling theorem reconstructs a realization from its samples, whereas the PSD describes ensemble power.

Using dimensionless digital frequency Ω , the sampled sequence has periodic PSD

$$S_d(e^{j\Omega}) = \frac{1}{T_s} \sum_{\ell=-\infty}^{\infty} S_X\left(\frac{\Omega + 2\pi\ell}{2\pi T_s}\right), \quad R_d[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_d(e^{j\Omega}) d\Omega.$$

Overlapping replicas produce aliasing. An exponential-correlation process has an unbounded spectral tail, so no finite sampling rate makes it exactly bandlimited; sampling its statistics remains well defined.

For flat Gaussian noise on $[-B, B]$, $R_X(\tau) = N_0 B \text{sinc}(2B\tau)$. Samples spaced $1/(2B)$ apart have zero nonzero-lag covariance and are independent because they are jointly Gaussian. Oversampling generally introduces correlation. Equal sampling rate alone does not establish independence for another spectral shape.

13.3 Exact and approximate quantization error

A quantizer assigns reproduction value q_i to decision region $\mathcal{D}_i$. For an input density f_X , its output probabilities, output power, and mean-square error are

$$P_i = \int_{\mathcal{D}_i} f_X(x) dx, \quad P_Q = \sum_i q_i^2 P_i, \quad D = \sum_i \int_{\mathcal{D}_i} (x - q_i)^2 f_X(x) dx.$$

These integrals remain the appropriate method for a few-level quantizer, a nonuniform input law, or saturation. Output power $E[Q(X)^2]$ and input power $E[X^2]$ are different quantities, so the SNR numerator must be identified explicitly.

For a sufficiently fine uniform quantizer with step Δ , negligible overload, and a smooth input density over each cell, the error is approximately uniform on $[-\Delta/2, \Delta/2]$. Therefore,

$$E[e_q] \approx 0, \quad D \approx \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} u^2 du = \frac{\Delta^2}{12}.$$

With Q levels spanning $[-U, U]$, $\Delta = 2U/Q$ and $D \approx U^2/(3Q^2)$. Independence and whiteness of quantization error are modeling approximations, not consequences of a deterministic staircase characteristic. A Gaussian input has unbounded support, so finite-range overload cannot literally vanish.

If an exercise instead supplies $D = C\sigma_v^2 Q^{-2}$ for quantizing a residual v , use that stated model and its constant C . Substituting $\Delta^2/12$ without establishing the matching range changes the problem. To enforce $D \leq D_{\max}$, compute

$$Q_{\min} = \left\lceil \sqrt{\frac{C\sigma_v^2}{D_{\max}}} \right\rceil.$$

Retain full precision until this ceiling and verify the resulting inequality. A value very close to an integer can cross the limit after premature rounding. For fixed-length binary words, $b = \lceil \log_2 Q_{\min} \rceil$ and a full b -bit quantizer commonly uses 2^b levels.

13.4 Linear prediction and differential coding

Center the sampled process if its mean is nonzero. A p th-order predictor uses $\widehat{X}[n] = \sum_{i=1}^p a_i X[n-i]$, with error $e[n] = X[n] - \widehat{X}[n]$. Minimizing $E[e^2]$ and differentiating with respect to every a_i gives the orthogonality conditions

$$E[e[n]X[n-j]] = 0, \quad \sum_{i=1}^p a_i R_d[j-i] = R_d[j], \quad j = 1, \dots, p.$$

Equivalently, $\mathbf{R}\mathbf{a} = \mathbf{r}$, where $R_{ji} = R_d[j-i]$ and $r_j = R_d[j]$. This symmetric Toeplitz system is the Yule-Walker predictor equation. When $\mathbf{R}$ is nonsingular,

$$\mathbf{a} = \mathbf{R}^{-1}\mathbf{r}, \quad \sigma_e^2 = R_d[0] - \mathbf{r}^T \mathbf{a}.$$

The full residual autocorrelation follows from $A(z) = 1 - \sum_i a_i z^{-i}$:

$S_e(e^{j\Omega}) = |A(e^{j\Omega})|^2 S_d(e^{j\Omega})$. For exponential covariance, the optimum coefficients are $a_1 = \rho$ and $a_i = 0$ for $i > 1$, giving $\sigma_e^2 = \sigma_X^2(1 - \rho^2)$. Its residual is white; for a Gaussian process it is an independent innovation sequence. General finite-order residuals need not be white. Differential coding quantizes this residual, reducing required levels when prediction reduces its variance.

13.5 From quantized samples to channel bandwidth

Uncompressed fixed-length PCM requires bit rate $R_b = b f_s$. For M -ary signaling, $R_s = R_b / \log_2 M$. Ideal Nyquist shaping has one-sided baseband bandwidth $R_s/2$; raised-cosine shaping with roll-off α has $(1 + \alpha) R_s/2$. After real carrier modulation, the positive-frequency RF passband has full width $(1 + \alpha) R_s$. Rectangular symbol pulses instead have first-null RF width $2R_s$. Thus channel-fit calculations must specify pulse shape and distinguish a baseband edge frequency from a complete RF-lobe width.

Solved Exercises for Lesson 13

Exercise 13.1. Prediction of a sampled Gaussian process and differential quantization

IMG_8587.pdf, PDF page(s) 1.

Problem. A zero-mean Gaussian process has power spectral density

$$S_X(f) = \frac{16}{1 + (16\pi f)^2}.$$

It is sampled at $f_s = 1$ Hz. A differential quantizer subtracts a three-tap linear predictor $a_1 X[n-1] + a_2 X[n-2] + a_3 X[n-3]$ from $X[n]$. The quantization noise power for a normal random variable of variance σ^2 , quantized using Q levels, is $2\sigma^2/Q^2$.

- a. Determine the number of levels needed to keep the quantization noise power at or below 4.32×10^{-4} . (b) Determine the values of the sampled process at $T_s, 2T_s, 3T_s, 4T_s$.

Solution.

1. Convert the PSD to autocorrelation. With the Fourier-transform convention $e^{-j2\pi f\tau}$,

$$R_X(\tau) = e^{-|\tau|/8}, \quad R_X(0) = 1.$$

At $T_s = 1$ s, define $\rho = e^{-1/8} = 0.882496903$. Then

$$r_X[m] = R_X(mT_s) = \rho^{|m|}.$$

2. Determine the minimum mean-square predictor. The orthogonality equations are

$$\begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} \rho \\ \rho^2 \\ \rho^3 \end{bmatrix}.$$

Their solution is $a_1 = \rho, a_2 = a_3 = 0$. Thus the Gaussian sampled process has a first-order innovation representation,

$$X[n] = \rho X[n-1] + D[n], \quad \sigma_D^2 = 1 - \rho^2 = 1 - e^{-1/4} = 0.221199217.$$

The innovations are independent zero-mean Gaussian variables with this variance.

3. Apply the stated quantization model to the prediction error.

$$\frac{2\sigma_D^2}{Q^2} \leq 4.32 \times 10^{-4} \Rightarrow Q \geq \sqrt{\frac{2(1 - e^{-1/4})}{4.32 \times 10^{-4}}} = 32.0011007.$$

The exact smallest integer satisfying the printed inequality is therefore $Q = 33$. If the number of levels must be a power of two, use 64 levels = 6 bits.

The common rounded calculation gives 32 levels. However, using the exact exponential, 32 levels yield 4.3202972×10^{-4} , slightly above the printed bound. This distinction is due to rounding, not a change to the noise model.

4. Interpret the request for sample values. A PSD specifies statistics, not a particular realization. Consequently the four numerical sample amplitudes cannot be determined from the supplied information. What can be determined is

$$\mathbb{E}[X[1]] = \dots = \mathbb{E}[X[4]] = 0, \quad \text{Var}(X[i]) = 1, \quad \text{Cov}(X[i], X[j]) = \rho^{|i-j|}.$$

Their joint distribution is Gaussian with covariance matrix $\left[\rho^{|i-j|} \right]_{i,j=1}^4$. An explicit realization would require an initial sample and particular innovation values:

$$X[n] = \rho^n X[0] + \sum_{k=1}^n \rho^{n-k} D[k], \quad n = 1, 2, 3, 4.$$

If the intended request was for autocorrelation samples, these are

$$R_X(T_s) = 0.882497, \quad R_X(2T_s) = 0.778801, \quad R_X(3T_s) = 0.687289, \quad R_X(4T_s) = 0.606531.$$

Exercise 13.2. Exact quantization powers for uniform and parabolic inputs

IMG_8587.pdf, PDF page(s) 7.

Problem. A four-level midrise quantizer has thresholds $-2, 0, 2$, reconstruction levels $-3, -1, 1, 3$, and operating input interval $[-4, 4]$:

$$q(x) = \begin{cases} -3, & -4 \leq x < -2, \\ -1, & -2 \leq x < 0, \\ 1, & 0 \leq x < 2, \\ 3, & 2 \leq x \leq 4. \end{cases}$$

(a) For an input uniform on the operating interval, calculate quantization-noise power and signal-to-quantization-noise ratio. (b) The scanned question gives a parabolic expression $f_X(x) = A(1 - x^2/c)$ on $|x| \leq 4$, and zero otherwise. The denominator c is faint and appears to be 10. Determine the normalization and requested powers only when this expression is a valid density. The solution analyzes the apparent reading $c = 10$, gives formulas for any valid positive denominator, and then provides the $c = 16$ calculation as an explicitly conditional interpretation.

Solution.

1. Uniform input. The input density is $1/8$, and the quantization step is $\Delta = 2$. With no overload,

$$P_q = E\{(X - q(X))^2\} = \frac{\Delta^2}{12} = \frac{1}{3}.$$

The input signal power is $E\{X^2\} = 4^2/3 = 16/3$. Therefore

$$\text{SQNR} = \frac{16/3}{1/3} = 16 \approx 12.04 \text{ dB}.$$

2. Check the scanned density before normalizing. If the denominator is 10, $1 - x^2/10 < 0$ on $\sqrt{10} < |x| \leq 4$, while it is positive near zero. No nonzero choice of A makes this sign-changing function a probability density. Thus the apparent printed version has no probabilistic solution

for part (b). Replacing 10 by 16 is a change to the model and must be identified explicitly.

For a general positive denominator c , validity on the complete stated support requires $c \geq 16$. Normalization then gives

$$1 = A \int_{-4}^4 \left(1 - \frac{x^2}{c}\right) dx = A \left(8 - \frac{128}{3c}\right), \quad A = \frac{3c}{8(3c - 16)}.$$

3. Exact quantization-noise power for the valid model. Symmetry gives

$$P_q = 2A \left[\int_0^2 (x-1)^2 \left(1 - \frac{x^2}{c}\right) dx + \int_2^4 (x-3)^2 \left(1 - \frac{x^2}{c}\right) dx \right] = \frac{5c - 28}{5(3c - 16)}.$$

If the intended denominator was $c = 16$, these formulas become

$$A = \frac{3}{16}, \quad P_q = \frac{13}{40} = 0.325.$$

The approximation $\Delta^2/12$ is not exact for this nonuniform density.

4. Quantized-signal power. For the valid general model,

$$P_1 = \Pr(0 < X < 2) = \frac{3c - 4}{4(3c - 16)}, \quad P_2 = \Pr(2 < X < 4) = \frac{3c - 28}{4(3c - 16)}.$$

The negative bins have the same probabilities, so

$$E\{q(X)^2\} = 2(P_1 + 9P_2) = \frac{15c - 128}{3c - 16}.$$

For the conditional interpretation $c = 16$, $P_1 = 11/32$, $P_2 = 5/32$, and

$$E\{q(X)^2\} = \frac{7}{2} = 3.5.$$

The corresponding unquantized input power is $E\{X^2\} = 16/5 = 3.2$. Output power need not equal input power minus quantization-noise power, because the exact quantization error can be correlated with the input.

The results in terms of c remain usable if a clearer source establishes a different valid denominator. The $c = 16$ numbers are conditional results, not a claim that the scan certainly specifies 16.

Lesson 14. Matched Filters and Binary Detection

Reference: Proakis and Salehi, Chapter 5, Gaussian filtering; Chapter 9, optimum reception in additive white Gaussian noise and binary signaling.

14.1 The detection objective

A receiver makes a discrete decision from a noisy continuous-time observation. The matched filter first extracts a scalar statistic with maximum signal-to-noise ratio at a chosen sampling instant. The decision rule then compares the statistic with a threshold determined by the signal alternatives, noise law, and prior probabilities. Filtering and threshold selection solve distinct parts of the problem.

Consider a known finite-energy real pulse $s(t)$ observed in additive zero-mean white Gaussian noise with two-sided PSD $N_0/2$. Define its energy as $E_s = \int_{-\infty}^{\infty} |s(t)|^2 dt$. For a real impulse response $h(t)$, the signal contribution at sampling time T is

$$y_s(T) = \int_{-\infty}^{\infty} s(u) h(T-u) du.$$

The output noise is Gaussian because linear filtering preserves Gaussianity. Its variance is

$$\sigma_n^2 = \frac{N_0}{2} \int_{-\infty}^{\infty} |h(u)|^2 du.$$

This expression requires a square-integrable filter. Ideal unfiltered continuous-time white noise has no finite pointwise variance, but the filtered statistic does.

14.2 Derivation of the matched filter

Cauchy–Schwarz gives

$$|y_s(T)|^2 \leq E_s \int_{-\infty}^{\infty} |h(u)|^2 du.$$

Dividing by the noise variance proves $\text{SNR}(T) \leq 2E_s/N_0$. Equality holds when the filter is proportional to the conjugated, time-reversed signal:

$$h_{\text{MF}}(t) = k s^*(T-t), \quad \text{SNR}_{\text{max}} = \frac{2E_s}{N_0}.$$

For a real pulse, conjugation changes nothing. The constant $k \neq 0$ changes output amplitude and noise variance together, leaving SNR unchanged. If a problem supplies the two-sided PSD height directly as D_n , then $D_n = N_0/2$ and the same result is E_s/D_n ; doubling an already two-sided density would introduce an error.

When s is supported on $[0, T]$, the matched filter is also supported on $[0, T]$ and is causal. A different sampling time can be implemented by adding delay. For complex signals, the same impulse-response form holds, but real and complex noise-density conventions must be kept distinct.

Define the pulse energy autocorrelation by $R_s(\tau) = \int s(u + \tau) s^*(u) du$. The noiseless output at every time is

$$y_s(t) = kR_s(t - T), \quad y_s(T) = kE_s.$$

Another Cauchy–Schwarz application yields $|R_s(\tau)| \leq E_s$, establishing the maximum output magnitude at $t = T$. For piecewise pulses, derive each convolution segment by locating the overlap between $s(u)$ and $h(t - u)$, splitting at every breakpoint, and integrating only the common support. Plotting h requires reversing the pulse first and then shifting it; merely shifting the original pulse is generally incorrect.

14.3 Correlators and sufficient statistics

At the sampling instant, matched filtering is equivalent to correlating the received waveform with $s(t)$:

$$y(T) = k \int r(u) s^*(u) du.$$

For multiple candidate waveforms, a bank of correlators or matched filters produces projections on their signal space. In white Gaussian noise, information orthogonal to that space does not distinguish the candidates. A channel filter must first be incorporated into the received candidate waveforms; matching the transmitted pulse while ignoring channel distortion can be suboptimal. In colored noise, the optimum filter additionally weights frequencies inversely with the noise PSD, provided the relevant integrals exist.

14.4 Gaussian binary decisions

For antipodal signals $\pm Ap(t)$, let $E_p = \int p^2 dt$ and normalize the matched-filter output by kE_p . The statistic becomes $Z = \pm A + N$, with $N \sim \mathcal{N}(0, N_0/(2E_p))$. Equal priors give a zero threshold, so

$$P_b = Q\left(\frac{A}{\sigma_N}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right), \quad E_b = A^2 E_p.$$

This is coherent BPSK error probability. For any two equally likely real waveforms, define squared distance $d^2 = \int [s_1(t) - s_0(t)]^2 dt$. The equivalent difference-waveform detector gives

$$P_b = Q\left(\frac{d}{\sqrt{2N_0}}\right).$$

For equal-energy waveforms with normalized correlation $\rho = \int s_1 s_0 dt / E_b$, $d^2 = 2E_b(1 - \rho)$. Orthogonal coherent binary signals have $\rho = 0$ and error $Q(\sqrt{E_b/N_0})$; antipodal signals have $\rho = -1$. For FSK, verify orthogonality over the actual symbol interval and with the specified phases. Distinct tone frequencies alone do not prove it.

For equally spaced M -PAM levels with adjacent spacing d_a and Gaussian scalar noise variance σ_N^2 , midpoint thresholds give

$$P_s = \frac{2(M-1)}{M} Q\left(\frac{d_a}{2\sigma_N}\right).$$

This is a symbol-error probability; converting it to bit error depends on the labeling and error pattern.

14.5 Prior probabilities and the noise law

For $H_1: Z = A + N$ and $H_0: Z = -A + N$, the minimum-error MAP rule chooses H_1 when

$$\frac{f_N(z - A)}{f_N(z + A)} > \frac{P_0}{P_1}.$$

Gaussian noise gives threshold $\gamma = (\sigma_N^2 / (2A)) \ln(P_0/P_1)$ and

$$P_e = P_0 Q\left(\frac{\gamma + A}{\sigma_N}\right) + P_1 Q\left(\frac{A - \gamma}{\sigma_N}\right).$$

The threshold moves toward the less likely signal, enlarging the decision region of the more likely one.

For Laplace noise $f_N(n) = (\lambda/2) e^{-\lambda|n|}$, the log-likelihood ratio is $\lambda(|z + A| - |z - A|)$. When the switching threshold lies between $-A$ and A ,

$$\gamma = \frac{1}{2\lambda} \ln \frac{P_0}{P_1}, \quad P_e = \sqrt{P_0 P_1} e^{-\lambda A}, \quad \text{SNR} = \frac{A^2}{2/\lambda^2}.$$

The interior-threshold condition is $|\ln (P_0/P_1)| < 2\lambda A$. Otherwise the optimal rule always selects the dominant prior, with error $\min (P_0, P_1)$. Laplace noise is not Gaussian, so a Gaussian Q -function result cannot be reused merely because the noise has a known variance. The appropriate decision rule follows from its actual density.

Solved Exercises for Lesson 14

Exercise 14.1. Matched filter for a flat-topped, linearly falling pulse

IMG_8587.pdf, PDF page(s) 1.

Problem. The input pulse is

$$s(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 2-t, & 1 \leq t < 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find and sketch a matched-filter impulse response. (b) Derive and sketch its signal output. (c) Find the time and value of the maximum absolute output. (d) If the two-sided AWGN PSD is $(2/3) \times 10^{-3}$ W/Hz, determine the maximum output SNR.

Solution.

1. Reverse and delay the pulse. Choose a causal matched filter with sampling time $T = 2$ and unit scale:

$$h(t) = s(2-t) = \begin{cases} t, & 0 < t \leq 1, \\ 1, & 1 < t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Thus $h(t)$ rises linearly from 0 to 1 on $(0, 1)$, remains at 1 on $(1, 2)$, and is zero elsewhere. An arbitrary nonzero gain would scale the signal and noise equally, leaving SNR unchanged.

2. Express the output as a shifted pulse autocorrelation.

$$y_s(t) = (s^* h)(t) = R_s(t-2), \quad R_s(\tau) = \int_{-\infty}^{\infty} s(u) s(u-\tau) du.$$

Let $a = |t-2|$. For $0 \leq a \leq 1$, divide the overlap into three intervals:

$$R_s(a) = \int_a^1 du + \int_1^{1+a} (2-u) du + \int_{1+a}^2 (2-u)(2-u+a) du.$$

For $1 \leq a \leq 2$, only $\int_a^2 (2-u) du$ remains. Therefore

$$y_s(t) = \begin{cases} \frac{4}{3} - \frac{1}{2}a - \frac{1}{2}a^2 + \frac{1}{6}a^3, & 0 \leq a \leq 1, \\ \frac{1}{2}(2-a)^2, & 1 < a \leq 2, \\ 0, & a > 2, \end{cases} \quad a = |t-2|.$$

The graph is nonnegative, symmetric about $t = 2$, and supported on $[0, 4]$. It passes through $(0, 0)$, $(1, 1/2)$, $(2, 4/3)$, $(3, 1/2)$, $(4, 0)$.

3. Identify the maximum. The pulse energy is

$$E_s = \int_0^1 1 dt + \int_1^2 (2-t)^2 dt = \frac{4}{3}.$$

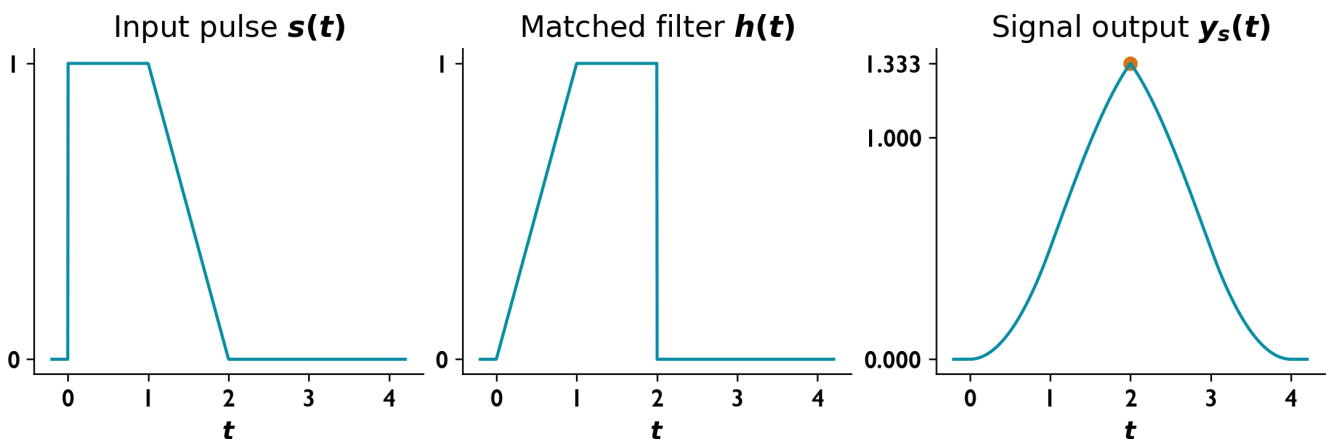
By the Cauchy-Schwarz bound, $|R_s(\tau)| \leq E_s$, with equality at $\tau = 0$. Hence

$$t_{\max} = 2, \quad |y_s(t_{\max})| = \frac{4}{3}.$$

4. Calculate the maximum output SNR. Write the two-sided noise PSD as

$N_0/2 = (2/3) \times 10^{-3}$. The output noise variance is $(N_0/2) E_s$. Therefore

$$\text{SNR}_{\max} = \frac{E_s^2}{(N_0/2) E_s} = \frac{4/3}{(2/3) 10^{-3}} = 2000 \approx 33.01 \text{ dB}.$$



Input pulse, causal matched-filter response, and signal output with peak $4/3$ at $t = 2$.

Exercise 14.2. Coherent FSK and PSK over a finite bandpass channel

IMG_8587.pdf, PDF page(s) 7.

Problem. A unity-gain bandpass channel passes $500 \leq |f| \leq 1500$ Hz and rejects other frequencies.

- a. Binary data are transmitted at 200 bit/s using coherent FSK tones at 1070 and 1270 Hz. Find the error probability when $A^2/\eta = 8000$.
- b. Determine the fastest PSK signaling rate on this channel and its coherent error probability.

Solution.

1. Establish the noise and signal-energy convention. Let the two-sided AWGN PSD be $\eta/2 = N_0/2$, and let each transmitted sinusoid have amplitude A . Its energy over a bit interval T_b is $E_b \simeq A^2 T_b/2$, under the conventional passband-envelope model.

2. Apply the coherent orthogonal-FSK formula. At 200 bit/s, $T_b = 0.005$ s. The tone spacing is 200 Hz, so $(f_2 - f_1) T_b = 1$, the usual orthogonal-spacing condition in the passband-envelope model. Thus

$$\frac{E_b}{N_0} = \frac{A^2 T_b}{\eta/2} = 8000 \frac{0.005}{2} = 20.$$

For equiprobable coherent orthogonal BFSK,

$$P_{e,\text{FSK}} = Q\left(\sqrt{E_b/N_0}\right) = Q\left(\sqrt{20}\right) = 3.8721 \times 10^{-6}.$$

This standard answer assumes the channel does not materially distort the shaped FSK signaling waveforms.

3. State the rate convention needed for PSK. The positive-frequency passband width is $B = 1000$ Hz, centered at 1000 Hz. The statement does not specify pulse shaping, modulation order, or a bandwidth definition, so there is no unique maximum bit rate from the channel plot alone.

For binary PSK with ideal Nyquist pulse shaping and roll-off zero, the passband width is $B = R_b$. Hence

$$R_{b,\text{max}} = 1000 \text{ bit/s}.$$

Keeping the stated average signal-power convention, $E_b/N_0 = A^2 / (2\eta R_b) = 4$. Coherent BPSK gives

$$P_{e,\text{BPSK}} = Q\left(\sqrt{2E_b/N_0}\right) = Q\left(\sqrt{8}\right) = 2.33887 \times 10^{-3}.$$

For raised-cosine roll-off α , the corresponding maximum is $R_b = B / (1 + \alpha)$.

Under the conventional rectangular-pulse first-null bandwidth approximation, $B \simeq 2R_b$, so $R_b \simeq 500$ bit/s. With ideal, undistorted coherent detection, this approximation gives

$$\frac{E_b}{N_0} = 8, \quad P_e \simeq Q(4) = 3.16712 \times 10^{-5}.$$

Rectangular pulses also have sidelobes outside the first-null band; a perfectly brick-wall channel truncates them. Therefore this 500-bit/s calculation is a bandwidth approximation, not an exact error analysis for truncated rectangular waveforms. Higher-order PSK would additionally require an explicitly specified modulation order.

Exercise 14.3. MAP detection of polar data in Laplace noise

IMG_8587.pdf, PDF page(s) 7.

Problem. At the sampling time of a baseband binary receiver,

$$V_0(T_b) = s_0(T_b) + N_0(T_b), \quad f_N(n) = \frac{\lambda}{2} e^{-\lambda|n|}.$$

The signaling is polar: $s_0(T_b) = +A$ for $b_k = 1$ and $-A$ for $b_k = 0$, with prior probabilities 0.25 and 0.75. An optimum receiver uses the optimum threshold. Determine the average signal-to-noise ratio required for $P_e = 10^{-4}$.

Solution.

1. Write the MAP likelihood test. Decide $+A$ when

$$\frac{f_{V|+}(v)}{f_{V|-}(v)} = \exp \{ \lambda(|v+A| - |v-A|) \} > \frac{0.75}{0.25} = 3.$$

For $-A < v < A$, the log-likelihood ratio is $2\lambda v$. Thus the optimum threshold is

$$\gamma = \frac{\ln 3}{2\lambda}, \quad \text{decide } +A \text{ if } v > \gamma.$$

This interior-threshold solution requires $\lambda A > (\ln 3)/2$, which will be satisfied by the requested error probability.

2. Evaluate the two conditional error probabilities. A positive symbol is in error when $N < \gamma - A$, and a negative symbol is in error when $N > \gamma + A$. The Laplace tail is $\Pr(N > x) = \frac{1}{2}e^{-\lambda x}$ for $x \geq 0$. Hence

$$P_e = \frac{11}{42}e^{-\lambda(A-\gamma)} + \frac{31}{42}e^{-\lambda(A+\gamma)}.$$

Substituting $\lambda\gamma = \frac{1}{2} \ln 3$ gives

$$P_e = \frac{\sqrt{3}}{4}e^{-\lambda A}.$$

3. Set the target error probability.

$$\lambda A = \ln \left(\frac{\sqrt{3}}{4 \times 10^{-4}} \right) = 8.37335216.$$

4. Convert to average SNR. Both polar levels have squared magnitude A^2 , so average signal power is A^2 , even with unequal priors. The Laplace noise variance is $\sigma_N^2 = 2/\lambda^2$. Therefore

$$\text{SNR} = \frac{A^2}{2/\lambda^2} = \frac{1}{2} \left[\ln \left(\frac{\sqrt{3}}{4 \times 10^{-4}} \right) \right]^2 = 35.0565 \approx 15.45 \text{ dB}.$$

The optimum threshold favors the more probable negative symbol. Using a zero threshold would fail to exploit the specified prior probabilities.

Lesson 15. Statistical Simulation, Spectral Estimation and Wiener Filtering

Reference: Proakis and Salehi, Chapter 5, Problems 5.55–5.56 and Computer Problems CP5.1–CP5.6, printed pp. 247–254.

15.1 Simulation as a test of a probability model

The computer problems in the reference connect the analytical distributions and spectra to finite numerical records. A simulation produces realizations from a specified model. Its histogram and estimated spectrum fluctuate from run to run; agreement should be assessed with the record length, transient behavior and estimator normalization in view.

Let U be uniform on $(0, 1)$. For a continuous increasing CDF, the inverse-CDF construction $X = F_X^{-1}(U)$ has CDF F_X . More generally, use the quantile $F_X^{-1}(u) = \inf \{x: F_X(x) \geq u\}$, which also handles jumps. Indeed, the event $F_X^{-1}(U) \leq x$ has probability $F_X(x)$. For the reference density $f_X(x) = x/2$ on $0 \leq x \leq 2$, the CDF is $x^2/4$, so $X = 2\sqrt{U}$. An exponential variable of rate λ can similarly be generated by $X = -\ln U/\lambda$.

A density-normalized histogram divides the number of observations in each bin by the total sample count and the bin width. Its approximate area is one. A raw count histogram has a different scale and cannot be compared directly with a theoretical PDF.

15.2 Gaussian generation and the polar construction

Take independent $U_1, U_2 \sim \mathcal{U}(0, 1)$ and form

$$R = \sigma \sqrt{-2 \ln U_1}, \quad \Theta = 2\pi U_2,$$

$$X = R \cos \Theta, \quad Y = R \sin \Theta.$$

The Rayleigh law for R and the uniform phase, together with the polar Jacobian, imply that X and Y are independent $\mathcal{N}(0, \sigma^2)$ variables. This is the Box–Muller method. An implementation must exclude $U_1 = 0$ before taking the logarithm. Add μ to obtain a Gaussian variable with mean μ ; changing the scale changes the standard deviation, not the variance directly.

15.3 Autocorrelation estimates from a finite record

Suppose $x[0], \dots, x[N-1]$ are samples of a zero-mean stationary process with sampling interval T_s . For $0 \leq k < N$, a common estimate is

$$\widehat{R}_X[k] = \frac{1}{N-k} \sum_{n=0}^{N-k-1} x[n+k] x[n].$$

With a known zero mean this estimate is unbiased at each fixed lag. Its uncertainty grows at large lags because fewer products remain. Dividing instead by N gives the biased estimate; its lag weighting has useful spectral positivity properties. Extend a real autocorrelation estimate by $\widehat{R}_X[-k] = \widehat{R}_X[k]$.

If the process has a nonzero mean, estimate or remove it before interpreting the result as covariance. Subtracting an estimated mean introduces additional finite-record bias. The mean-square estimate and variance estimate also differ by the square of the estimated mean. Compare estimates with the ensemble formulas only under an appropriate ergodic or convergence assumption.

15.4 DFT, FFT and periodogram normalization

Define the unnormalized N -point DFT

$$X_m = \sum_{n=0}^{N-1} x[n] e^{-j2\pi mn/N}.$$

The two-sided periodogram at the DFT frequencies is

$$\widehat{S}_X(f_m) = \frac{T_s}{N} |X_m|^2, \quad \Delta f = \frac{1}{NT_s}.$$

After centering the frequency bins, the displayed interval is $[-f_s/2, f_s/2)$, where $f_s = 1/T_s$.

Parseval's identity gives the useful check

$$\sum_m \widehat{S}_X(f_m) \Delta f = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2.$$

The PSD therefore has units of amplitude squared per hertz. The FFT computes the same DFT efficiently; it does not create a different spectral definition. For a one-sided PSD of a real record, double the positive interior bins, while leaving the DC and, when present, Nyquist bins undoubled.

A finite observation window causes spectral leakage. Averaging spectra of several records or segments reduces variability, with a corresponding choice of frequency resolution. A single periodogram can remain noisy even when the record is long. Truncating or windowing an autocorrelation before Fourier transformation also changes the spectral estimate; the window and normalization must be reported.

The inverse Fourier transform of sampled PSD values instead approximates the autocorrelation, rather than a particular noise realization:

$$R_X(\tau_k) \approx \Delta f \sum_{j=0}^{N-1} S_X(f_j) e^{j2\pi f_j \tau_k}, \quad \tau_k = \frac{k}{N\Delta f}.$$

This is an inverse-DFT computation with the factor Δf retained. Frequency-bin ordering and any centering shifts must be handled consistently. A zero-lag check compares the recovered $R_X(0)$ with the sampled spectral area.

15.5 Simulating filtered and bandpass noise

Generate a discrete white Gaussian sequence with a stated variance, then convolve it with a discrete filter. Its sample-domain white PSD is flat over one Nyquist interval; its height in hertz is $\sigma_w^2 T_s$. This finite-band model must be distinguished from ideal continuous-time white noise of infinite bandwidth.

The chapter's computer exercises also use independent uniform inputs: a draw from $(-4, 4)$ has mean zero and variance $16/3$, while a draw from $(-1/2, 1/2)$ has variance $1/12$. Such an i.i.d. sequence is discrete-time white without being Gaussian. Its expected covariance is zero at nonzero lag; a finite estimated covariance fluctuates around that result. Filtering it changes correlation and spectrum, but does not establish Gaussianity merely by linearity.

Frequency-domain filtering multiplies the DFT by a desired response and applies an inverse FFT. For a real output, the transformed record must retain conjugate symmetry. Direct FFT multiplication implements circular convolution; use sufficient zero padding when the intended operation is linear convolution. A rectangular averaging impulse response smooths a white input and produces a colored output with a squared-sinc spectral envelope.

The reference recursion

$$y[n] = 0.9y[n-1] + w[n]$$

has a stable steady-state variance $\sigma_w^2 / (1 - 0.9^2)$ for a zero-mean independent white input of variance σ_w^2 . Its steady-state autocorrelation is

$$R_Y[k] = \frac{\sigma_w^2}{1 - 0.9^2} (0.9)^{|k|}.$$

Its spectrum over one Nyquist interval is $\sigma_w^2 T_s / |1 - 0.9e^{-j2\pi f T_s}|^2$. Starting with $y[-1] = 0$ produces a transient; exclude a suitable initial record or initialize with the stationary distribution before comparing with these formulas.

For a bandpass simulation, generate independent lowpass Gaussian components $n_c[n]$ and $n_s[n]$ with the required common PSD and form

$$n[n] = n_c[n] \cos(2\pi f_c n T_s) - n_s[n] \sin(2\pi f_c n T_s).$$

Choose a sampling rate that accommodates the complete bandpass spectrum. Compare the measured PSD and correlation with the shifted-spectrum and I/Q formulas in Lesson 11. Independent components generate the symmetric case; an asymmetric target generally requires the specified cross spectrum as well.

15.6 Wiener estimation and the orthogonality principle

Let $X(t)$ and $Z(t)$ be jointly WSS zero-mean processes, with $Z(t)$ observed, and seek an LTI estimate $\widehat{X}(t) = h * Z(t)$ of the target $X(t)$. For nonzero means, first center both processes and restore the known target mean afterward. The objective is to minimize $E\left[\left(X - \widehat{X}\right)^2\right]$. At the optimum, the error is orthogonal to every observation used by the estimator. Writing $e(t) = X(t) - \widehat{X}(t)$, the orthogonality condition is

$$E[e(t)Z(t-\tau)] = 0 \quad \text{for every } \tau, \quad R_{XZ}(\tau) = h(\tau) * R_Z(\tau).$$

For any other LTI estimate $g * Z$, orthogonality eliminates the error cross term and gives

$$\text{MSE}(g) = \text{MSE}(h) + E\left[\left((g-h) * Z\right)^2\right] \geq \text{MSE}(h).$$

Thus the condition is sufficient as well as necessary within the stated linear class. For the unconstrained, generally noncausal filter, frequency-by-frequency minimization gives

$$H_{\text{opt}}(f) = \frac{S_{XZ}(f)}{S_Z(f)}, \quad S_Z(f) > 0.$$

The orientation is $R_{XZ}(\tau) = E[X(t+\tau)Z(t)]$, as elsewhere in this course. On frequency sets where $S_Z = 0$, the observations carry no power and the filter value is immaterial to the mean-square error. The minimum error is

$$\text{MSE}_{\min} = \int_{-\infty}^{\infty} \left[S_X(f) - \frac{|S_{XZ}(f)|^2}{S_Z(f)} \right] df,$$

with the quotient used where the denominator is positive. For $Z = X + N$ with zero-mean noise uncorrelated with the target,

$$H_{\text{opt}} = \frac{S_X}{S_X + S_N}, \quad \text{MSE}_{\min} = \int \frac{S_X S_N}{S_X + S_N} df.$$

The filter suppresses frequencies dominated by noise and retains those dominated by the target. A causal filter adds a realizability constraint and generally has a different solution. These formulas complete the estimation concepts introduced in the chapter's exercises without assuming that a spectral ratio is automatically causal.

If $Z = X + N$ and the signal and noise are correlated, then $S_{XZ} = S_X + S_{XN}$ and $S_Z = S_X + S_N + 2 \text{Re } S_{XN}$. Therefore the general form is

$$H_{\text{opt}}(f) = \frac{S_X(f) + S_{XN}(f)}{S_X(f) + S_N(f) + 2 \text{Re } S_{XN}(f)}.$$

The independent-noise formula is recovered by setting the cross spectrum to zero. Omitting the cross terms without checking their statistical assumptions would solve a different estimation problem.

References

John G. Proakis and Masoud Salehi. Fundamentals of Communication Systems. Second edition. Pearson, 2014. Chapter 5 is the core reference; the application lessons identify the additional relevant chapters.

GJWHC course PDFs supplied with the task, including their probability, Gaussian, stationarity, correlation, filtering, binary-sequence and narrowband-noise material.

Chapter 2, supplied notation reference. Taylor Series, supplied presentation and typography reference.

Worked-question sources: IMG_8587.pdf, IMG_8623.pdf, IMG_8641.pdf, and the University of Tripoli EE416 Spring 2025 midterm photograph. The second message-tone frequency in Spring 2025 Q4 is 10^4 Hz, as confirmed by the user.